



# Set Final Specifications

# Setting Final Specifications



Once the concept is finalised, the specifications that originally were only targets, expressed as broad ranges of values, will be **refined** and made **more precise**.

The most challenging part is handling “**Trade-offs**”, which often due to **inverse relationship between two specifications**.

For example,

- (1) Cost Vs. Mass: Lighter is costly
- (2) Power Vs. Mileage: Higher power consumes more fuel

# Process to Set Final Specifications



Five step process for setting Final Specifications:

- (1) Develop **Technical Models** of the Product  
*To predict the value of the metrics for particular set of design decisions*
- (2) Develop a **Cost Model** of the Product  
*To make sure that target cost can be met*
- (3) **Refine** Specification
- (4) **Flow Down** the Specification
- (5) **Reflect** on the Result and the Process



# **Step 1: Develop Technical Model of the Product**

*Set Final Specifications*

# #1: Technical Model

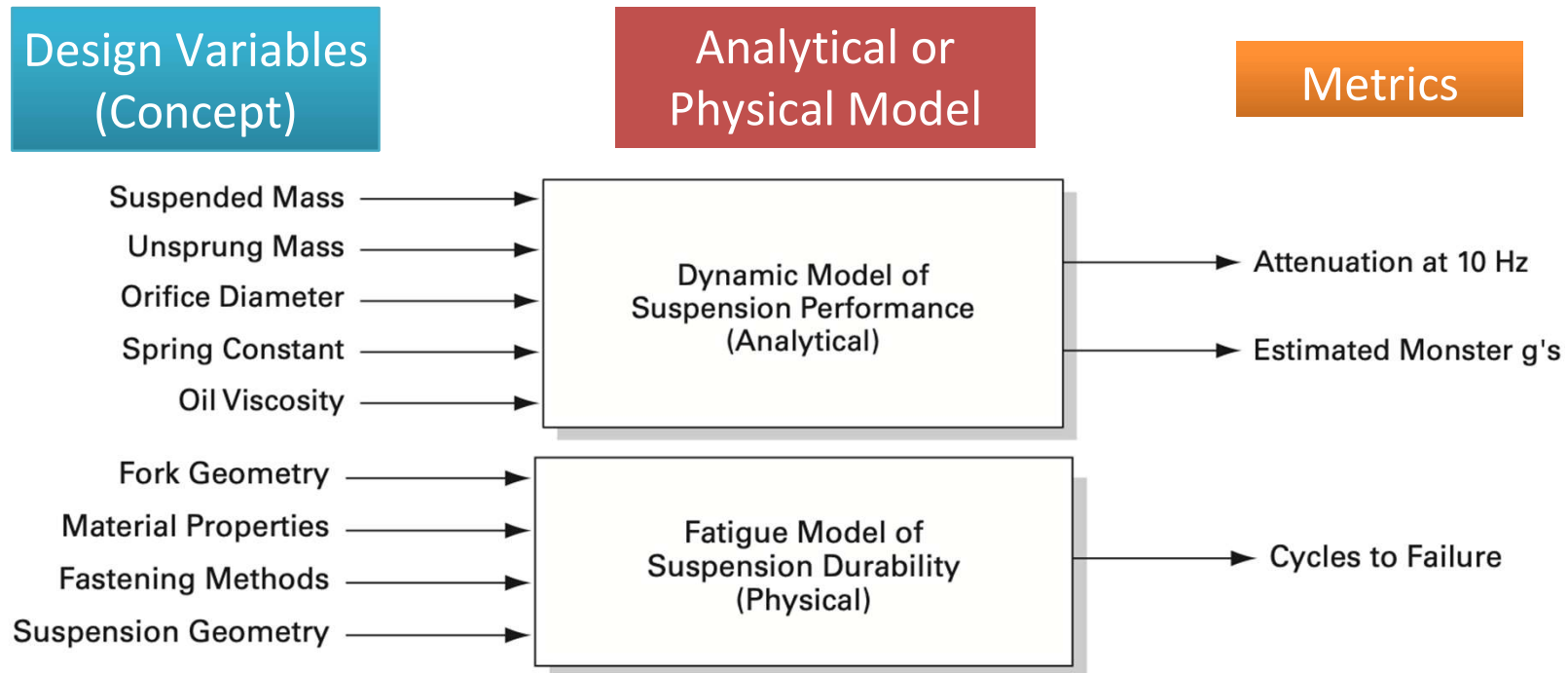


A technical model (analytical or physical) of the product is a tool for **predicting the values of the metrics for a particular set of design decisions**.

Ideally, the team will be able to accurately model the product analytically, perhaps by implementing the model equations in a spreadsheet or computer simulation.

Such a model allows the team to **predict rapidly what type of performance can be expected from a particular choice of design variables**, without costly physical experimentation.

# #1: Technical Model



Technical Models helps the PD team to predict whether any particular set of specifications (such as the ideal target values) is technically feasible by exploring different combinations of design variables



## **Step 2: Develop Cost Model of the Product**

*Set Final Specifications*

## #2: Target Cost



The goal of this step of the process is to make sure that the product can be produced at the target cost.

The first estimates of manufacturing costs are completed by drafting a bill of materials (a list of all the parts) and estimating a purchase price or fabrication cost for each part.



**Think about how much cost to add up to  
increase the mileage by 0.5 Km/l**

## #2: Cost Model - Bill of Materials



| Component                      | Qty/<br>Fork | High<br>(\$ ea.) | Low<br>(\$ ea.) | High Total<br>(\$/fork) | Low Total<br>(\$/fork) |
|--------------------------------|--------------|------------------|-----------------|-------------------------|------------------------|
| Steertube                      | 1            | 2.50             | 2.00            | 2.50                    | 2.00                   |
| Crown                          | 1            | 4.00             | 3.00            | 4.00                    | 3.00                   |
| Boot                           | 2            | 1.00             | 0.75            | 2.00                    | 1.50                   |
| Lower tube                     | 2            | 3.00             | 2.00            | 6.00                    | 4.00                   |
| Lower tube top cover           | 2            | 2.00             | 1.50            | 4.00                    | 3.00                   |
| Main lip seal                  | 2            | 1.50             | 1.40            | 3.00                    | 2.80                   |
| Slide bushing                  | 4            | 0.20             | 0.18            | 0.80                    | 0.72                   |
| Slide bushing spacer           | 2            | 0.50             | 0.40            | 1.00                    | 0.80                   |
| Lower tube plug                | 2            | 0.50             | 0.35            | 1.00                    | 0.70                   |
| Upper tube                     | 2            | 5.50             | 4.00            | 11.00                   | 8.00                   |
| Upper tube top cap             | 2            | 3.00             | 2.50            | 6.00                    | 5.00                   |
| Upper tube adjustment knob     | 2            | 2.00             | 1.75            | 4.00                    | 3.50                   |
| Adjustment shaft               | 2            | 4.00             | 3.00            | 8.00                    | 6.00                   |
| Spring                         | 2            | 3.00             | 2.50            | 6.00                    | 5.00                   |
| Upper tube orifice cap         | 1            | 3.00             | 2.25            | 3.00                    | 2.25                   |
| Orifice springs                | 4            | 0.50             | 0.40            | 2.00                    | 1.60                   |
| Brake studs                    | 2            | 0.40             | 0.35            | 0.80                    | 0.70                   |
| Brake brace bolt               | 2            | 0.25             | 0.20            | 0.50                    | 0.40                   |
| Brake brace                    | 1            | 5.00             | 3.50            | 5.00                    | 3.50                   |
| Oil (liters)                   | 0.1          | 2.50             | 2.00            | 0.25                    | 0.20                   |
| Misc. snap rings, o-rings      | 10           | 0.15             | 0.10            | 1.50                    | 1.00                   |
| Decals                         | 4            | 0.25             | 0.15            | 1.00                    | 0.60                   |
| Assembly at \$20/hr            |              | 30 min           | 20 min          | 10.00                   | 6.67                   |
| Overhead at 25% of direct cost |              |                  |                 | 20.84                   | 15.74                  |
| <b>Total</b>                   |              |                  |                 | <b>\$104.19</b>         | <b>\$78.68</b>         |

### Aspects of BoM:

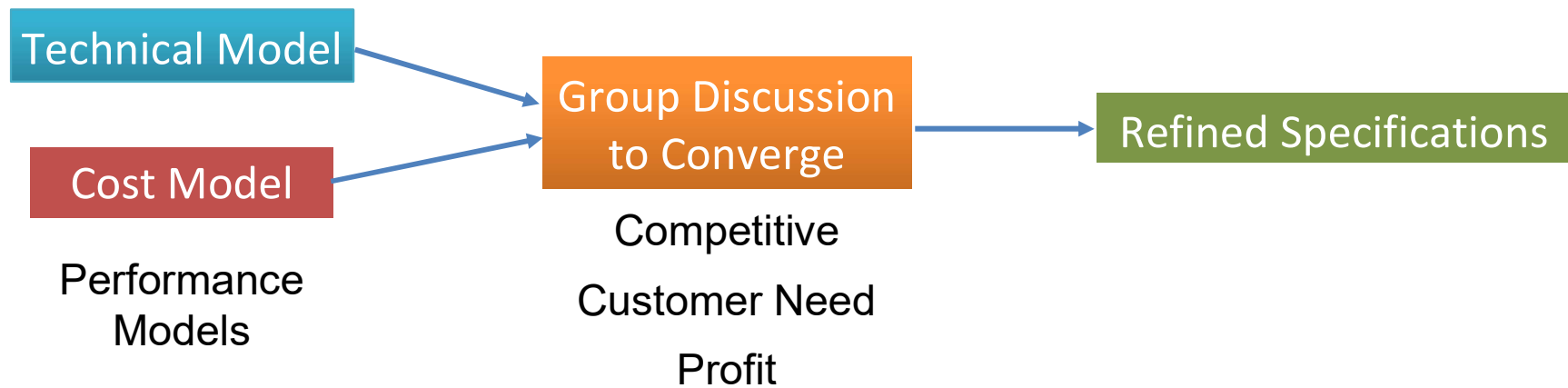
- Rough estimate of manufacturing & assembly cost
- Involves vendors
- Facilitated by Purchaser and Production engineers
- Iterative
- Updated regularly
- Cost performance
- Range = Uncertainty in estimate



# **Step 3: Refine the Specifications, Making Trade-offs Where Necessary**

*Set Final Specifications*

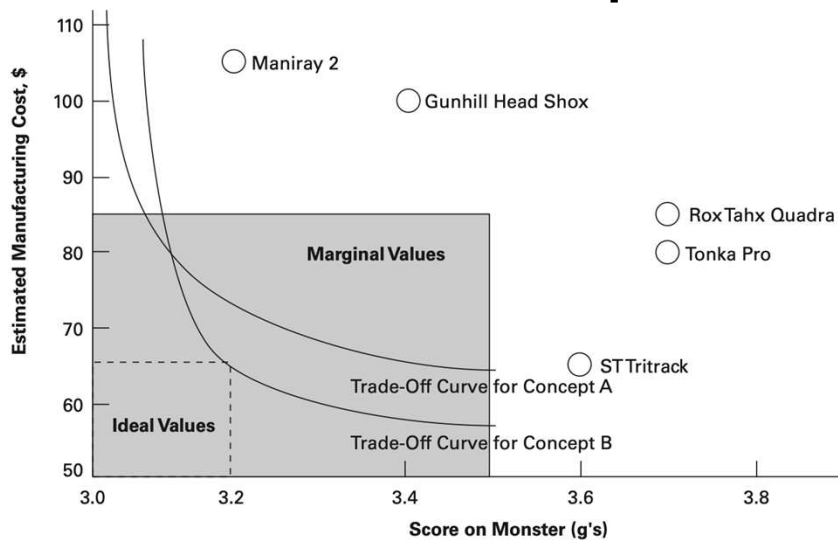
# #3: Refining Specifications



# #3: Trade-off Map & Conjoint Analysis



## Trade-off map



Typically for new products

## Conjoint Analysis

- A model constructed using Customer survey data
- Hypothetical products evaluated
- Identifies specification values that maximise market share

Typically for mature products

# #3: List of Final Specifications



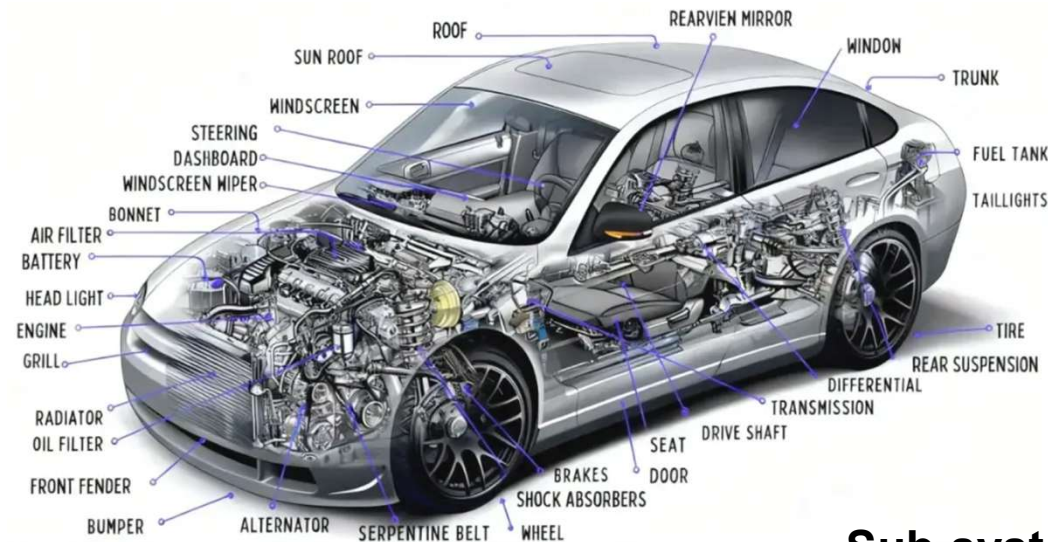
| Metric No. | Need Nos. | Metric                                         | Imp. | Units | Marginal Value | Ideal Value | Value   |
|------------|-----------|------------------------------------------------|------|-------|----------------|-------------|---------|
| 1          | 1, 3      | Attenuation from dropout to handlebar at 10 Hz | 3    | dB    | >10            | >15         | 12      |
| 2          | 2, 6      | Spring preload                                 | 3    | N     | 480–800        | 650–700     | 600–650 |
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| 6          | 5         | Maximum travel (26-in. wheel)                  | 3    | mm    | 33–50          | 45          | 43      |
| 7          | 5         | Rake offset                                    | 3    | mm    | 37–45          | 38          | 38      |
| 8          | 6         | Lateral stiffness at the tip                   | 3    | kN/m  | >65            | >130        | >75     |
| 9          | 7         | Total mass                                     | 4    | kg    | <1.4           | <1.1        | <1.4    |



# Step 4: Flow Down the Specifications

*Set Final Specifications*

# #4: Flow Down to Sub-system



## Car Specification

Fuel economy  
Acceleration  
Turning radius

*Reflect overall spec.  
Equally challenging specs.*

## Sub-system

## Engine Metrics

Power  
Torque  
Fuel consumption

<https://www.xavier-parts.com/automotive-parts-manufacturing/>



# **Step 5: Reflect on the Results and the Process**

*Set Final Specifications*

# #5: Reflection



- (1) Is the product a winner?  
*Will it meet the customer needs and excel competitively*
- (2) Uncertainty in the models
- (3) Concept suited for the target market?  
*Dramatically superior product?*
- (4) Scope of improvement in the future



# Quality Function Deployment (QFD)

# Overview of QFD

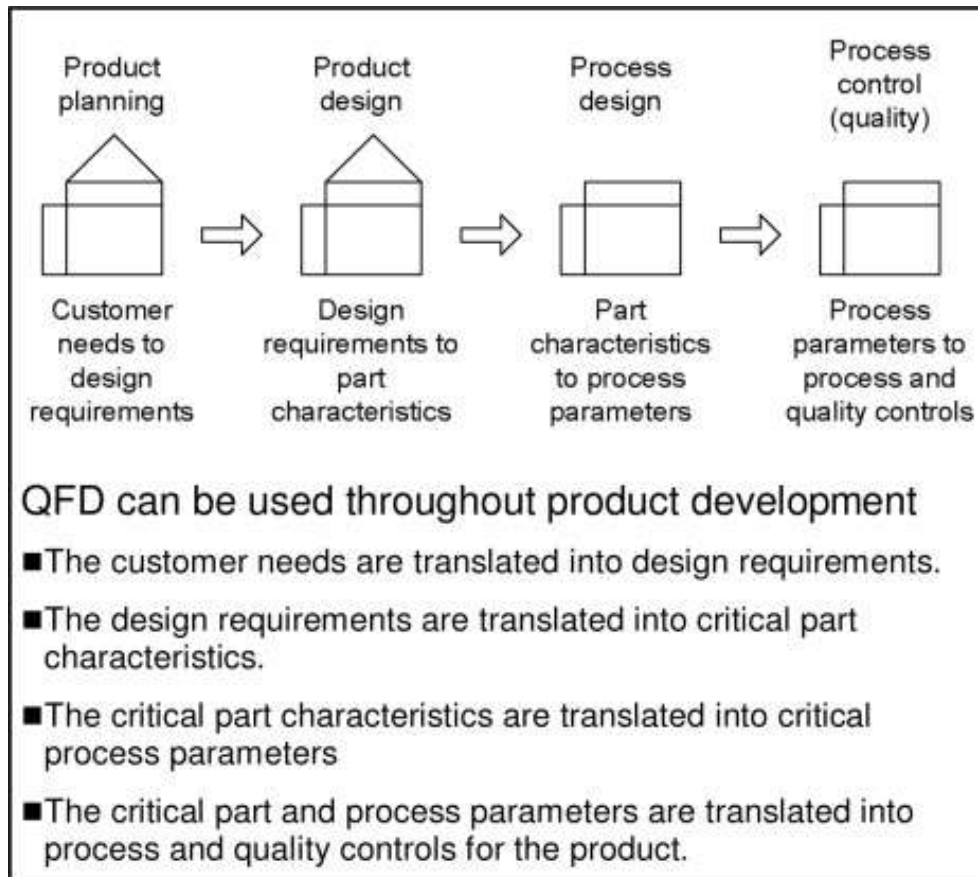


Quality Function Deployment (QFD) is one of the popular technique used to generate engineering specifications. It was developed in Japan in the mid-1970s. The goal of QFD is:

- Increase customer satisfaction
- Reduce the cycle time of product development
- Increase competitiveness

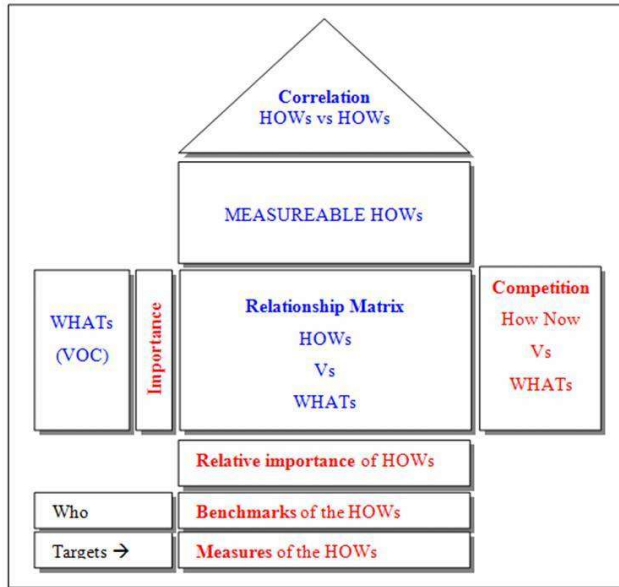
**Toyota use case:** Toyota was able to reduce the costs of bringing a new car model to market by over 60%, and decrease the time required for its development by one-third.

# QFD Approach

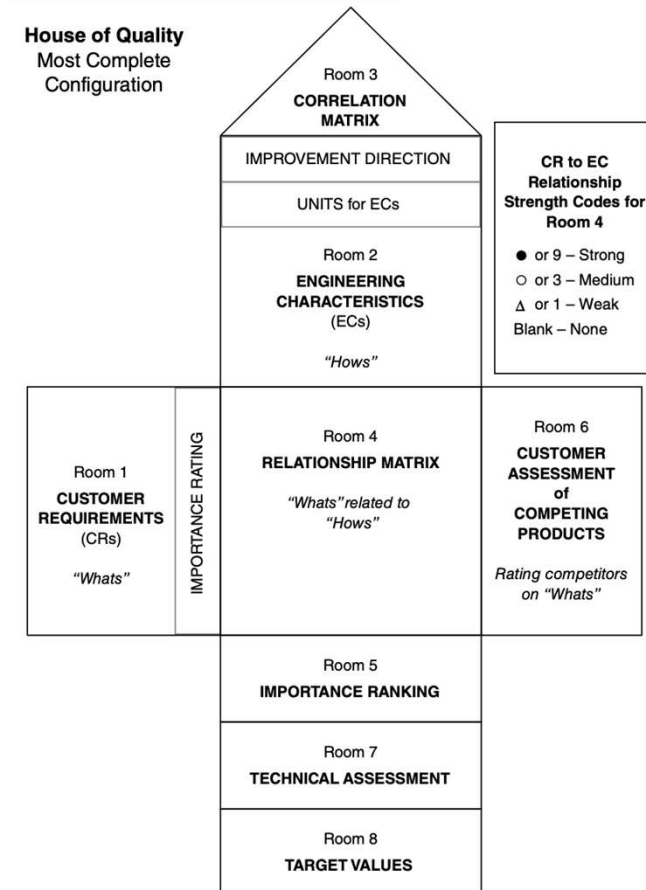


QFD is a **planning** and **team problem-solving tool** that has been adopted by a wide variety of companies as the tool of choice for focusing a design team's attention on satisfying customer needs throughout the product development process

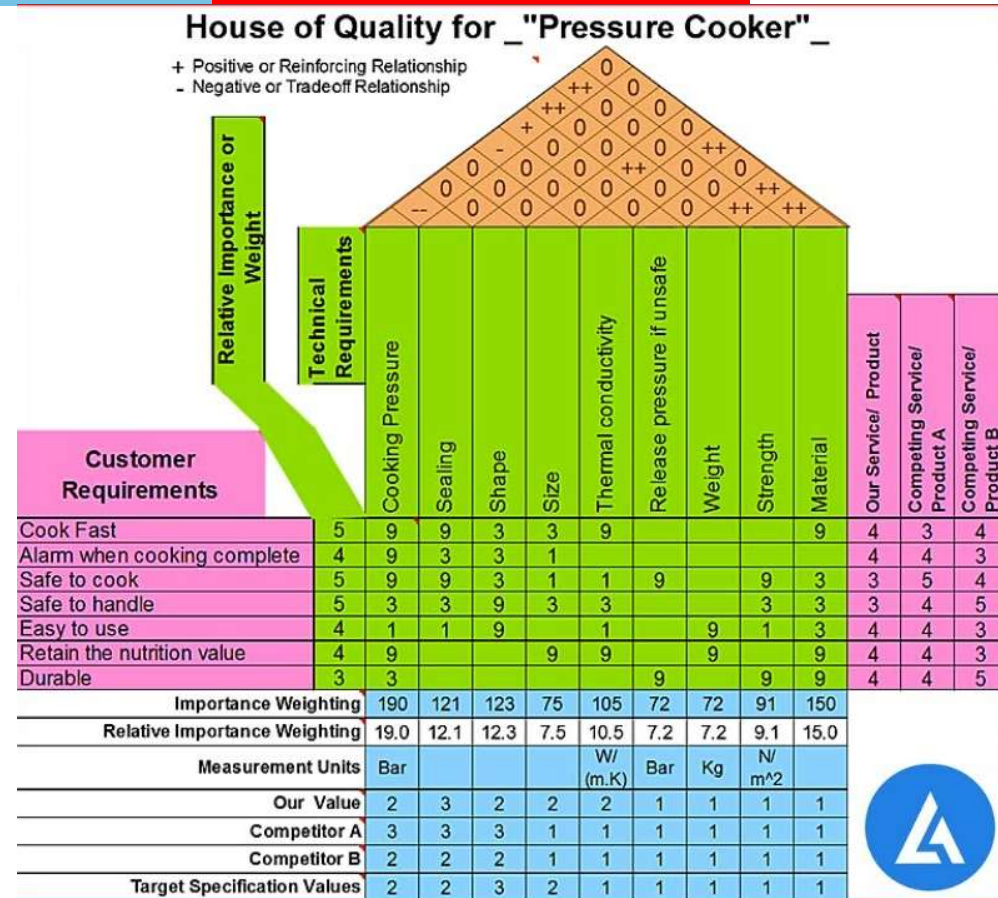
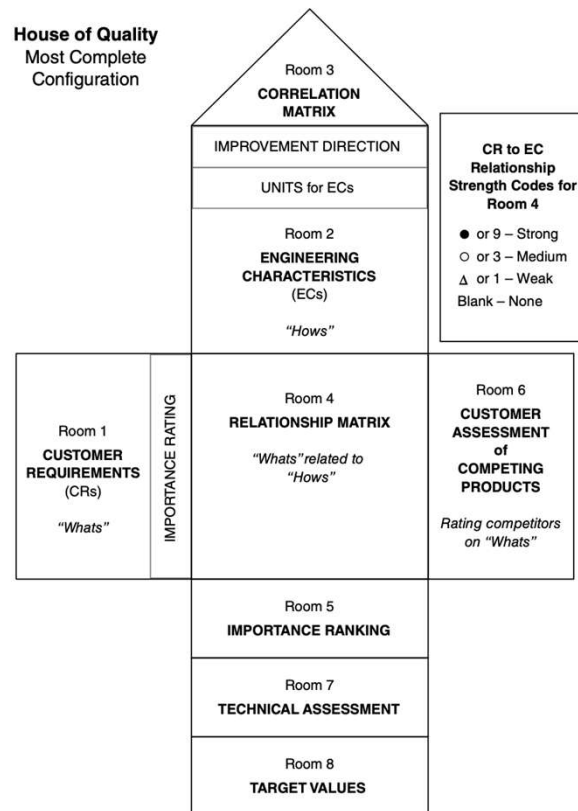
# House Of Quality (HOQ)



The HOQ develops the relationships between what the customer wants from a product and which of the product's features and overall performance parameters are most critical to fulfilling those wants.



# Example: Pressure Cooker



# Advantages of QFD



- (1) **Reduces the time** required for the planning process
- (2) Focuses completely on **customer needs** and requirements
- (3) **Reduces design changes** and alterations
- (4) Improved house of **quality** of the products
- (5) High customer **satisfaction**
- (6) Decreased design and manufacturing **costs** and overheads
- (7) Reduces **time to market** the product
- (8) Helps in **prioritising** various design parameters
- (9) Aids in **benchmarking** the processes

# Disadvantages of QFD



- Ambiguity of Categories
- Complex Matrix
- Difference of Opinion
- Not Universally Applicable
- Labour-Intensive Process
- Comprehensive and Extensive Method
- Time Consuming
- Based on Qualitative Data



# Summary

# Summary



- (1) Customer needs are generally expressed in the “language of the customer”
- (2) Specification is a precise and measurable detail of what the product has to do to be commercially successful
- (3) Specifications are typically established at least twice.  
*Immediately after identifying the customer needs, the team sets target specifications*  
*After concept selection and testing, the team develops final specifications*
- (4) Target specifications represent the hopes and aspirations of the team
- (5) Final specifications are developed by assessing the actual technological constraints and the expected production costs using analytical and physical models

# Summary



- (6) Tools such as the list of metrics, the needs-metrics matrix, the competitive benchmarking charts, and the competitive/Trade-offs maps provides the team with a way to represent and discuss the specifications
- (7) The specifications process requires active participation from team members representing the marketing, design, and manufacturing functions



**Thank You!**



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# **Product Design**

**(AE ZG531/ AEL ZG531/ DE ZG541/DM ZG541/MM ZG541)**



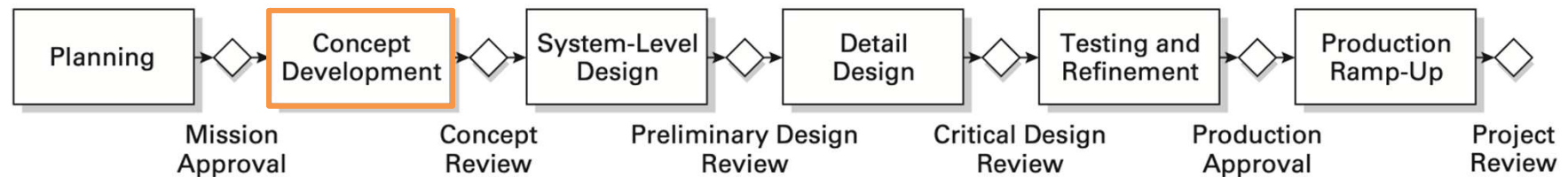
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# Session 5: Establishing Product Function

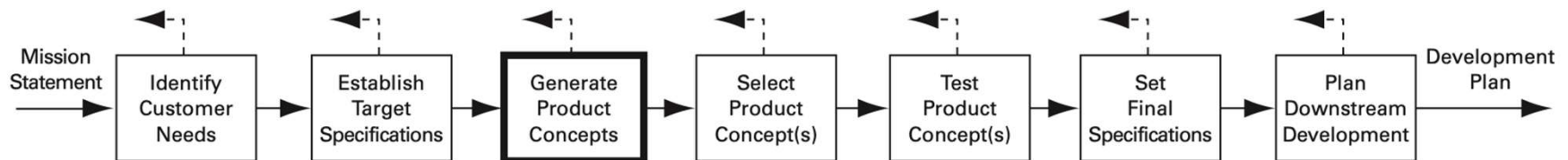
# When Does it Happen?



## Generic PD Process

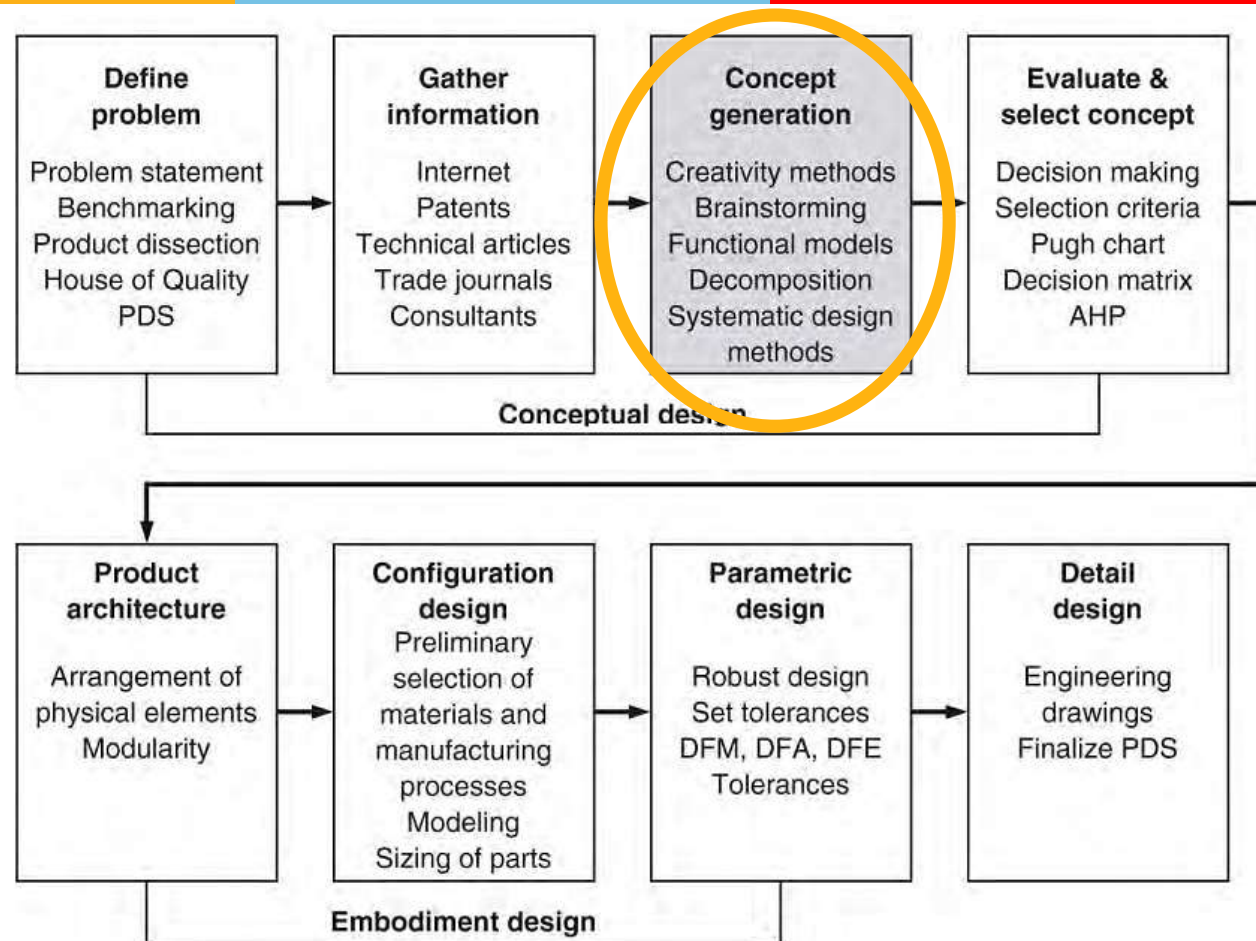


## Concept Development

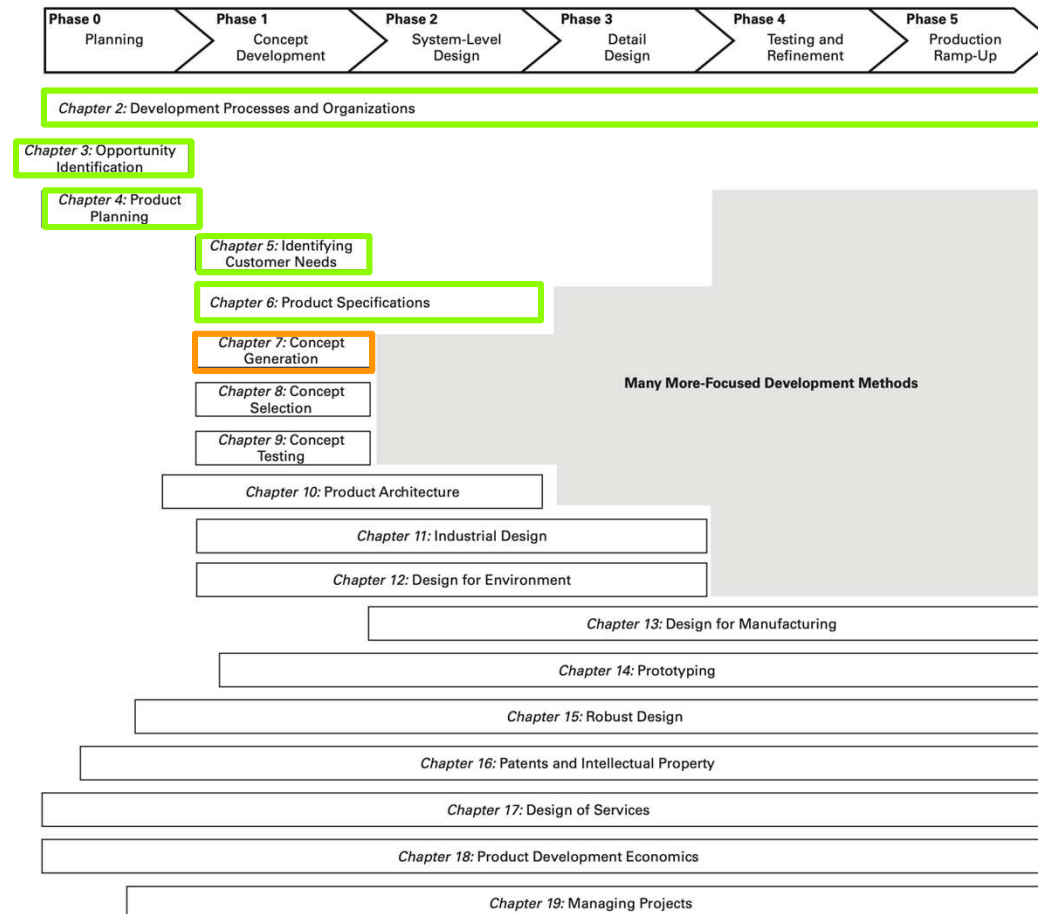


Inexpensive & Quick, e.g. 5% of budget, 15% of PD time

# Product Development Process



# Course Roadmap



# Product Specification



| Metric No. | Need Nos. | Metric                                         | Imp. | Units | Marginal Value | Ideal Value | Value   |
|------------|-----------|------------------------------------------------|------|-------|----------------|-------------|---------|
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## What next?

- Identity existing solution concepts
- Identify new concepts
- Approach to concept generation

# What is a Concept?

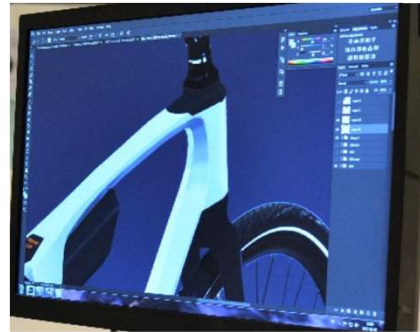
Concept is an approximate description of:

- Technology
- Working principles
- Form

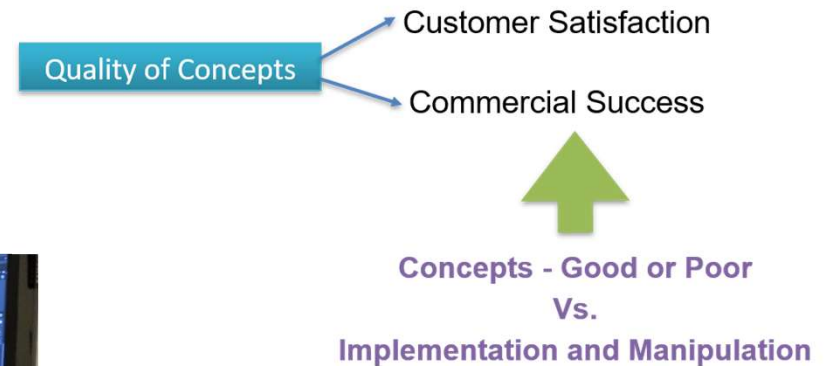
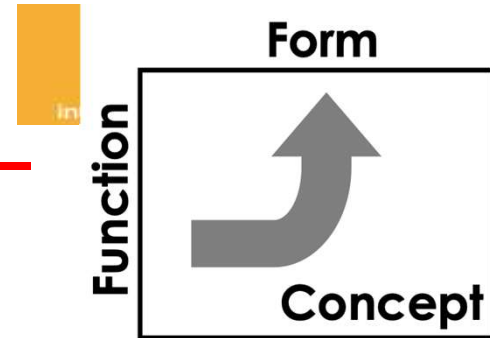
Often expressed as:



Sketch



3D Model



**Is not a product/system attribute, but a mapping**

# Thinking Processes that Lead to Creative Ideas

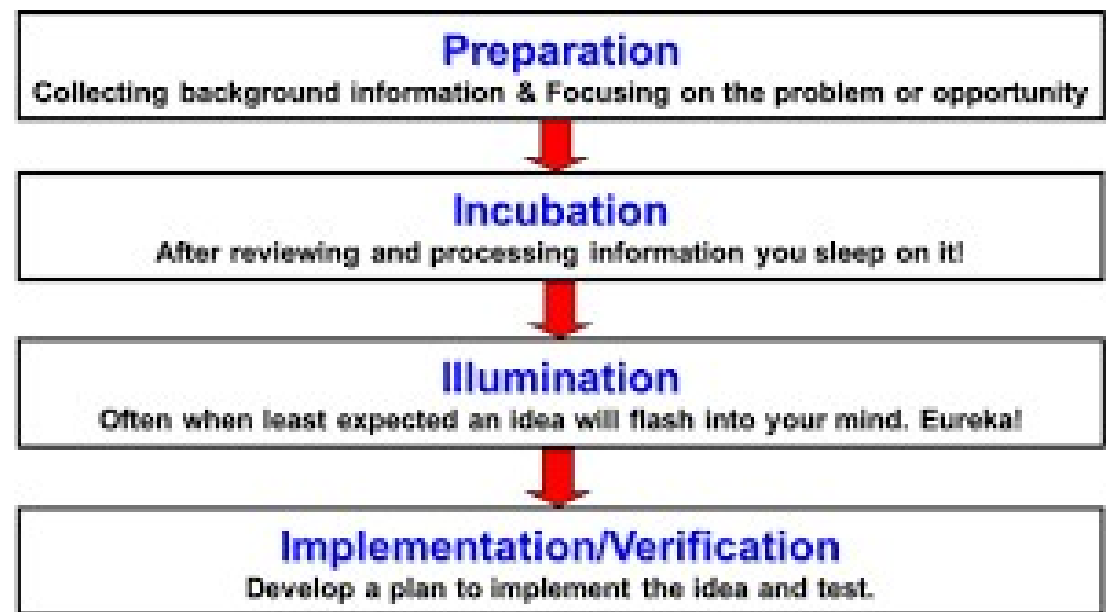


- Aids to Creative Thinking:

- Develop a creative attitude
- Unlock your imagination
- Be persistent
- Develop an open mind
- Suspend your judgment
- Set problem boundaries



- Creative thinking process

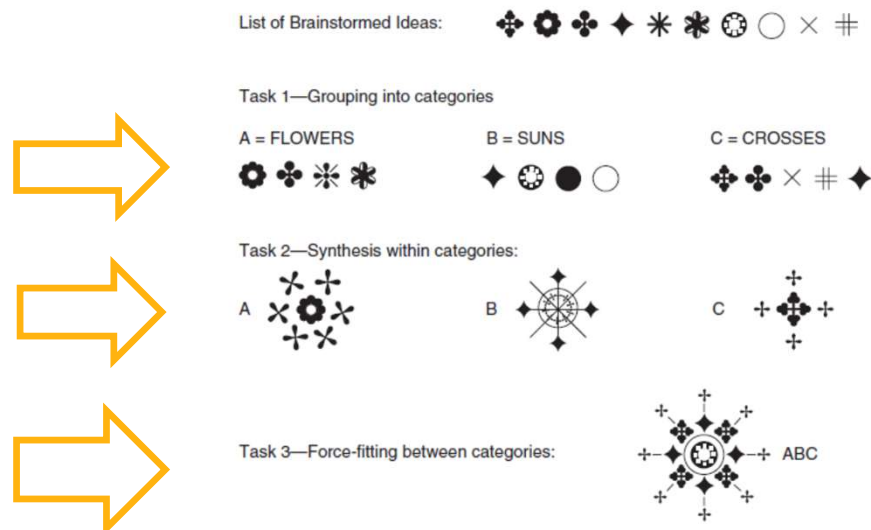


# Creative Methods for Design



- **Refinement and Evaluation of Ideas:**

- Ideas that are feasible.
- Ideas that may have potential after more thought or research are applied.
- Ideas that are very unfeasible and have no chance of becoming good solutions.



# Types of Mental Blocks



Perceptual  
Blocks

Not properly defining / not recognizing

Emotional  
Blocks

Psychological

Cultural Blocks

Thought process

Intellectual  
Blocks

Strategy, background and knowledge

Environmental  
Blocks

physical / social

# Perceptual Blocks



- **Stereotyping**
  - Thinking **conventionally** or in a formulaic way about an event, person, or way of doing something
- **Information overload**
  - You become so **overloaded with minute details** that you are unable to sort out the critical aspects of the problem
- **Limiting the problem unnecessarily**
  - Broad statements of the problem help keep the mind open to a wider range of ideas
- **Fixation**
  - People's thinking can be influenced so greatly by their **previous experience** or some other bias that they are not able to sufficiently recognize alternative ideas
- **Priming or provision of cues**
  - If the thinking process is started by giving examples or solution cues, it is possible for thinking **to stay** within the realm of solutions suggested by those initial starting points

# Emotional Blocks



- **Fear of risk taking**

- This is the fear of proposing an idea that is ultimately found to be faulty

- **Unease with chaos**

- People in general, and many engineers are uncomfortable with highly unstructured situations

- **Inability or unwillingness to incubate new ideas**

- In our busy lives, we often don't take the time to let ideas lie dormant so they can incubate properly

- **Motivation**

- People differ considerably in their motivation to seek creative solutions to challenging problems

# Cultural Blocks



- **Thought patterns**
  - People acquire a set of **thought patterns** from living in a culture
- **Tradition bound**
  - Certain industries are **tradition bound** and are **reluctant to change**, even though profitability is decreasing
- **Attitudes**
  - Countries even **differ in their attitudes** toward creative problem solutions (**political, educational & business culture**)
  - bankruptcy

# Intellectual Blocks



- **Poor choice of problem-solving language or problem representation**
- **Memory block**
  - Hindrance due to variations in remembering based on different types of memory
- **Insufficient knowledge base**
  - Generally, ideas are generated from a person's education and experience
- **Incorrect information**
  - It is obvious that using incorrect information can lead to poor results

# Environmental Blocks



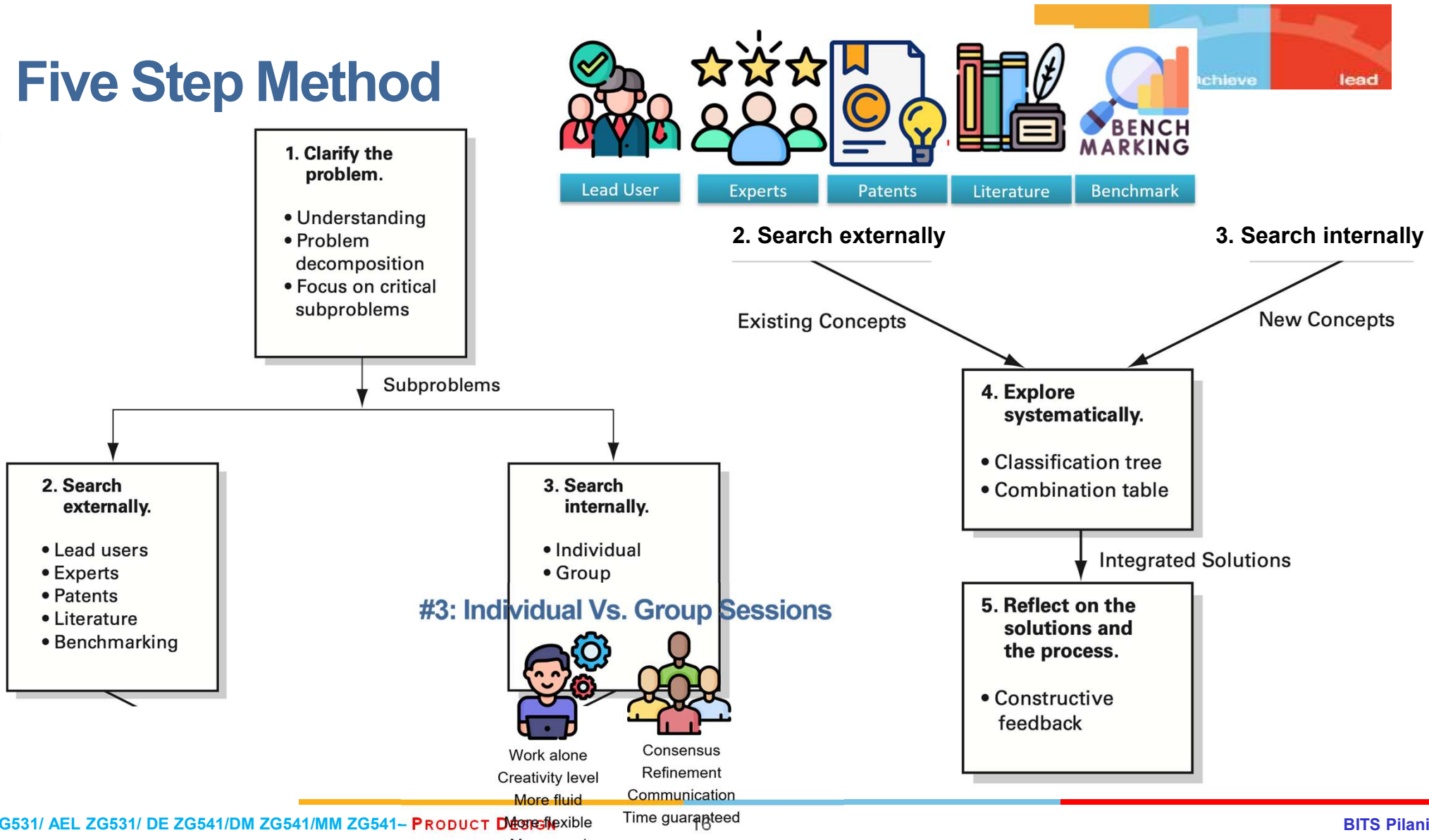
- **Physical environment**

- Ambiance, weather, space etc

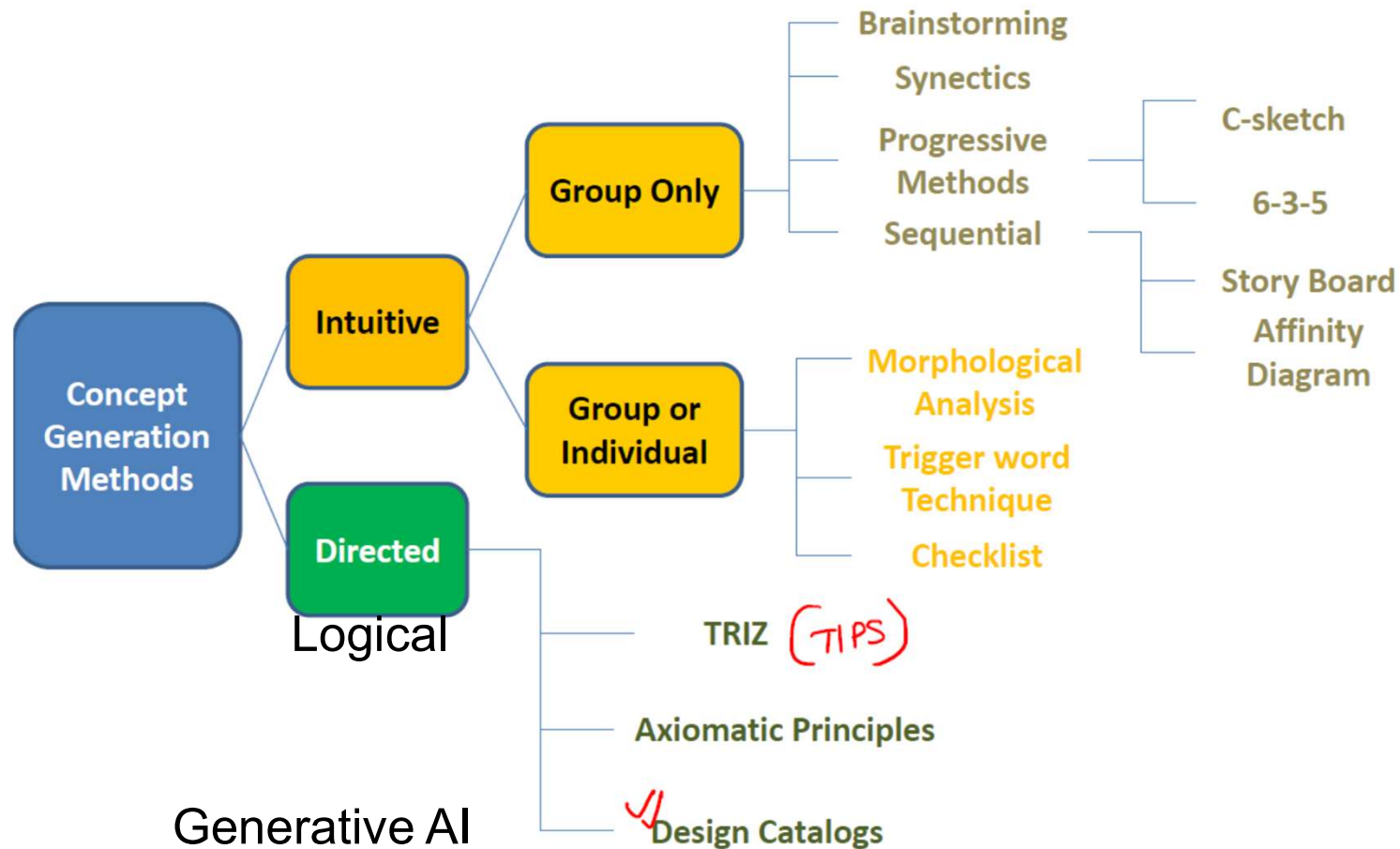
- **Criticism**

- Nonsupportive remarks about your ideas can be personally hurtful and harmful to your creativity
- Important for the team to maintain an atmosphere of support and trust

# Five Step Method



# Creative Thinking Methods



# Intuitive Methods



- Brainstorming
- 6-3-5 Method / Brainwriting
- C-Sketch method
- Concept maps
- Affinity diagram



# #1: Enhance Creative Thinking



How to enhance one's creative thinking?



Creative  
Attitude

Self-confidence



Unlock  
Imagination

Why?  
What-if?



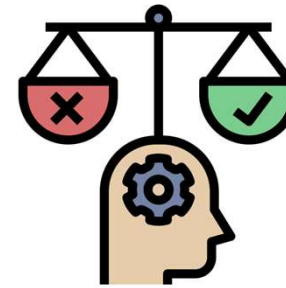
Persistent

Hard work



Open Mind

Receptive



Suspend  
Judgement

Do not Criticise



Set  
Boundaries

Problem  
boundaries

[www.flaticon.com](http://www.flaticon.com)

innovate      achieve      lead



- Defer judgment
- Reach for quantity

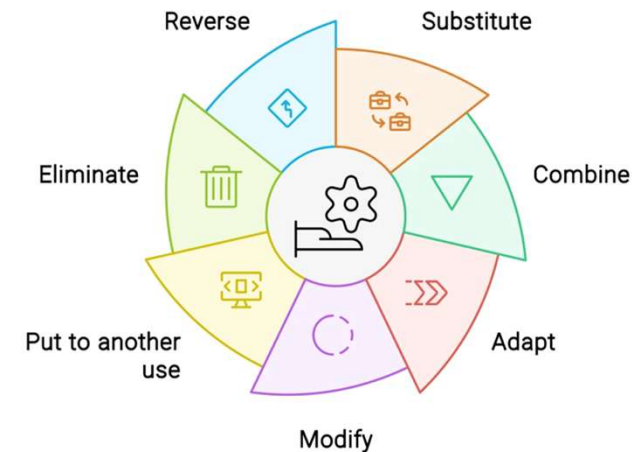
1. Go for quantity
2. Withhold criticism
3. Welcome wild ideas
4. Combine and improve ideas

# Creative Thinking Methods



- **Brainstorming:**

- The overall aim of brainstorming is to obtain several creative ideas that might work as solution principles for development of product.
- All team members are encouraged to be open and uninhibited during the initial session of brainstorming.
- Generally, group with 5-15 people participate in brainstorming session. Usually, session will continue for 30-45 minutes.
- Avoid hierarchically structured groups.



SCAMPER checklist is used as a stimulant  
for new ideas.

# Brainstorming Method



- **Rules to be observed during a session:**

- Members must shed their inhibitions: avoid rejecting ideas as absurd, stupid, embarrassing or false Members must not criticize ideas generated (killer phrases forbidden)
- Members are encouraged to freely change, develop further or combine other ideas.
- All ideas are to be recorded and displayed, and the practicality of ideas is ignored.
- The wilder the idea, the better it is (easier to tone down wilder ideas than to improve mediocre ideas).

# #1: SCAMPER Success Stories



- **LEGO:** The iconic toy company has consistently relied on the SCAMPER technique to develop new product lines and explore unique building concepts. By applying the “Combine” prompt, LEGO created the highly successful LEGO Movie product line, combining the physical building blocks with a storytelling experience.
- **Apple:** Known for its innovative products, Apple has utilized the SCAMPER technique to reimagine existing technologies and create disruptive innovations. The development of the iPod, for instance, can be attributed to the “Put to another use” prompt, where Apple reimaged the way users could store and access digital music.
- **Procter & Gamble (P&G):** As a consumer goods giant, P&G has embraced the SCAMPER technique to drive product innovation. The development of the Tide Pod, a compact and convenient laundry detergent solution, stemmed from the “Modify” and “Eliminate” prompts, where P&G sought to simplify the traditional detergent packaging and application process.

<https://www.6sigma.us/lean-tools/scamper-technique/>

# Checklist for Brainstorming Session

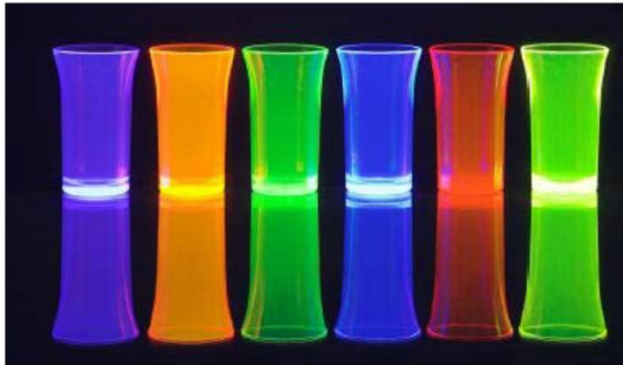


| Proposed Change         | Description                                                                                                                                                                                                        |
|-------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Substitute              | What if used in a different material, process, person, power source, place, or approach?                                                                                                                           |
| Combine                 | Could I combine units, purposes, or ideas?                                                                                                                                                                         |
| Adapt                   | What else is like this? What other idea does it suggest? Does the past offer a parallel? What can I copy?                                                                                                          |
| Modify, magnify, minify | Could I add a new twist? Could I change the meaning, color, motion, form, or shape? Could I add something? Make stronger, higher, longer, thicker? Could I subtract something?                                     |
| Put to other uses       | Are there new ways to use this as is? If I modify it, does it have other uses?                                                                                                                                     |
| Eliminate               | Can I remove a part, function, person without affecting outcome?                                                                                                                                                   |
| Rearrange, reverse      | Could I interchange components? Could I use a different layout or sequence? What if I transpose cause and effect? Could I transpose positive and negative? What if I turn it backward, upside down, or inside out? |

# Checklist Example



- **Modify?** New twist? Change meaning? Change colour, motion, odour, form, shape? Other changes?  
[colour TV; coloured plastics; variously coloured toothbrushes, deodorants, scented erasers, acoustical aids]



# Checklist Example



- **Rearrange?** Interchange components? Other pattern? Other layout? Other sequence? Transpose cause and effect? Change place? Change schedule?

[Rear engine automobiles, idea of bonus or incentive pay, modern housing plans]

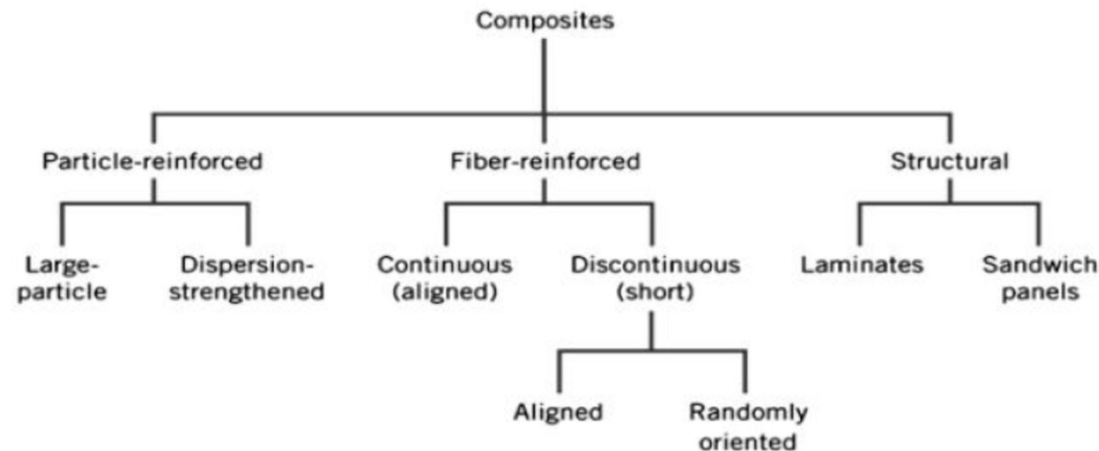


# Checklist Example



- **Substitute?** What else instead? Who else instead? Other ingredient? Other material? Other process? Other power? Other place? Other approach?

[FRP bodies of automobiles, converting manually operated systems to fuel or electronically operated ones and vice versa, alternative fuels, synthetic fibres replacing cotton]



# Checklist Example



- **Reverse?** Transpose positive and negative? How about opposites? Turn it backward? Turn it upside down? Reverse roles? Change shoes? Turn tables? Turn other cheek?

[Escalator after staircase, shaper vs planer, reversible dresses, soft shoes to be worn on either foot, deriving new linkage mechanisms by inversion]



# Checklist Example



- **Combine?** How about a blend, an alloy, an assortment, an ensemble, Combine units? Combine purposes? Combine ideas?

[Various alloys and new materials like composites, hybrid plants and crops, two in one's]



# Checklist Example



- **Magnify?** What to add? More time? Greater frequency? Stronger? Higher? Longer? Extra value? Plus ingredient? Duplicate? Multiply? Exaggerate?  
[Economy size packages, photo-enlargements, king-size cigarettes, 3 stockings – a pair with a spare, long-playing records, filter tipped cigarettes, deluxe versions]

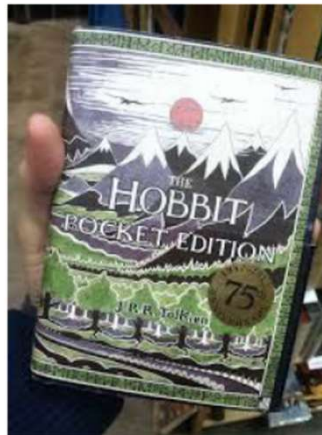


# Checklist Example



- **Minify?** What to subtract? Smaller? Condensed? Miniaturize? Lower? Shorter? Lighter? Omit? Streamline? Split up? Understate?

[Miniaturize components in electronics, microfilm, pocket editions of books, mini dresses, split-level highways]



# Brainstorming sessions



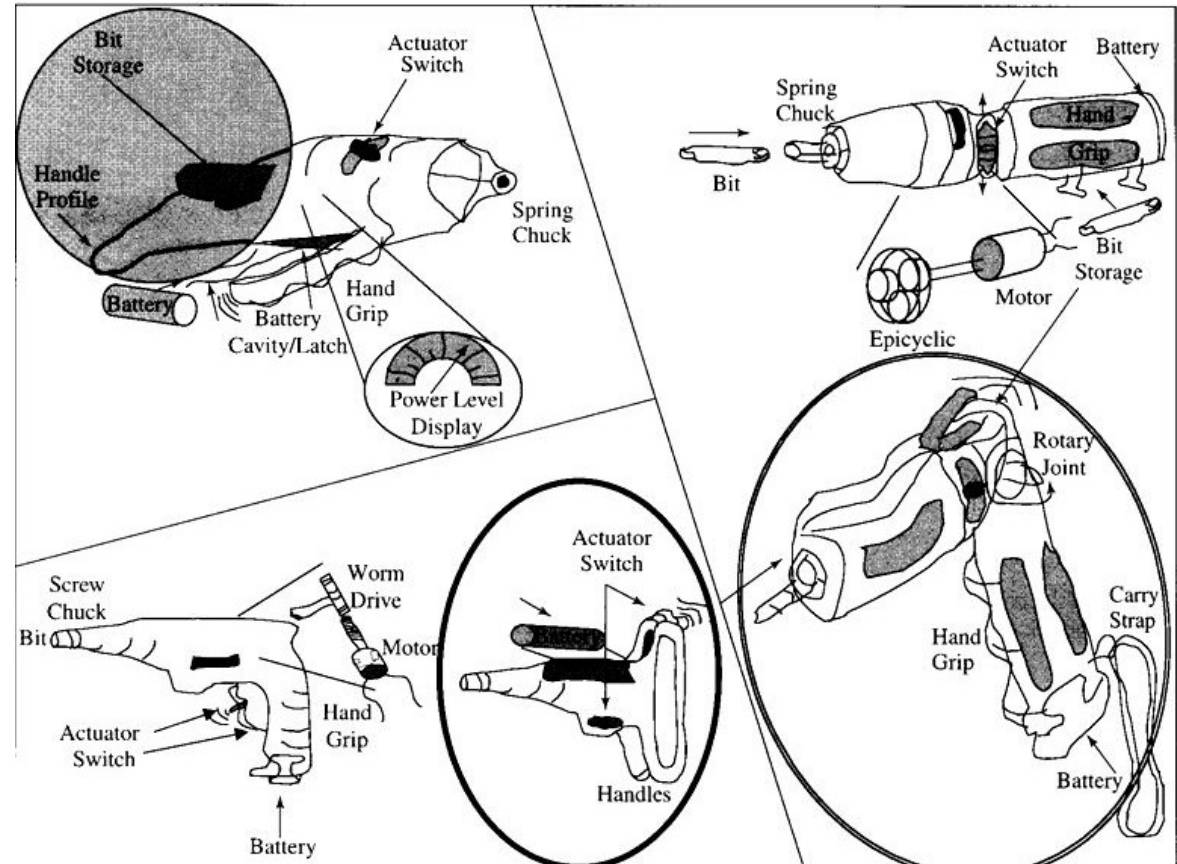
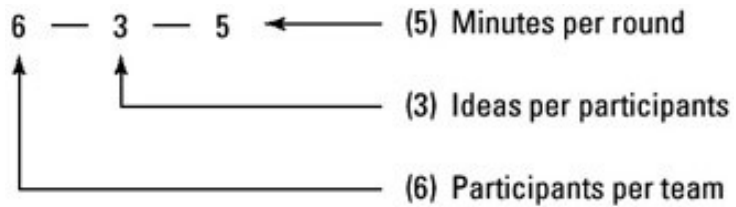
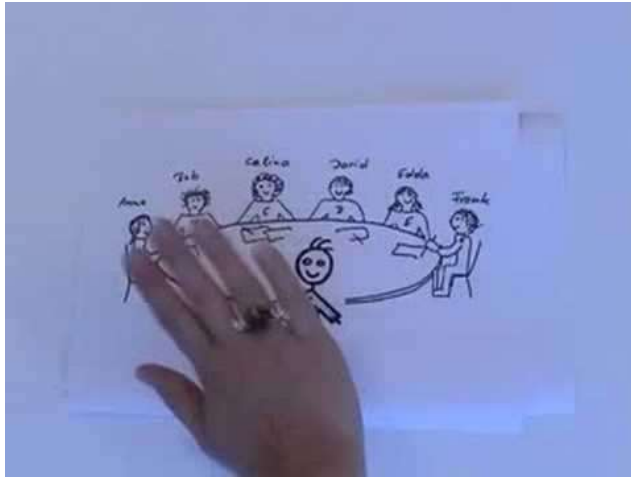
## Methods to improving brainstorming sessions

- Stick to the rules
- Pay attention to everyone's ideas
- Include both individual and group approaches
- Take breaks
- Do not rush
- Stay persistent
- Facilitate the session

## Drawbacks:

- Right idea may not come at right time
- Group conventions may sidetrack original ideas
- Distractions by misdirected focus
- Certain team members may dominate the discussion
- The team may not be open to new ideas

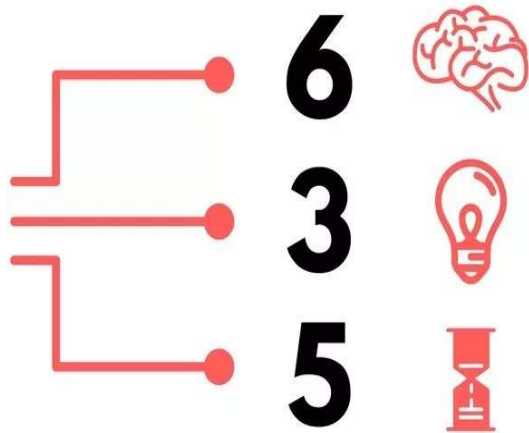
## 6-3-5 Method (Brainwriting Method)



## 6-3-5 Brainwriting



### BRAINWRITING



6 participants proposent  
3 idées en 5 minutes

- It is fundamental to assure that all participants share a deep background knowledge on the topic of the brainwriting session.
- One of the main advantages of using 6-3-5 brainwriting is that it is a very straightforward method and therefore is easy and quick to learn.
- Issues in clarity of ideas.
- Stress due to time constraints might cause quality of ideas to decrease.
- Risk of clash of similar ideas.

# Brainwriting Method Advantages and Disadvantages

---



- **Advantages:**

- The procedure and rules are easy to communicate.
- The non-verbal brainstorming – promotes a creative and open exchange.
- Introverted employees are also encouraged to participate in concept generation.
- No minute-taker is needed, no idea is lost due to lack of documentation.

- **Disadvantages:**

- Lack of spontaneity,
- The need for concrete formulation, and
- The possibility of multiple entries due to the simultaneous, anonymous writing down of ideas

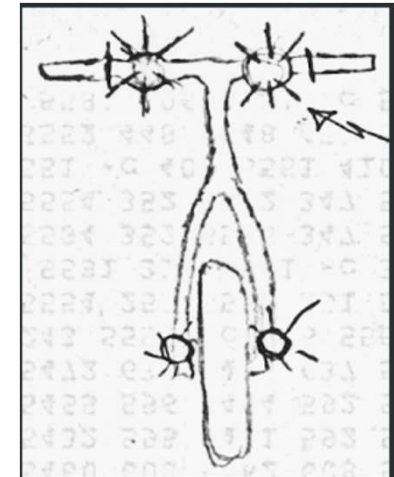
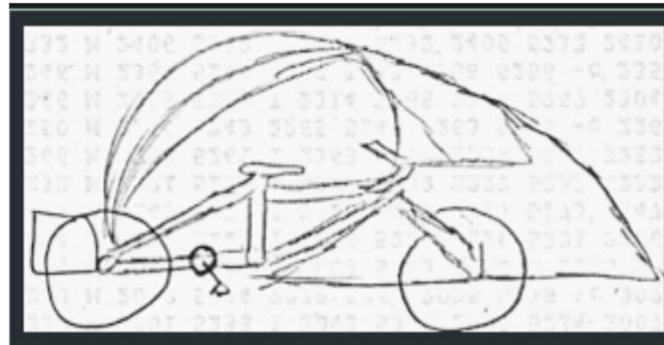
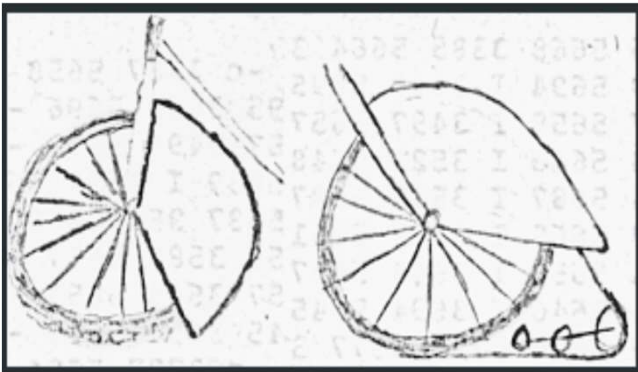
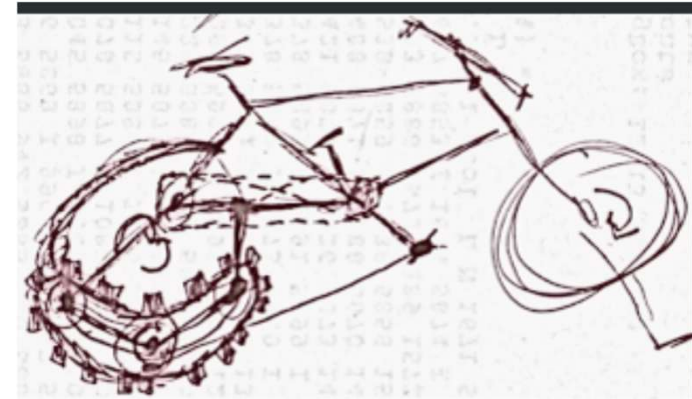
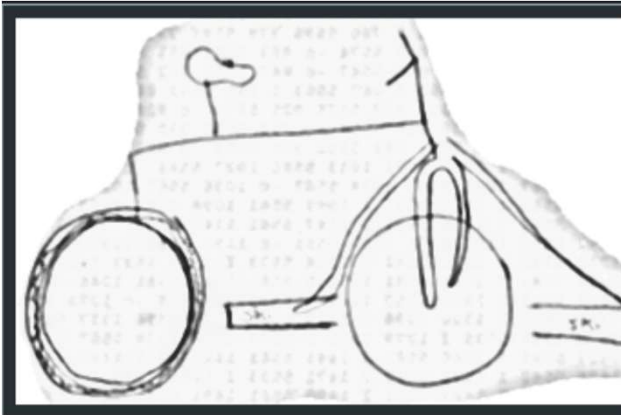
# Collaborative Sketching (C-Sketch) Method

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- The C-Sketch method of concept development requires team members to draw sketches of possible design solutions.
- Subsequent rounds are used to add to each sketch in attempts to create a progressive sketch.
- Each team member starts by drawing a sketch, and then every couple minutes, that sketch is passed to a different team member. This is done until each drawing has made it to each team member.
- The ideas that are fostered in this stage will be used to drive the final design.

# C-Sketch Method Example

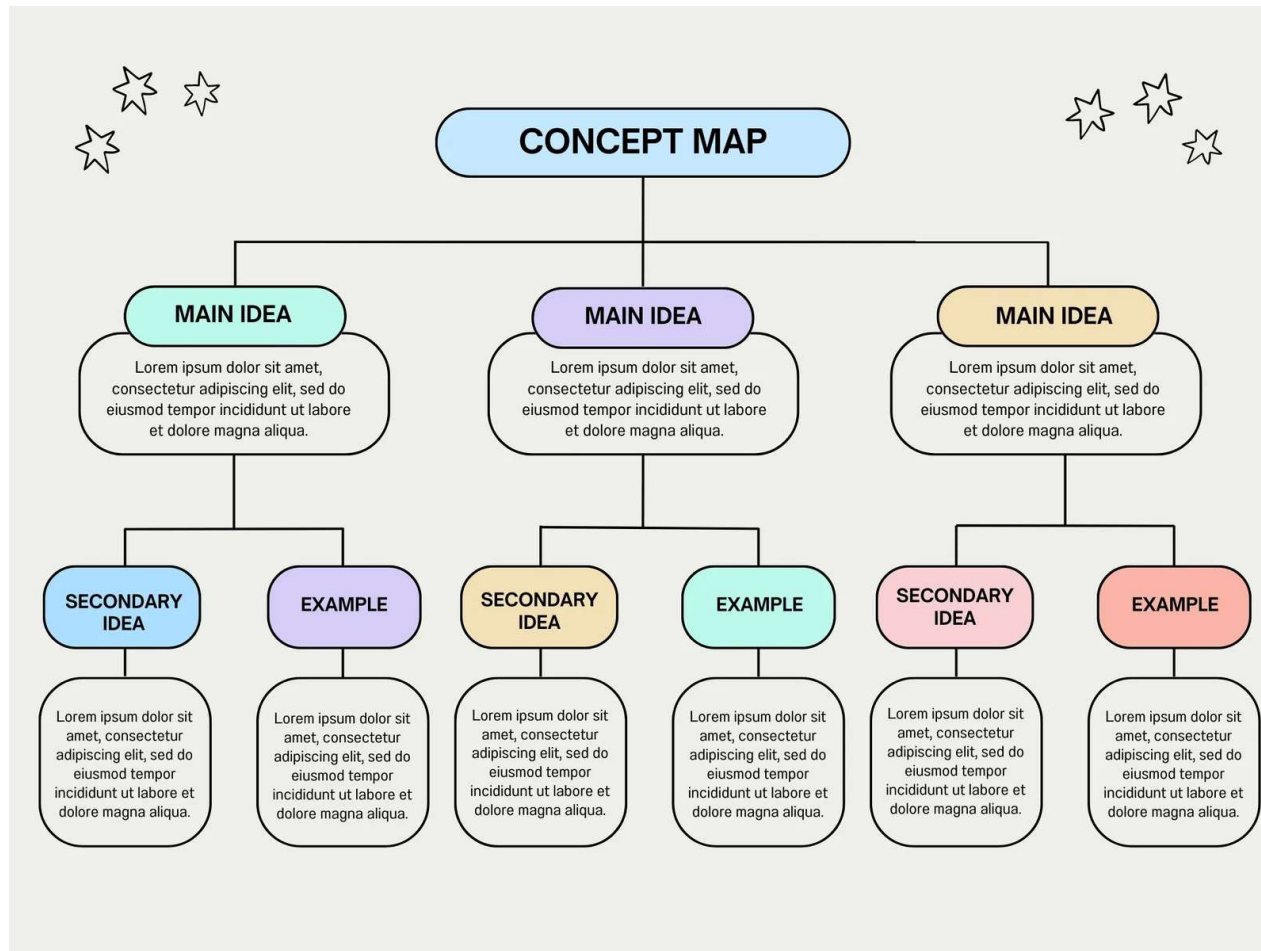


# Concept Maps

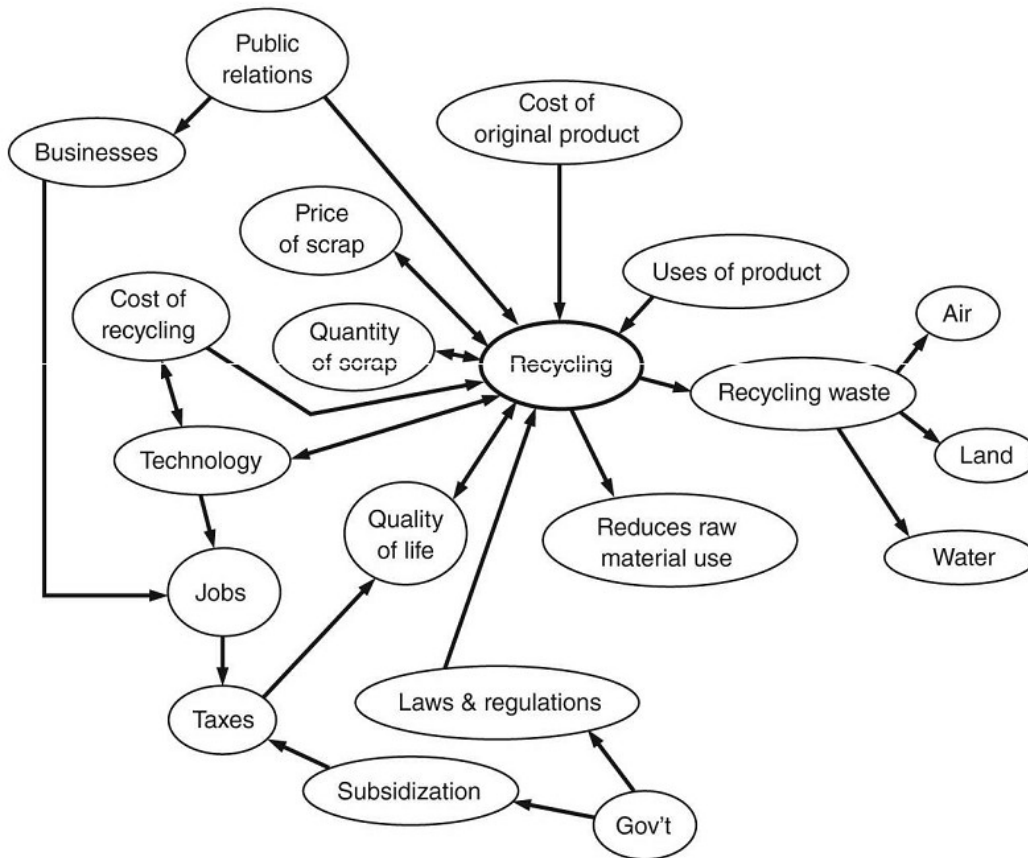


- One of the effective way to record the results of brainstorming session as it happens is by memory mind mapping.
- The facilitator starts with a clean sheet of paper, writes the problem statement in the middle of the paper as two words and draws a box around it.
- Then the ideas generated to solve the problem is then recorded quickly, with circle around it
- Each new idea to solve problem is connected to the original problem statement.
- The concept maps servers as an effective visual documentation of the brainstorming session.

# Concept Map



# Concept Map Example



## Concept Maps Pros and Cons

- **Pros**

- More detailed information
- Illustrate multiple problems
- Activate the critical thinking process
- Optimal tool for analyzing homework
- Good interface
- Explicit relationship among factors
- Solving various problems at the same time

- **Cons**

- Unable to memorize information
- Take much effort
- Complicated layout
- Difficult to change one block

# Affinity diagram

[10 Affinity Diagram Examples to Organize Ideas + Free Templates | Creately](#)



- The affinity diagram is a business tool used to organize ideas and data.
- The term affinity diagram was devised by **Jiro Kawakita** in the 1960s and is sometimes referred to as the **KJ Method**.
- The tool is commonly used within project management and allows large numbers of ideas stemming from brainstorming to be sorted into groups, based on their natural relationships, for review and analysis.
  - Record each idea on cards or notes.
  - Look for ideas that seem to be related.
  - Sort cards into groups until all cards have been used.

# AI tools in Concept Generation

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- [Featured AI Image Generator : Turn Text to Images | PromeAI](#)
- [AI Image Generator - Create Art, Images & Video | Leonardo AI](#)
- [Vizcom](#)
- [\(48\) Vizcom - Make it real - YouTube](#)

# #4: Concept Combination Table



Concept Combination Table >> Combinations of solution fragments systematically

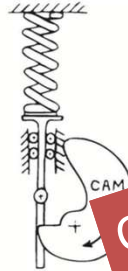
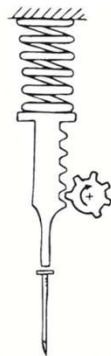
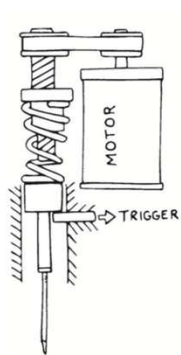
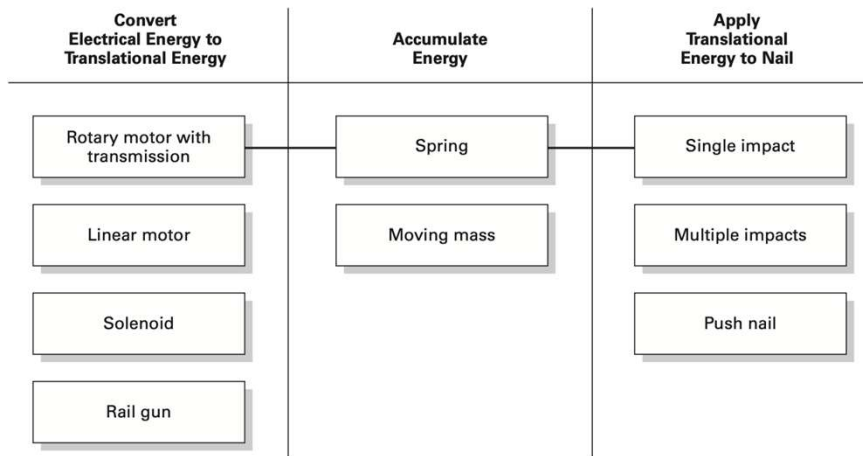
| Subproblems        | Convert<br>Electrical Energy to<br>Translational Energy | Accumulate<br>Energy | Apply<br>Translational<br>Energy to Nail |
|--------------------|---------------------------------------------------------|----------------------|------------------------------------------|
|                    | Rotary motor with<br>transmission                       | Spring               | Single impact                            |
|                    | Linear motor                                            | Moving mass          | Multiple impacts                         |
|                    | Solenoid                                                |                      | Push nail                                |
|                    | Rail gun                                                |                      |                                          |
| Solution Fragments |                                                         |                      |                                          |

$4 \times 2 \times 3 = 24$  combinations

# #4: Potential Solutions

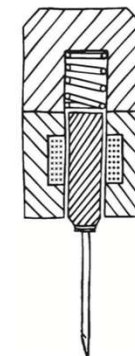
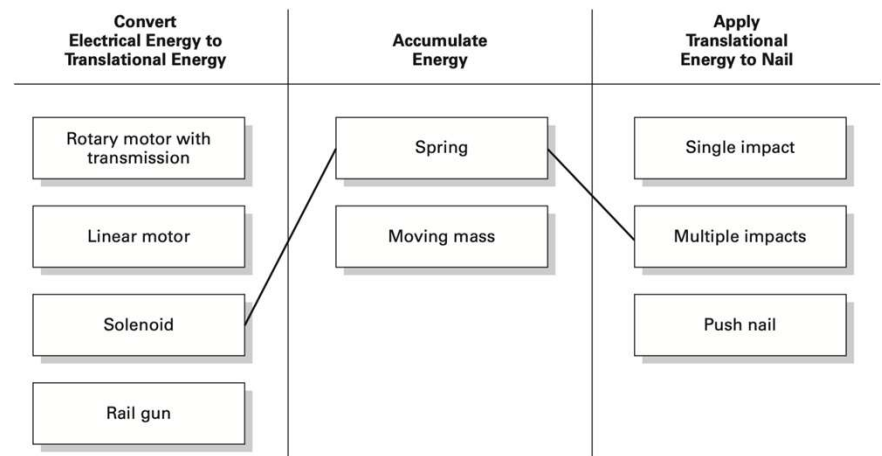


## Combination-1



Check for infeasible fragments and coupled subproblems

## Combination-2



# Direct or Logical Method

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- Functional decomposition:
- Morphological Method:
- Theory of Inventive Problem Solving (TRIZ / TIPS):



# Why Functional Decomposition

- Concentrates on “**What**” has to be achieved by a new concept & not “**How**” it is to be achieved
- Basis for organizing design team, tasks & process
- **Creativity** enhancement
- More acceptable solutions can be generated with mapping **Needs** to **Function** & then to **Form**
- This process removes excess reliance on **designer’s previous experience**
- Avoids jumping to conclusions (Form) too quickly



# Functional Modeling Basics

- **Function**

- Statement of clear, reproducible relationship between available input & desired output of product, **independent** of any particular form – **Input Electricity and hand force, Output – Ironing of clothes**

- **Product Function**

- Intended overall function of product just an **active verb & noun**- **Wash clothes, Cut Beans, Cools Air, Heats water**

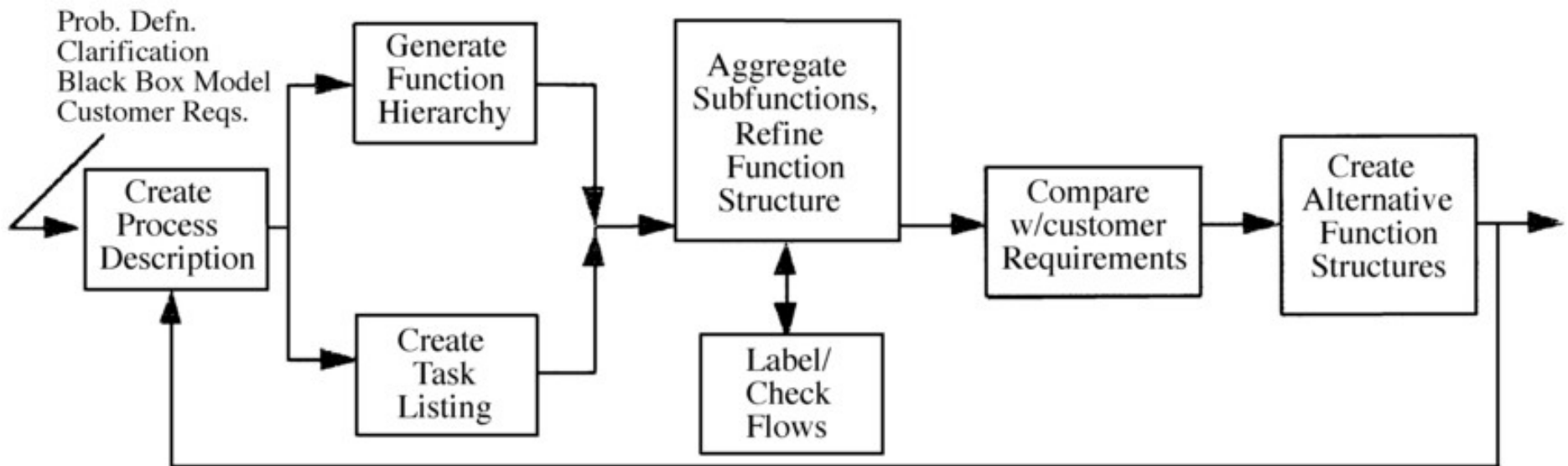
- **Sub function**

- Component of product function corresponding to subtasks

- **Abstraction**

- Process of **ignoring** what is **particular** or **incidental** & **emphasizing** what is **essential** and **general**

# Functional Decomposition and Synthesis



# Black Box



# Functions and Constraints



- **Constraints**

- Is a statement of clear criterion that must be satisfied by a product & requires consideration of entire product to determine the criterion value
  - *Cost of product is a constraint*
- **Functions** are satisfied through **operation** of product
- **Constraints** are satisfied by **properties** of entire product
  - Includes criteria of cost, compactness, mass & reliability

# Functional Trees



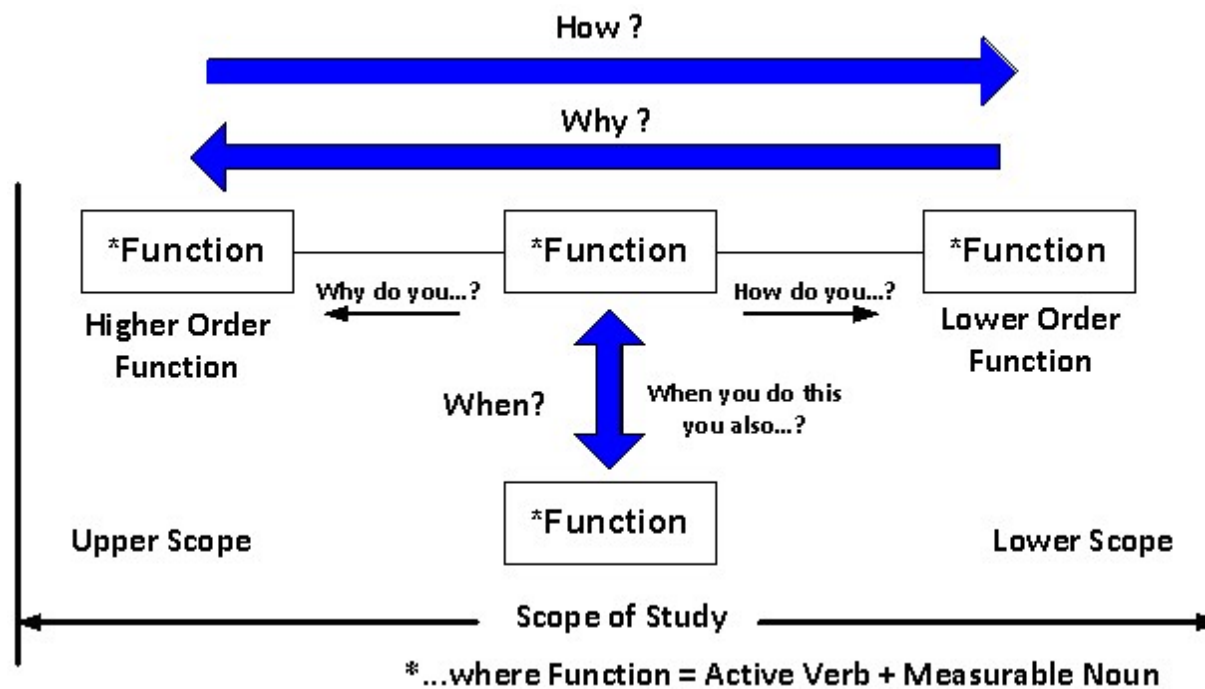
- **Decompose** the Prime function hierarchically into sub functions
- **Fast & Simple** to construct
  - **FAST** Method (top-down approach)
  - **Subtract & Operate** Method (bottom-up approach)

# Function Analysis System Technique (FAST)

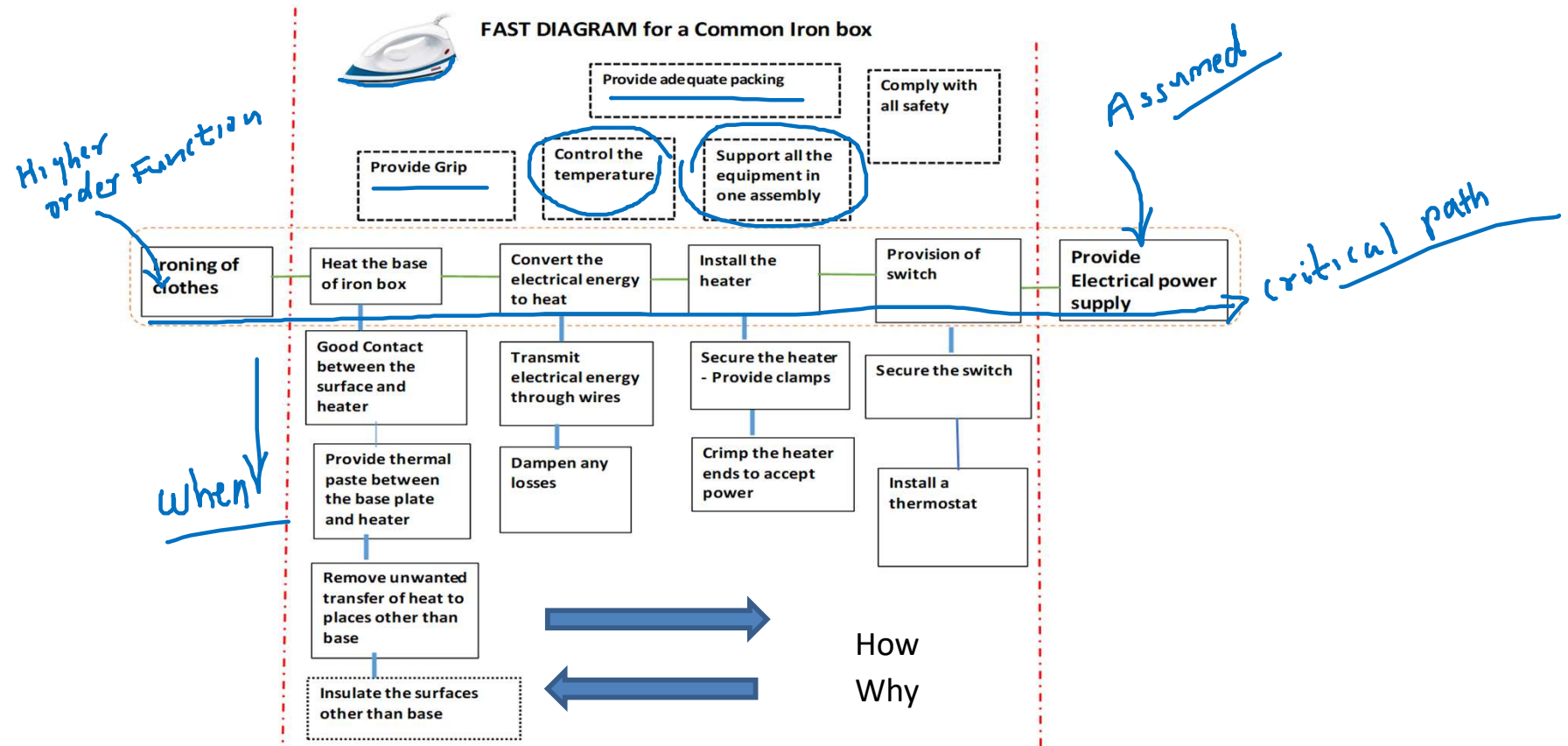


- Displays function in a **logical sequence**, **Prioritize** them and tests their **dependency**
- **HOW-WHY** analysis
- Higher order function? – “Iron the clothes”
- Basic function? – “Heats the base of the iron box”
- Secondary function - 3 types (Required, Aesthetic and unwanted)
  - Required ---- “Heater assembly which converts electricity to heat”
  - Aesthetic -----” Styling of outer grip or handle etc.”
  - Unwanted ----- “ Heating elements loses heat to handles, handles need to be Insulated”
- Critical path?- Higher order      Basic function      Required secondary function
- Each **right node** in critical path must answer **how the left** node is achieved, every **left node** should answer **why right** node is being achieved

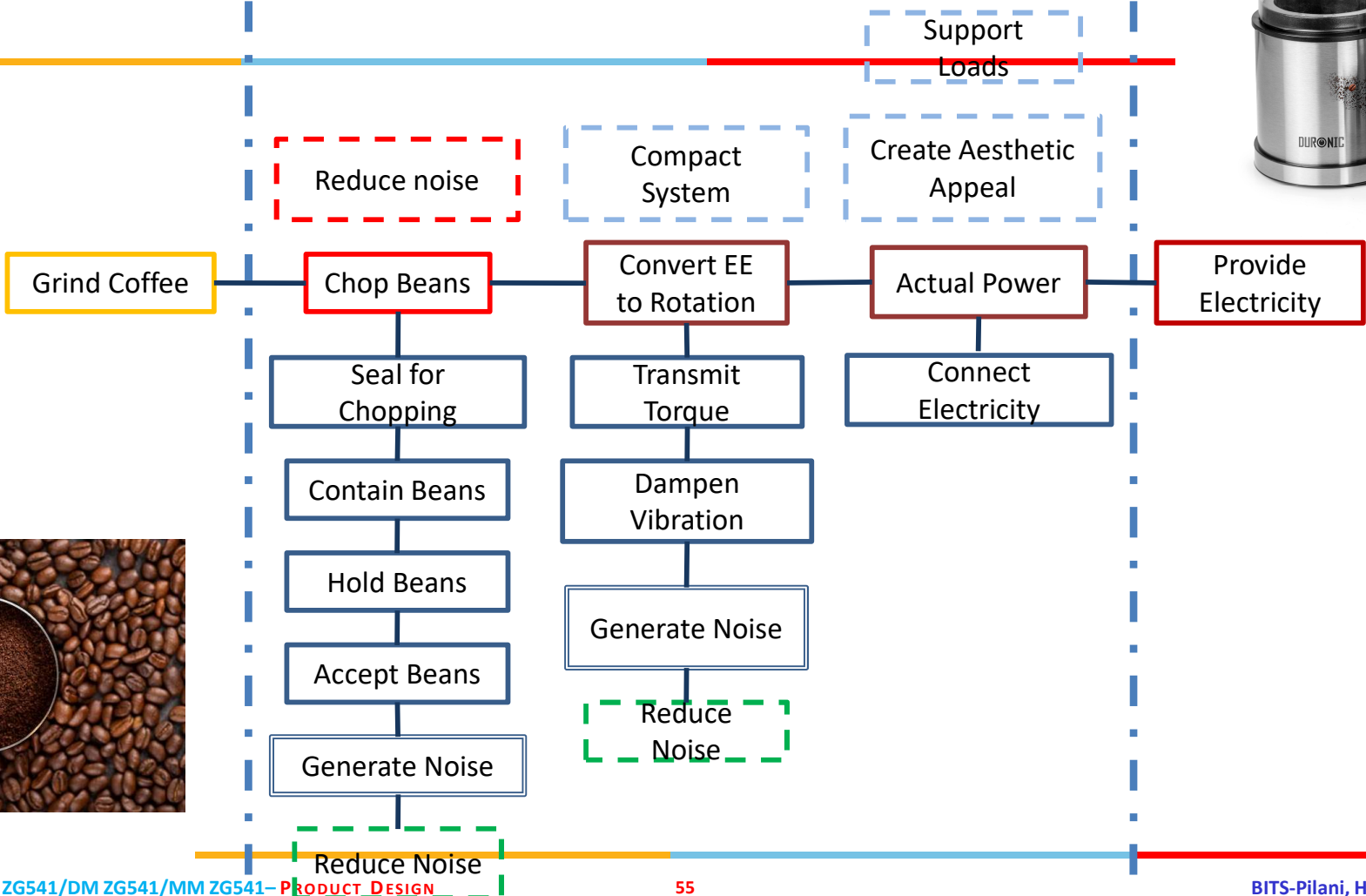
# FAST Method



# Iron Box Example



# Coffee Bean Chopper Example



# Subtract and Operate Method

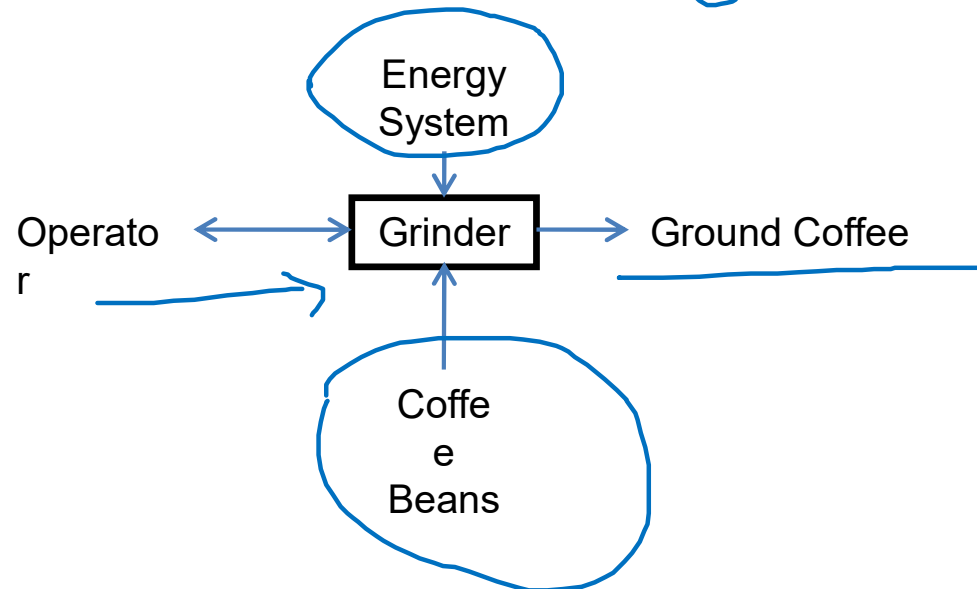
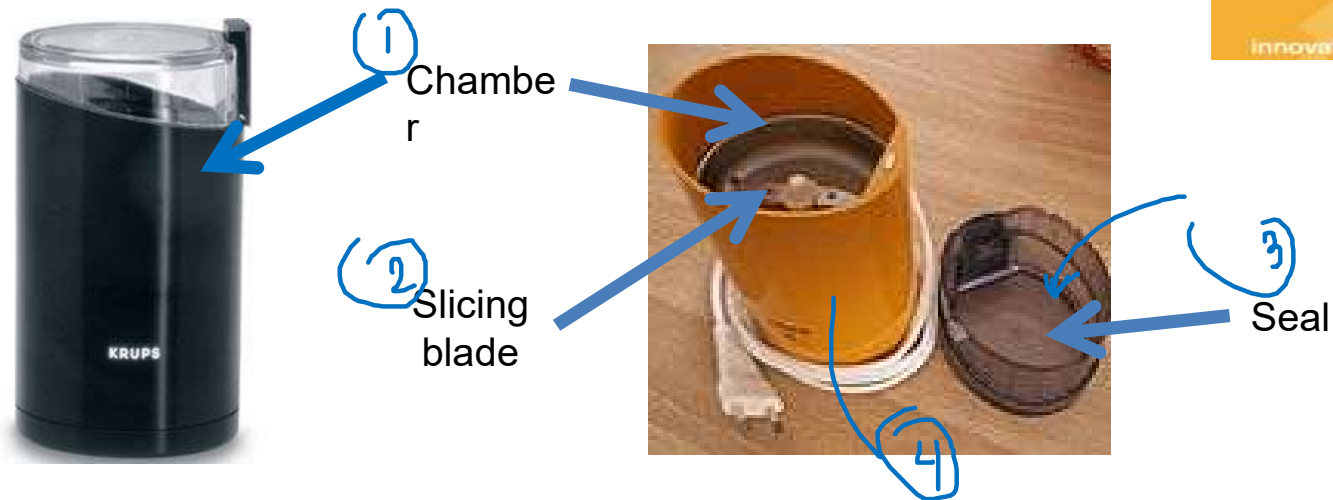


- **Bottom-up** approach to develop a function tree
- Assumption: a **form concept** or **actual product exists**, this concept/product is then **reverse engineered**
- Better way to figure out function of component; **removing it, operating system** without it conceptually
- Starts by considering **smallest isolatable function of feature** (which are not further decomposable)

# Procedure



- Disassemble one component of assembly
- Operate the system through it's full range
- Analyze the effect
- Deduce the sub function of the missing component
- Replace the component and repeat the procedure “n” times, n- being the number of components in the assembly
- Translate the collection of sub functions into a function tree



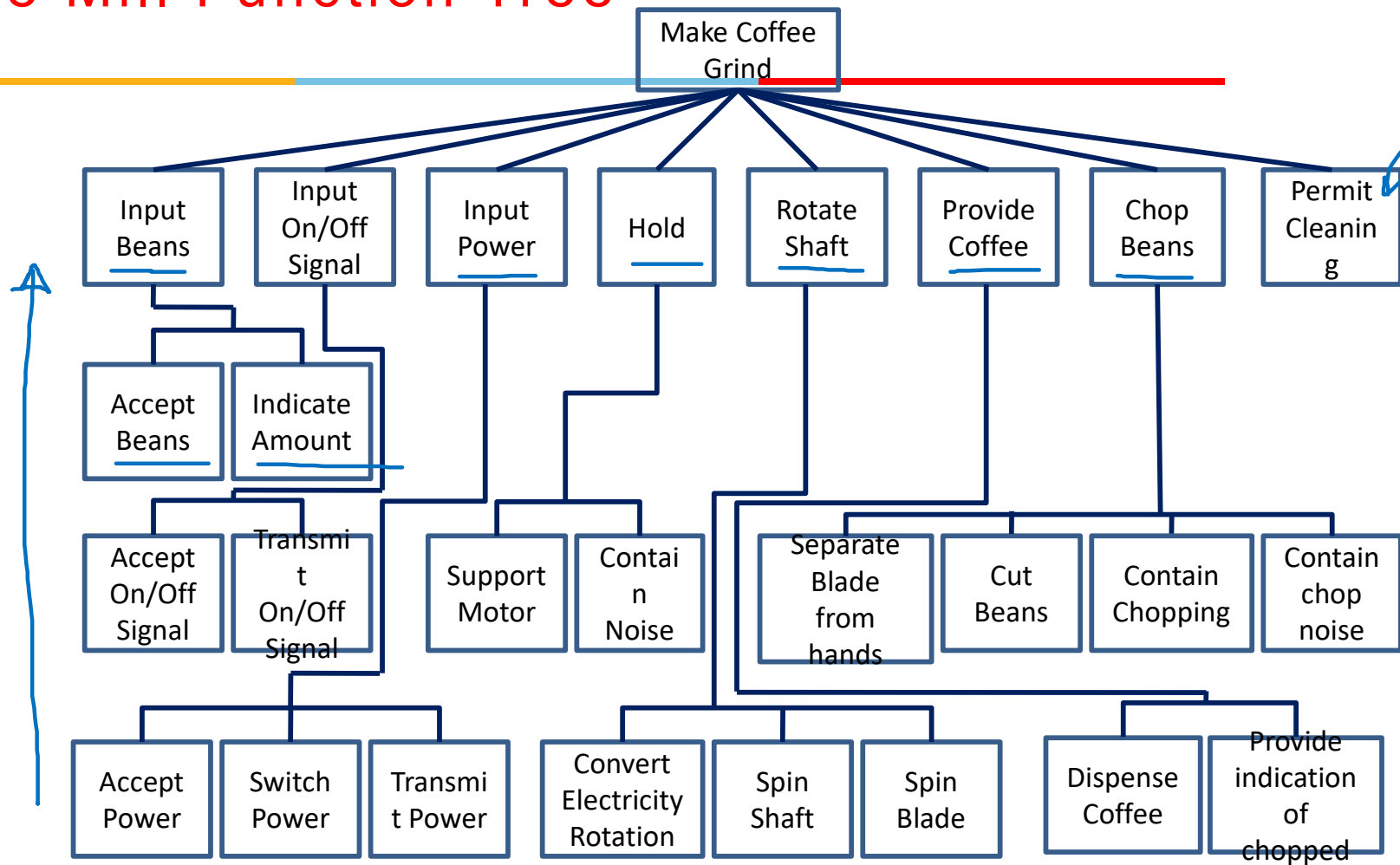
# Coffee Mill



| Chamber                             | Seal                                                           | Slicing blade                | Shaft                                | Motor                                                 |
|-------------------------------------|----------------------------------------------------------------|------------------------------|--------------------------------------|-------------------------------------------------------|
| ✓ No defined way of holding content | ✓ No protection against contents splattering                   | Contents will not be chopped | Slicing blade will not be attachable | Shaft does not spin                                   |
| ✓ No measurable volume              | No protection against spinning blade                           | No resistance to torque      | Contents will not be chopped         | Electricity is not transformed into mechanical energy |
| ✓ No body to measure contents       | Safety issues will fail                                        |                              | No resistance to torque              |                                                       |
| ✓ No body to contain contents       | Chamber cannot be closed                                       |                              |                                      |                                                       |
| ✓ No body to hold the apparatus     | Power can't be actuated because electric circuit is not closed |                              |                                      |                                                       |
| ✓ Aesthetic appearance reduced      | Impact noise will not be enclosed                              |                              |                                      |                                                       |
| ✓ Difficult to clean undefined body |                                                                |                              |                                      |                                                       |
| ✓ Pour out contents                 |                                                                |                              |                                      |                                                       |



# Coffee Mill Function Tree



*key words*



# THANK YOU!



# **BITS Pilani**

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# **Product Design**

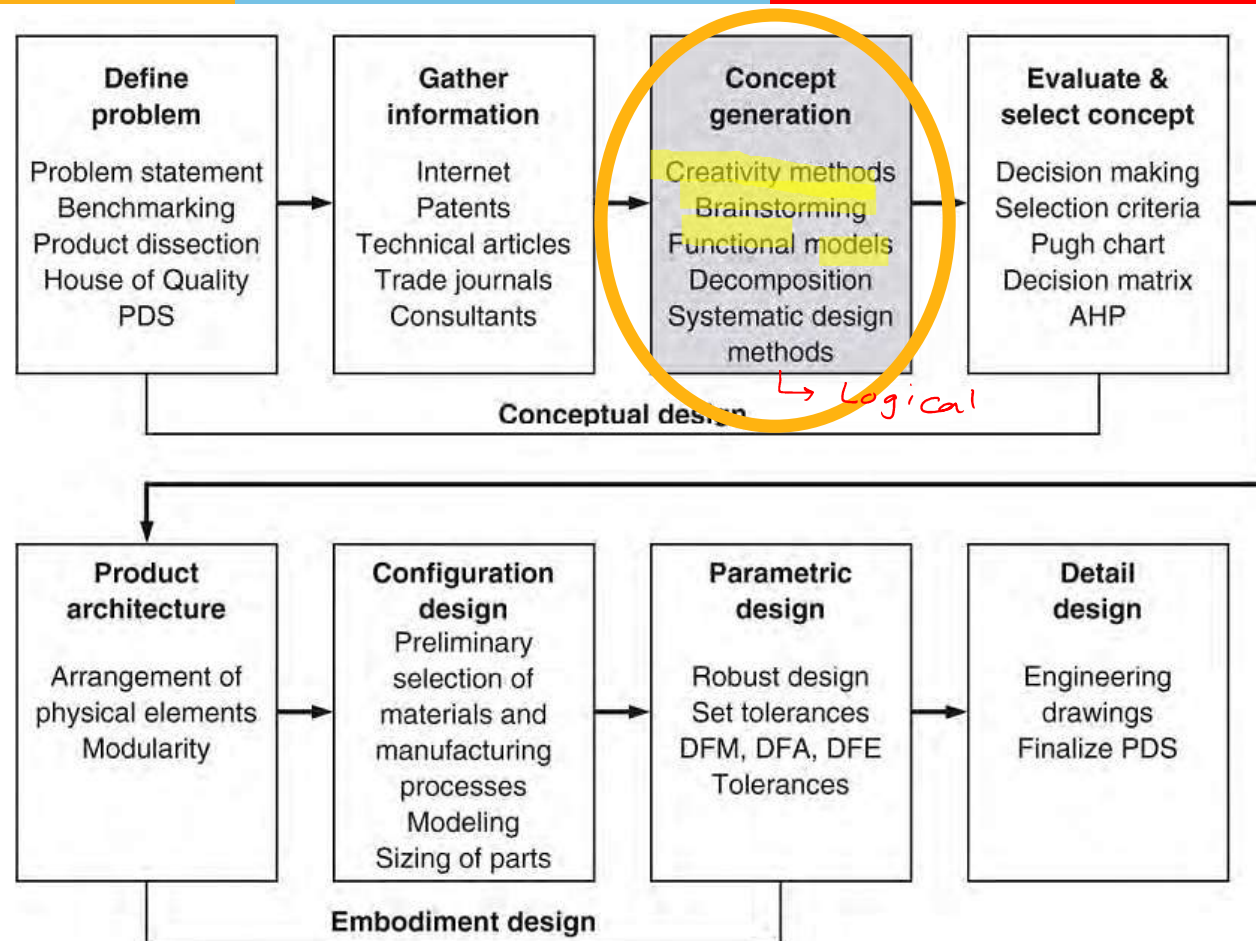
**(AE ZG531/ AEL ZG531/ DE ZG541/DM ZG541/MM ZG541)**



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# Session 6: Concept Development

# Product Development Process



# Content

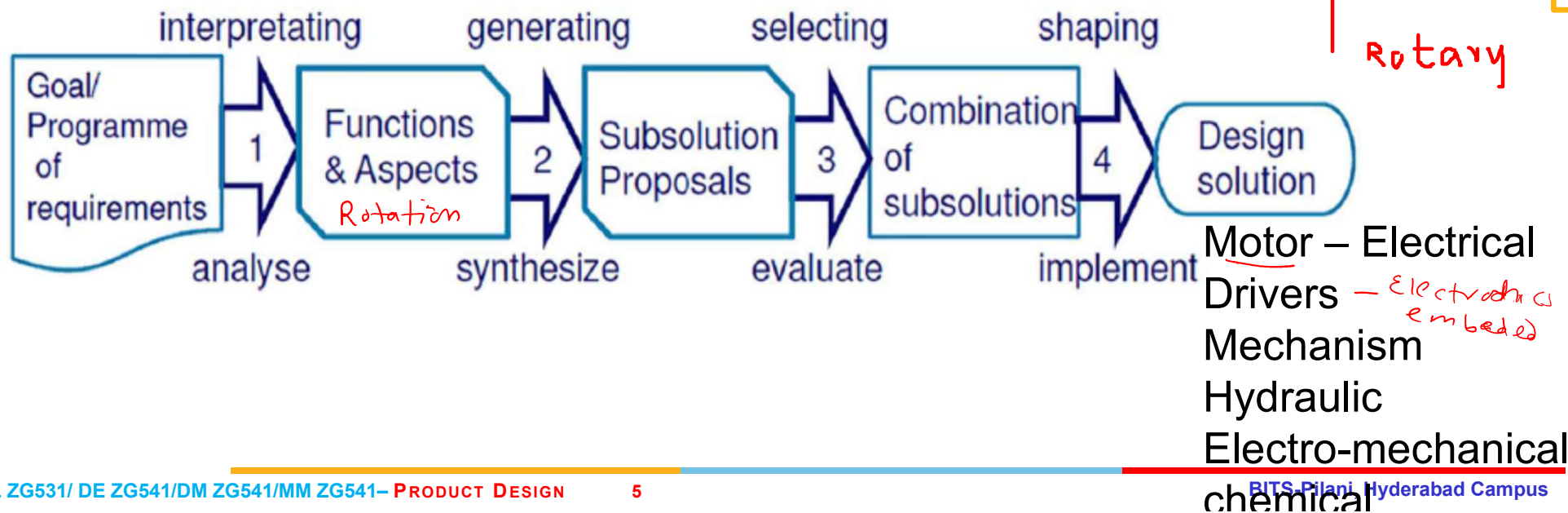
→ Direct Concept Generation method



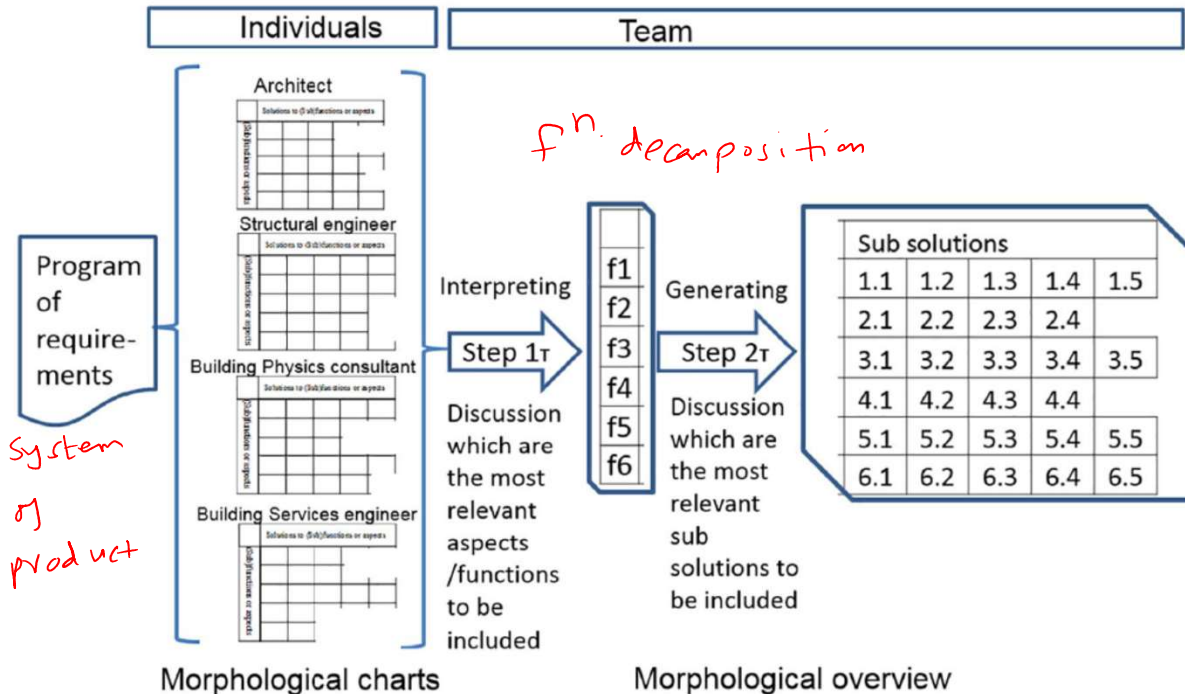
- Morphological Analysis
- Theory of Inventive Problem Solving  
(TIPS or TRIZ)
- Axiomatic Design
- manual / standards / Rules & Regulation / Laws.
- Summary

# Morphological Analysis

- The morphological method essentially is nothing more than an orderly way of looking at things.
- Morphological charts originate from the concept of the n-dimensional morphological box (Zwicky box-1948).



# Morphological Analysis

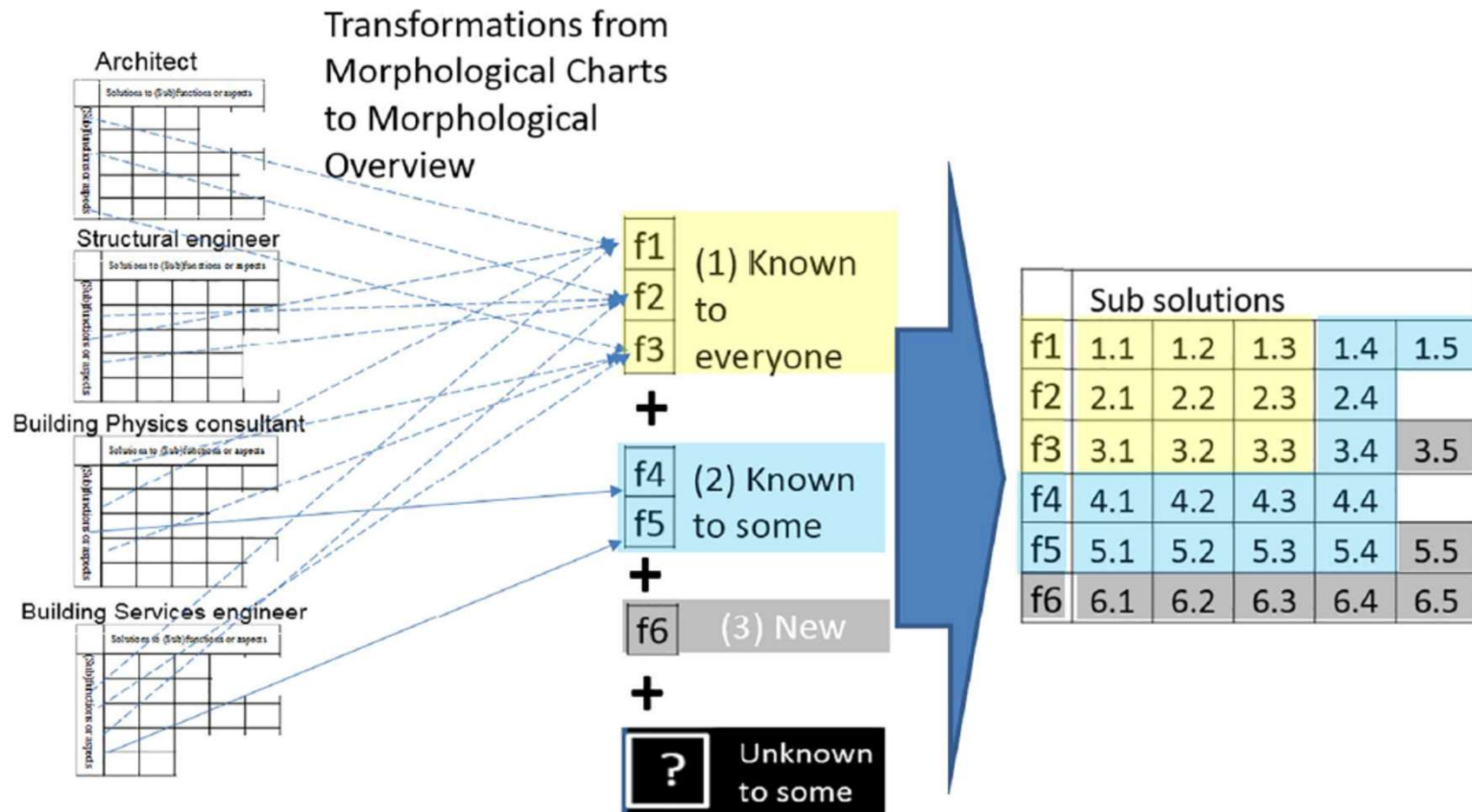


- In the first step of the individual designer makes a list of what he/she considers to be the most important functions that need to be fulfilled based on the design brief.
- The morphological charts are formed as each designer translates the main goals of the design task, derived from the program of demands, into functions and aspects. This is then inserted into the first column of the morphological chart.

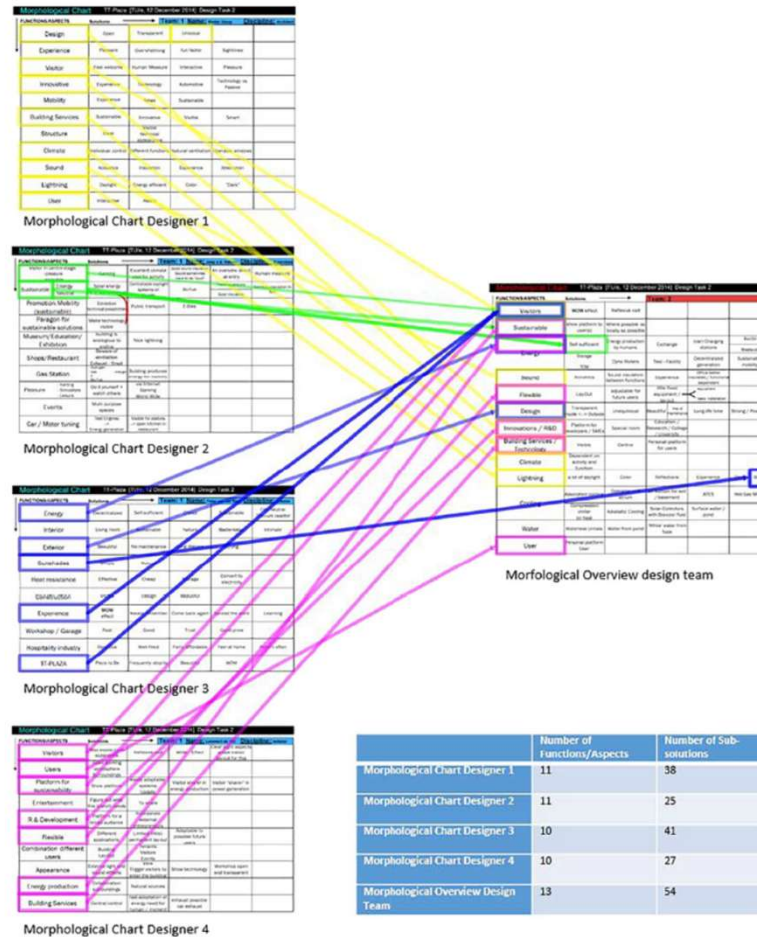
# Steps Involved in Morphological Analysis

- In the second step of the process, the designer adds the possible part-solutions to the related rows of the functions/aspects of the first column.
- Based on the given design task, each design team member approaches the problem according to his/her active perception, memory, knowledge, and needs.
- These individual morphological charts can then be combined by the design team to form one morphological overview.
- First, in step three, functions and aspects are discussed and then the team decides which functions and aspects will be placed in the morphological overview.
- Then, in step 4, all participants of the design team can contribute their solutions for these functions and aspects by filling in the rows within the morphological overview.
- Putting the morphological charts together enables individual perspectives from each discipline to be put forward, which in turn highlights the implications of design choices for each discipline.

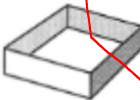
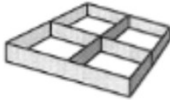
# Morphological Analysis



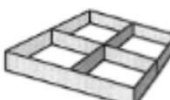
# Morphological Analysis



# Morphological Method: Example

|                          | Option 1                                                                            | Option 2                                                                            | Option 3                                                                            | Option 4                                                                           |
|--------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|
| Vegetable picking device |    |    |    |   |
| Vegetable placing device |    |    |    |   |
| Dirt sifting device      |    |    |    |                                                                                    |
| Packaging device         |   |   |   |  |
| Method of transportation |  |  |  |                                                                                    |
| Power source             | Hand pushed                                                                         | Horse drawn                                                                         | Wind blown                                                                          | Pedal driven                                                                       |

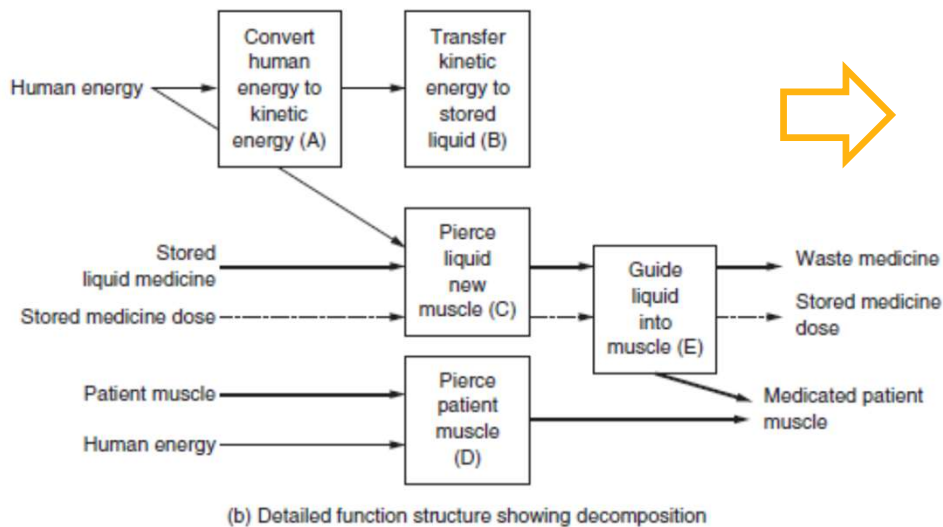
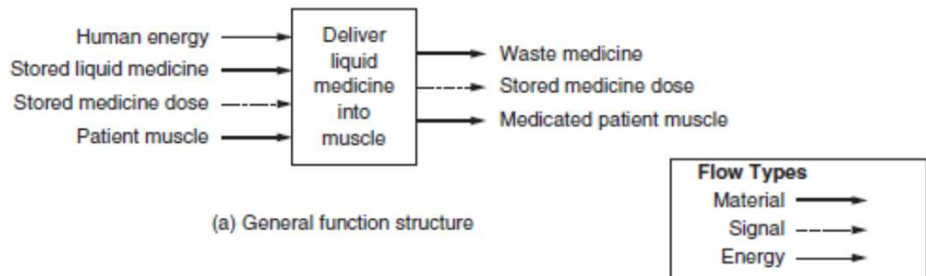
Concept 2

|                          | Option 1                                                                              | Option 2                                                                              | Option 3                                                                              | Option 4                                                                             |
|--------------------------|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| Vegetable picking device |    |    |    |   |
| Vegetable placing device |    |    |    |   |
| Dirt sifting device      |    |    |    |                                                                                      |
| Packaging device         |   |   |   |  |
| Method of transportation |  |  |  |                                                                                      |
| Power source             | Hand pushed                                                                           | Horse drawn                                                                           | Wind-blown                                                                            | Pedal driven                                                                         |

Concept 1

# Morphological Method: Example

Medical / Pharma



Morphological Chart for Disposable Syringe Problem

| Row Number | Subproblem Solution Concepts           |                                          |                                     |                         |                          |
|------------|----------------------------------------|------------------------------------------|-------------------------------------|-------------------------|--------------------------|
|            | Convert Human Energy to Kinetic Energy | Transfer Kinetic Energy to Stored Liquid | Position Liquid Near Muscle         | Pierce Patient Muscle   | Guide Liquid into Muscle |
| 1          | Hand pump                              | Heating liquid                           | Manual method                       | Sharp pointed tool      | Rigid tube               |
| 2          | Piston and cylinder                    | Physically displacing liquid             | Suction device                      | Shearing tool           | Flexible tube            |
| 3          | Crank                                  | Pressure differential                    | Adhesive                            | Hole punch              | Misting sprayer          |
| 4          | Fan                                    | Mechanical stirrer                       | Physical connector attached to skin | Multiple puncture sites | Osmosis                  |
| 5          |                                        | Radiation                                | Strap and cuff                      |                         | Funnel                   |

# Theory of Inventive Problem Solving (TIPS or TRIZ)

- A Russian acronym: Theoria Resheneyva Isobretatelskehuh Zadach
- (Theory of Inventive Problem Solving)



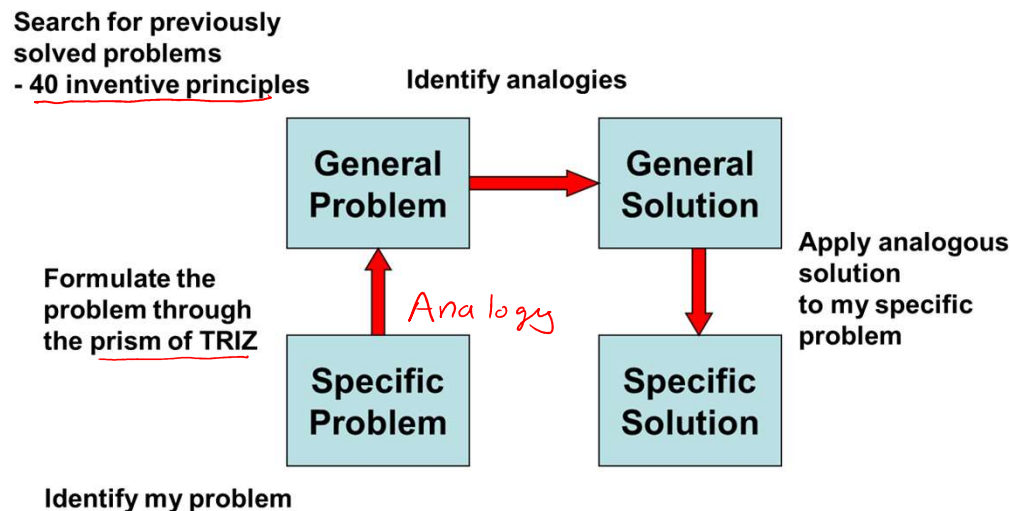
# TRIZ

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- These three categories are “ identifying conflicts and solving them with known physical principles,” “ identifying new principles,” and identifying new product functions and solving them with known or new principles.”
- Altshuller observed a number of trends in historical invention.
- Evaluation of engineering system is done by same pattern, independent of engineering discipline.
- Conflicts (contradictions) are the key drivers for product invention.
- Compromises is unacceptable.

# TRIZ

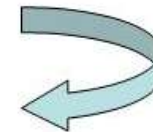
- TIPS (TRIZ) is based on two basic principles
  1. Somebody, sometime, somewhere has already solved your problem or one similar to it.  
Creativity means finding that solution and adapting it to the current problem.
  2. Don't accept contradictions. Resolve them.



# Engineering Parameters (39)

1. Weight of moving object
2. Weight of nonmoving object
3. Length of moving object
4. Length of nonmoving object
5. Area of moving object
6. Area of nonmoving object
7. Volume of moving object
8. Volume of nonmoving object
9. Speed
10. Force
11. Tension, pressure
12. Shape
13. Stability of object
14. Strength
15. Durability of moving object
16. Durability of nonmoving object
17. Temperature
18. Brightness
19. Energy spent by moving object
20. Energy spent by nonmoving object

21. Power
22. Waste of energy
23. Waste of substance
24. Loss of information
25. Waste of time
26. Amount of substance
27. Reliability
28. Accuracy of measurement
29. Accuracy of manufacturing
30. Harmful factors acting on object
31. Harmful side effects
32. Manufacturability
33. Convenience of use
34. Repairability
35. Adaptability
36. Complexity of device
37. Complexity of control
38. Level of automation
39. Productivity



# Inventive Principles (40)

## 1. Segmentation

## 2. Extraction, Separation, Removal, Segregation

3. Local Quality

4. Asymmetry

5. Combining, Integration, Merging

6. Universality, Multi-functionality

7. Nesting

8. Counterweight, Levitation

9. Preliminary anti-action, Prior counteraction

## 10. Prior action

11. Cushion in advance, compensate before

12. Equipotentiality, remove stress

13. Inversion, The other way around

14. Spheroidality, Curvilinearity

15. Dynamicity, Optimization

16. Partial or excessive action

17. Moving to a new dimension

18. Mechanical vibration/oscillation

19. Periodic action

20. Continuity of a useful action

21. Rushing through

22. Convert harm into benefit, "Blessing in disguise"

23. Feedback

24. Mediator, intermediary

25. Self-service, self-organization

## 26. Copying

## 27. Cheap, disposable objects

28. Replacement of a mechanical system with 'fields'

29. Pneumatics or hydraulics:

30. Flexible membranes or thin film

31. Use of porous materials

32. Changing color or optical properties

33. Homogeneity

34. Rejection and regeneration, Discarding and recovering

35. Transformation of the physical and chemical states of an object, parameter change, changing properties

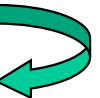
36. Phase transformation

## 37. Thermal expansion

38. Use strong oxidizers, enriched atmospheres, accelerated oxidation

39. Inert environment or atmosphere

## 40. Composite materials



| Feature to Improve |                             | Weight object | Weight moving | Length object | Length moving | Area of object | Area of object | Volume object | Volume moving | Speed        | Force        | Tension      | Shape        | Stability    |
|--------------------|-----------------------------|---------------|---------------|---------------|---------------|----------------|----------------|---------------|---------------|--------------|--------------|--------------|--------------|--------------|
| 1                  | Weight of moving object     |               |               | 15,8, 29,34   |               | 29,17, 38,34   |                | 29,2, 40,28   |               | 2,8, 15,38   | 8,10, 18,37  | 10,36, 37,40 | 10,14, 35,40 | 1,35, 19,39  |
| 2                  | Weight of non-moving object |               |               |               | 10,1, 29,35   |                | 35,30, 13,2    |               | 5,35, 14,2    |              | 8,10, 19,35  | 13,29, 10,18 | 13,10, 29,14 | 26,39, 1,40  |
| 3                  | Length of moving object     | 8,15, 29,34   |               |               |               | 15,17, 4       |                | 7,17, 4,35    |               | 13,4, 8      | 17,10, 4     | 1,8, 35      | 1,8, 10,29   | 1,8, 15,34   |
| 4                  | Length of non-moving object |               | 35,28, 40,29  |               |               |                | 17,7, 10,40    |               | 35,8, 2,14    |              | 28,10        | 1,14, 35     | 13,14, 15,7  | 39,37, 35    |
| 5                  | Area of moving object       | 2,17, 29,4    |               | 14,15, 18,4   |               |                |                | 7,14, 17,4    |               | 29,30, 4,34  | 19,30, 35,2  | 10,15, 36,28 | 5,34, 29,4   | 11,2, 13,39  |
| 6                  | Area of non-moving object   |               | 30,2, 14,18   |               | 26,7, 9,39    |                |                |               |               |              | 1,18, 35,36  | 10,15, 36,37 |              | 2,38         |
| 7                  | Volume of moving object     | 2,26, 29,40   |               | 1,7, 4,35     |               | 1,7, 4,17      |                |               |               | 29,4, 38,34  | 15,35, 36,37 | 6,35, 36,37  | 1,15, 29,4   | 28,10, 1,39  |
| 8                  | Volume of non-moving object |               | 35,10, 19,14  | 19,14         | 35,8, 2,14    |                |                |               |               |              | 2,18, 37     | 24,35        | 7,2, 35      | 34,28, 35,40 |
| 9                  | Speed                       | 2,28, 13,38   |               | 13,14, 8      |               | 29,30, 34      |                | 7,29, 34      |               |              | 13,28, 15,19 | 6,18, 38,40  | 35,15, 18,34 | 28,33, 1,18  |
| 10                 | Force                       | 8,1, 37,18    | 18,13, 1,28   | 17,19, 9,36   | 28,10         | 19,10, 15      | 1,18, 36,37    | 15,9, 12,37   | 2,36, 18,37   | 13,28, 15,12 |              | 18,21, 11    | 10,35, 40,34 | 35,10, 21    |
| 11                 | Tension, pressure           | 10,36, 37,40  | 13,29, 10,18  | 35,10, 36     | 35,1, 14,16   | 10,15, 36,25   | 10,15, 35,37   | 6,35, 10      | 35,24         | 6,35, 36     | 36,35, 21    |              | 35,4, 15,10  | 35,33, 2,40  |
| 12                 | Shape                       | 8,10, 29,40   | 15,10, 26,3   | 29,34, 5,4    | 13,14, 10,7   | 5,34, 4,10     |                | 14,4, 15,22   | 7,2, 35       | 35,15, 34,18 | 35,10, 37,40 | 34,15, 10,14 |              | 33,1, 18,4   |
| 13                 | Stability of object         | 21,39, 2,39   | 26,39, 1,40   | 13,15, 1,28   | 37            | 2,11, 13       | 39             | 28,10, 19,39  | 34,28, 35,40  | 33,15, 28,18 | 10,35, 21,16 | 2,35, 40     | 22,1, 18,4   |              |
| 14                 | Strength                    | 1,8, 40,15    | 40,28, 27,1   | 1,15, 3,35    | 15,14, 28,26  | 3,34, 40,29    | 9,40, 28       | 10,15, 14,7   | 9,14, 17,15   | 8,13, 26,14  | 10,18, 3,14  | 10,3, 18,40  | 10,30, 35,40 | 13,17, 35    |
| 15                 | Durability of moving object | 19,5, 34,31   |               | 2,19, 9       |               | 3,17, 19       |                | 10,2, 19,30   |               | 3,35, 5      | 19,2, 16     | 19,3, 27     | 14,26, 28,25 | 13,3, 35     |

# TRIZ Example

- **Statement:** Bullet proof vests should be strong, but not heavy.
- **Step 1 –** Identify the contradiction(s): Strength (improves) versus Weight (worsens)
- **Step 2 –** Look at the list of features and identify those important to your contradiction.



**Strength – #14**

**Weight – #2**



|                                | 2: Weight of stationary object |
|--------------------------------|--------------------------------|
| 1: Weight of moving object     | *                              |
| 2: Weight of stationary object | -                              |
| 3: Length of moving object     | 8, 15<br>29, 34                |
| 4: Length of stationary object | -                              |
| 5: Area of moving object       | 2, 17<br>29, 4                 |
| 6: Area of stationary object   | -                              |
| 7: Volume of moving object     | 2, 26<br>29, 40                |
| 8: Volume of stationary object | -                              |
| 9: Speed of object             | 2, 28<br>13, 38                |
| 10: Force (Intensity)          | 8 1<br>37 18                   |
| 11: Stress or pressure         | 10 36<br>37 40                 |
| 12: Shape                      | 8 10<br>29 40                  |
| 13: Stability of the object    | 21 35<br>2 39                  |
|                                | 40, 26                         |
| 14: Strength                   | 27, 1                          |

# TRIZ Example

|    |                     |
|----|---------------------|
| 1  | Segmentation        |
| 26 | Copying             |
| 27 | Cheap short living  |
| 40 | Composite materials |

## Principle 1: Segmentation ✓

- Divide an object into independent parts.
- Make an object easy to disassemble.

## Principle 26: Copying

- Instead of an unavailable, expensive, fragile object, use simpler and inexpensive copies.
- Replace an object, or process with optical copies.

## Principle 27: Cheap short-living objects

- Replace an inexpensive object with a multiple of inexpensive objects, comprising certain qualities (such as service life, for instance).

## ~~Principle~~ 40: Composite materials

- For lighter-weight, stronger vests, the use of composites is an active area of research.
- Polymers (Kevlar) reinforced with carbon nanofibers are currently being investigated as a strong lightweight alternative to steel for structural materials.

# TRIZ Example

- **Iron product to remove the wrinkles:**



Conflicts is we desired heavy iron to remove the wrinkles from the cloths but we do not want a heavy iron due to the impact on ergonomics.

- ✓ **8** – Principle of counterweight: Attach an object with lifting power or use the interactions with the environment, e.g. aerodynamic lift.
- ✓ **1** – Principle of segmentation: Divide the object into independent parts that are easy to disassemble, increase the degree of segmentation as much as possible.
- ✓ **37** – Application of thermal expansion: Use expansion or contraction of material by heat. Use materials with different coefficient of thermal expansion.
- ✓ **18** – Use of mechanical vibration: Make the object vibrate. Increase the frequency of vibration.

## TRIZ Example

---

- Airbags need to inflate before contacting occupants to prevent forward motion. We would like to inflate the air bags faster while decreasing the adverse effects.
  - Principle 16: Partial or Excessive Action Use a lower powered air bag. By using less power the acceleration of the bag is less, and injuries will be reduced. Use smaller air bags with higher power. These bags will reach full inflation sooner.
  - Principle 21: Rushing Through Inflate the air bag faster than current practice.
  - Principle 40: Composite materials Airbag material that can't grab skin as it is deployed

# Refrigerator

- Contradiction –
  - Size – Volume – increasing
  - Energy consumption - Low

Energy usage and refrigerator volume are directly correlated, not contradictory; a larger-volume refrigerator requires more energy to cool a greater space than a smaller one, though other factors like energy efficiency, ambient temperature, and usage patterns also significantly impact overall energy consumption. 

## Volume Vs Productivity

1. Segmentation
  2. Extraction, Separation, Removal, Segregation
  3. Local Quality
  4. Asymmetry
  5. Combining, Integration, Merging
  6. Universality, Multi-functionality
  7. Nesting
  8. Counterweight, Levitation
  9. Preliminary anti-action, Prior counteraction
  10. Prior action
  11. Cushion in advance, compensate before
  12. Equipotentiality, remove stress
  13. Inversion, The other way around
  14. Spheroidality, Curvilinearity
  15. Dynamicity, Optimization
  16. Partial or excessive action
  17. Moving to a new dimension
  18. Mechanical vibration/oscillation
  19. Periodic action
  20. Continuity of a useful action
  21. Rushing through
  22. Convert harm into benefit, "Blessing in disguise"
  23. Feedback
  24. Mediator, intermediary
  25. Self-service, self-organization
  26. Copying
  27. Cheap, disposable objects
  28. Replacement of a mechanical system with 'fields'
  29. Pneumatics or hydraulics
  30. Flexible membranes or thin film
  31. Use of porous materials
  32. Changing color or optical properties
  33. Homogeneity
  34. Rejection and regeneration, Discarding and recovering
  35. Transformation of the physical and chemical states of an object, parameter change, changing properties
  36. Phase transformation
  37. Thermal expansion
  38. Use strong oxidizers, enriched atmospheres, accelerated oxidation
  39. Inert environment or atmosphere
  40. Composite materials
- 

- Volume of moving part - less
  - Strength – high
  - 10, 15, 14, 7
1. Segmentation
  2. Extraction, Separation, Removal, Segregation
  3. Local Quality
  4. Asymmetry
  5. Combining, Integration, Merging
  6. Universality, Multi-functionality
  7. Nesting
  8. Counterweight, Levitation
  9. Preliminary anti-action, Prior counteraction
  10. Prior action
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  15. Dynamicity, Optimization
  16. Partial or excessive action
  17. Moving to a new dimension
  18. Mechanical vibration/oscillation

#### 19. Periodic action

21. Rushing through
22. Convert harm into benefit, "Blessing in disguise"
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24. Mediator, intermediary
25. Self-service, self-organization
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27. Cheap, disposable objects
28. Replacement of a mechanical system with 'fields'
29. Pneumatics or hydraulics:
30. Flexible membranes or thin film
31. Use of porous materials
32. Changing color or optical properties
33. Homogeneity
34. Rejection and regeneration, Discarding and recovering
35. Transformation of the physical and chemical states of an object, parameter change, changing properties
36. Phase transformation
37. Thermal expansion
38. Use strong oxidizers, enriched

- High speed
- Low Noise
- 14,20,19, 35

1. Segmentation
2. Extraction, Separation, Removal, Segregation
3. Local Quality
4. Asymmetry
5. Combining, Integration, Merging
6. Universality, Multi-functionality
7. Nesting
8. Counterweight, Levitation
9. Preliminary anti-action, Prior counteraction
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33. Homogeneity
34. Rejection and regeneration, Discarding and recovering
35. Transformation of the physical and chemical states of an object, parameter change, changing properties
36. Phase transformation
37. Thermal expansion
38. Use strong oxidizers, enriched

high water volume,  
less power consumption  
0, 6

1. Segmentation
2. Extraction, Separation, Removal, Segregation
3. Local Quality
4. Asymmetry
5. Combining, Integration, Merging
6. **Universality, Multi-functionality**
7. Nesting
8. Counterweight, Levitation
9. Preliminary anti-action, Prior counteraction
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15. Dynamicity, Optimization
16. Partial or excessive action
17. Moving to a new dimension
18. Mechanical vibration/oscillation

#### 19. Periodic action

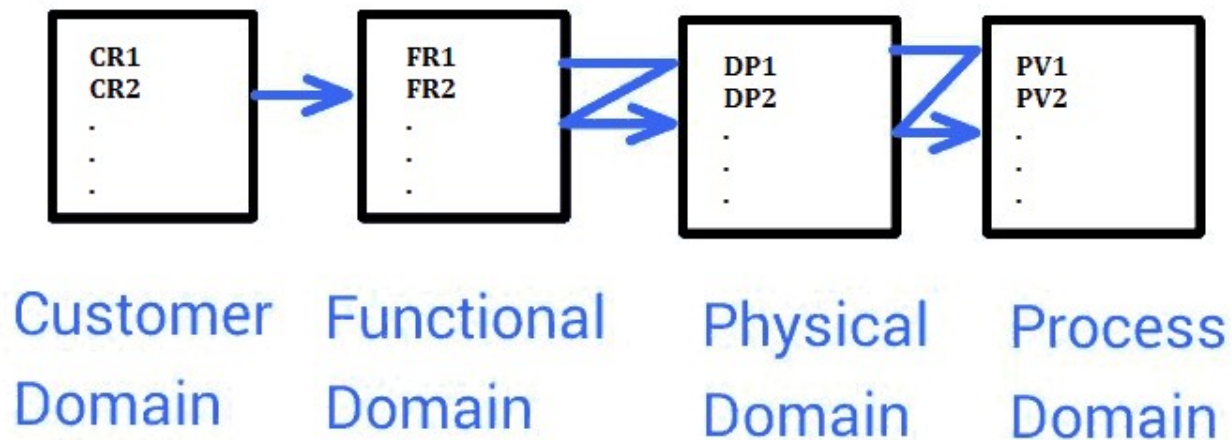
21. Rushing through
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35. Transformation of the physical and chemical states of an object, parameter change, changing properties
36. Phase transformation
37. Thermal expansion
38. Use strong oxidizers, enriched

# Axiomatic Design



- Design models that claim legitimacy from the context of “first principles” include Suh’s texts on Axiomatic Design that articulate and explicate Design Independence and Information Axioms (i.e., maintain functional independence and minimize information content).
- This methods provide a means to translate a design task into functional requirements (the engineering equivalent of what the customer wants) and use those to identify design parameters, the physical components of the design.
- Suh’s principles lead to theorems and corollaries that help designers diagnose a candidate solution now represented as a matrix equation with function requirements and design parameters.
- *Nam P. Suh , Axiomatic Design, Oxford University Press, New York , 2001 ; Nam P. Suh , The Principles of Design, Oxford University Press, New York , 1990 .*

# Axiomatic design



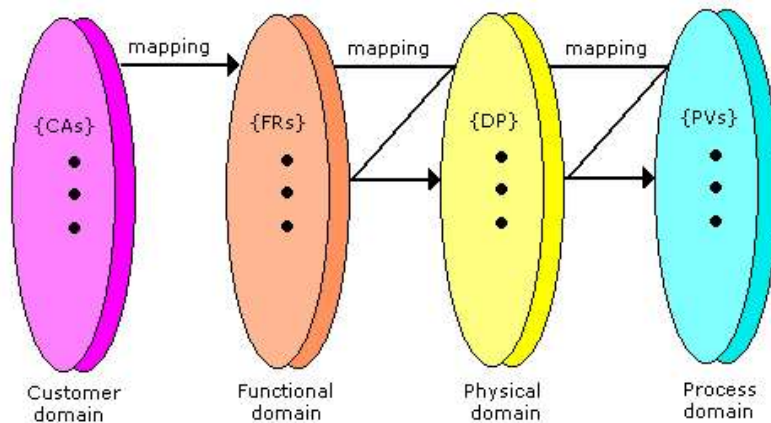
CR -> Customer Requirement

FR -> Functional Requirement

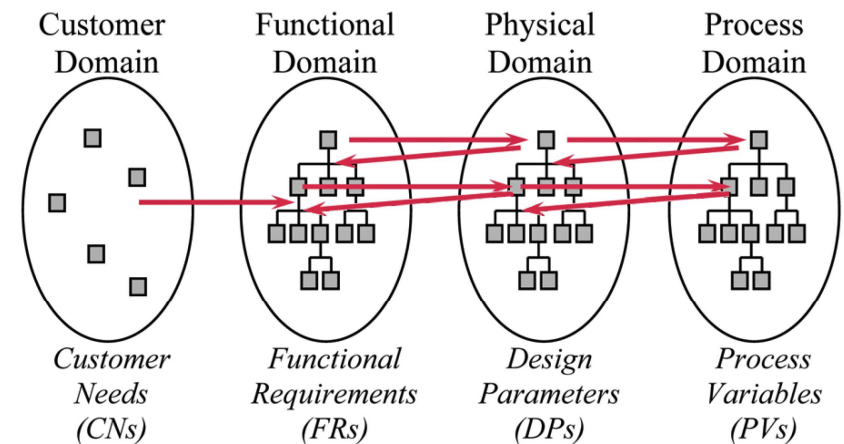
DP -> Design Parameter

PV -> Process Variable

# Axiomatic design



**Customer**  
**Functional**  
**Physical**  
**Process**



The benefits customers seek  
Functional requirements of the design solution  
Design parameters of the design solution  
Process variables

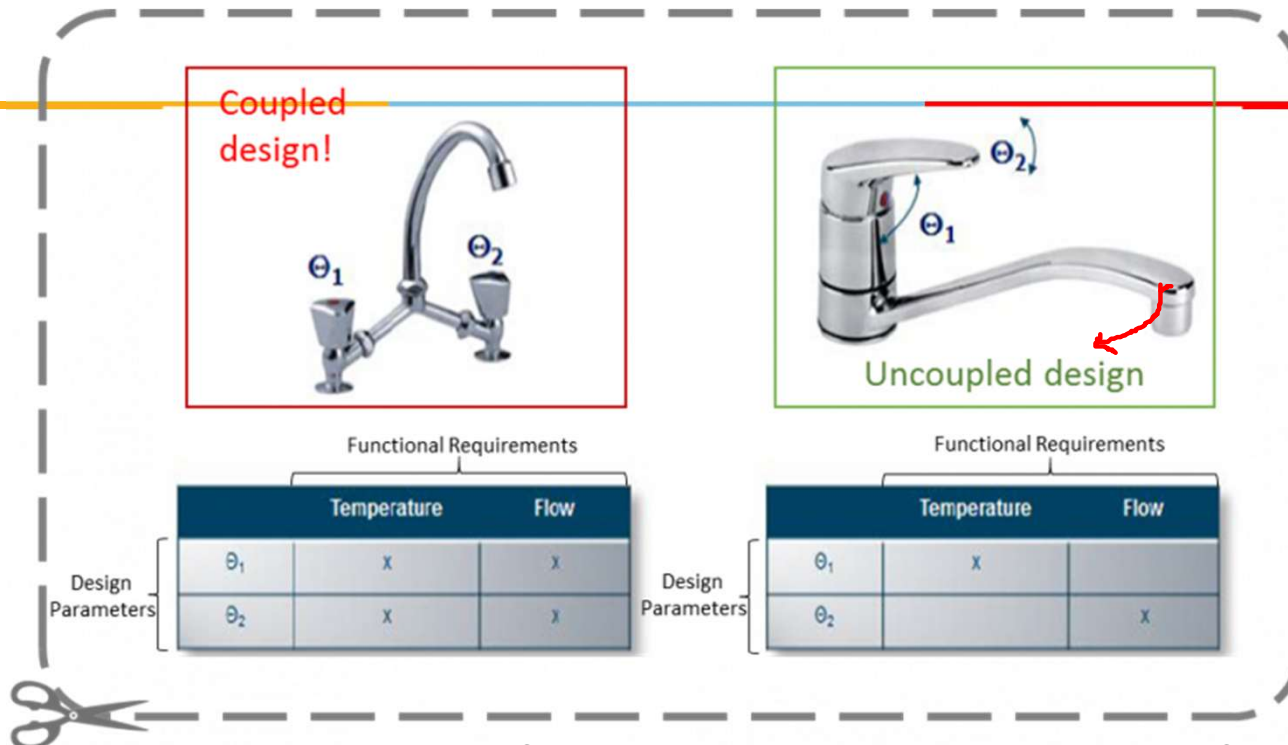
## [Chapter 1. Introduction to Axiomatic Design](#)

# Characteristics of the four domains of the design world



| Domains Character Vectors | Customer Domain {CAs}                   | Functional Domain {FRs}                           | Physical Domain {DPs}                                            | Process Domain {PVs}                                       |
|---------------------------|-----------------------------------------|---------------------------------------------------|------------------------------------------------------------------|------------------------------------------------------------|
| Manufacturing             | Attributes which consumers desire       | Functional requirements specified for the product | Physical variables which can satisfy the functional requirements | Process variables that can control design parameters (DPs) |
| Materials                 | Desired performance                     | Required Properties                               | Micro-structure                                                  | Processes                                                  |
| Software                  | Attributes desired in the software      | Output Spec of Program codes                      | Input Variables or Algorithms Modules Program codes              | Sub-routines machine codes compilers modules               |
| Organization              | Customer satisfaction                   | Functions of the organization                     | Programs or Offices or Activities                                | People and other resources that can support the programs   |
| Systems                   | Attribute desired of the overall system | Functional requirements of the system             | Machines or components, sub-components                           | Resources (human, financial, materials, etc.)              |
| Business                  | ROI                                     | Business goals                                    | Business structure                                               | Human and financial resource                               |

# Axiomatic design



Maintain the independence of FRs: In an acceptable design, the DPs and the FRs are related in such a way that a specific DP can be adjusted to satisfy its corresponding FR without affecting other FRs.

Minimize the information content: Among alternative designs which satisfy Axiom 1, the best has the minimum information content which means the maximum probability of success.

# Summary



- The concept generation process begins with a set of customer needs and target specifications and results in a set of product concepts from which the team will make a final selection.
- Although concept generation is an inherently creative process, teams can benefit from using a structured method. Such an approach allows full exploration of the design space and reduces the chance of oversight in the types of solution concepts considered.
- It also acts as a map for those team members who are less experienced in design problem-solving.
- Concept generation is a skill that can be learned and developed.

formal concept  
intuitive " } generation



# THANK YOU!



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## **Product Design**

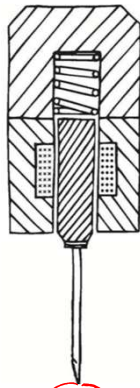
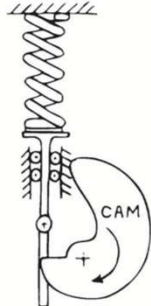
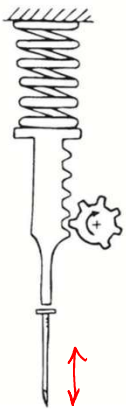
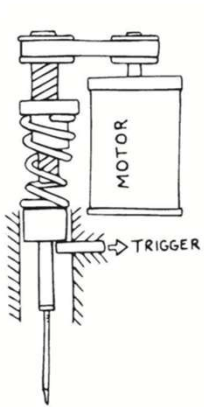
**(AE ZG531/ AEL ZG531/ DE ZG541/DM ZG541/MM ZG541)**



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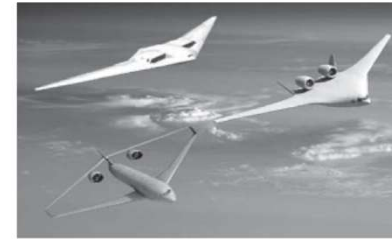
# Session 6: Concept Selection and Testing

# Examples of Concepts Generated



Handheld  
Nailer

1 ✓ product

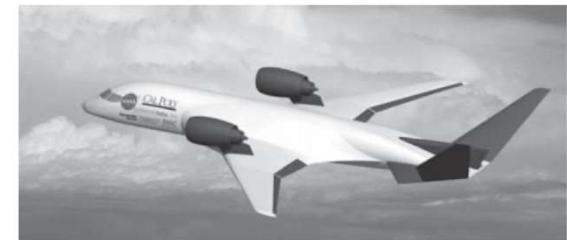


(a)



(b)

Futuristic  
Aircraft



(c)

What next?

- Decide and Choose the best concept  $\Rightarrow$  product
- Desirable attributes
- Process and documentation

Ref: Conceptual Aircraft Design, Wiley

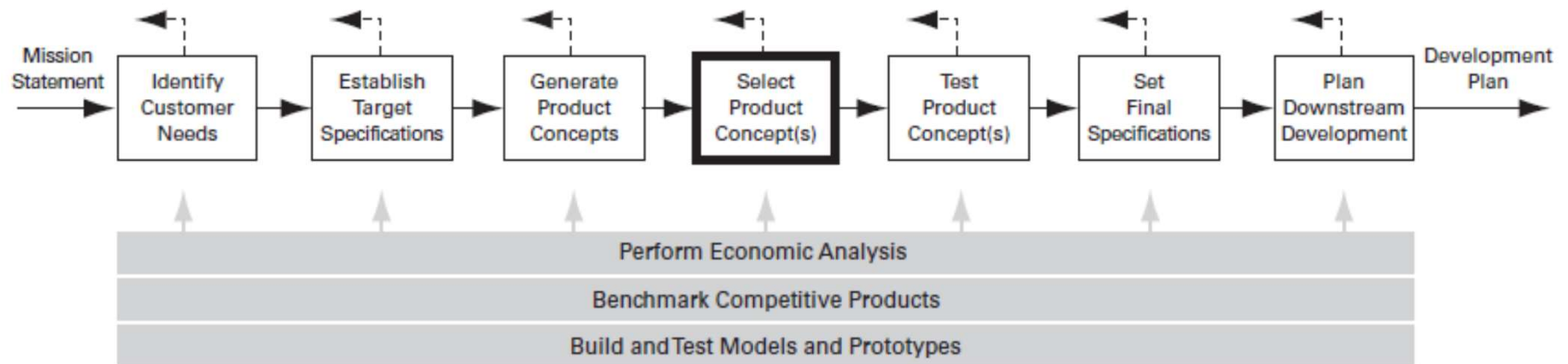
# Concept Selection Process



“Selecting” concepts/design requires:

- (1) A set of design **selection criteria** (Specifications)
- (2) A set of design **alternatives** (Generate Concepts) → *scoring / scaling*
- (3) A means to **evaluate** the alternatives against each criteria (Select Concept)

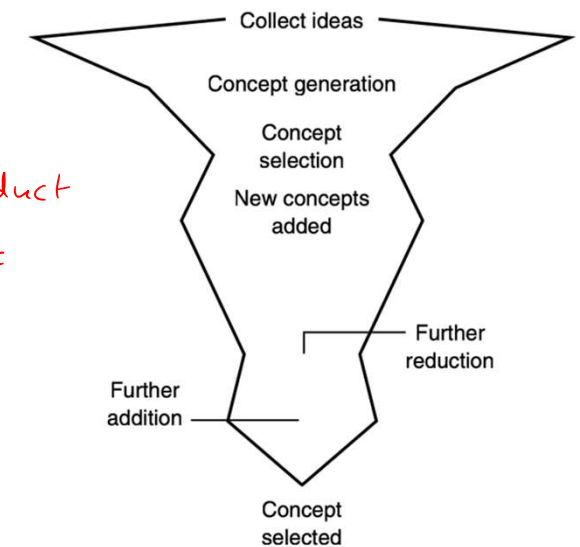
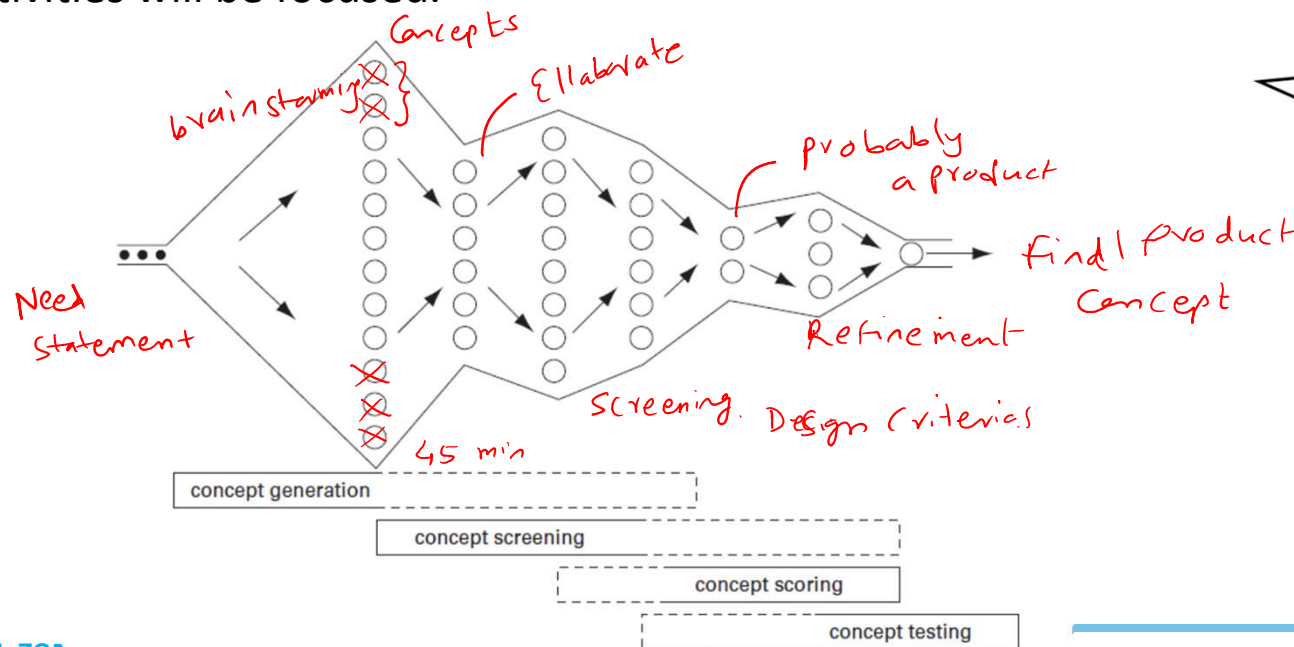
*Most prominent Concept*



# Concept Selection Process — funelling technique.



- Concept selection is an iterative process closely related to concept generation and testing.
- The concept screening and scoring methods help the team refine and improve the concepts, leading to one or more promising concepts upon which further testing and development activities will be focused.



# Patterns Observed in People



CFT  
/  
Product champion

Decide to continue with current action,  
and ignore information about risk of losses

fn. - 11 -  
new & old exp.  
Creative & lateral thinking  
Engg. Principles

Uncritically adopt whichever course of  
action is most strongly recommended

Search painstakingly for relevant information that  
is assimilated in an unbiased manner, and  
appraised carefully before a decision is made

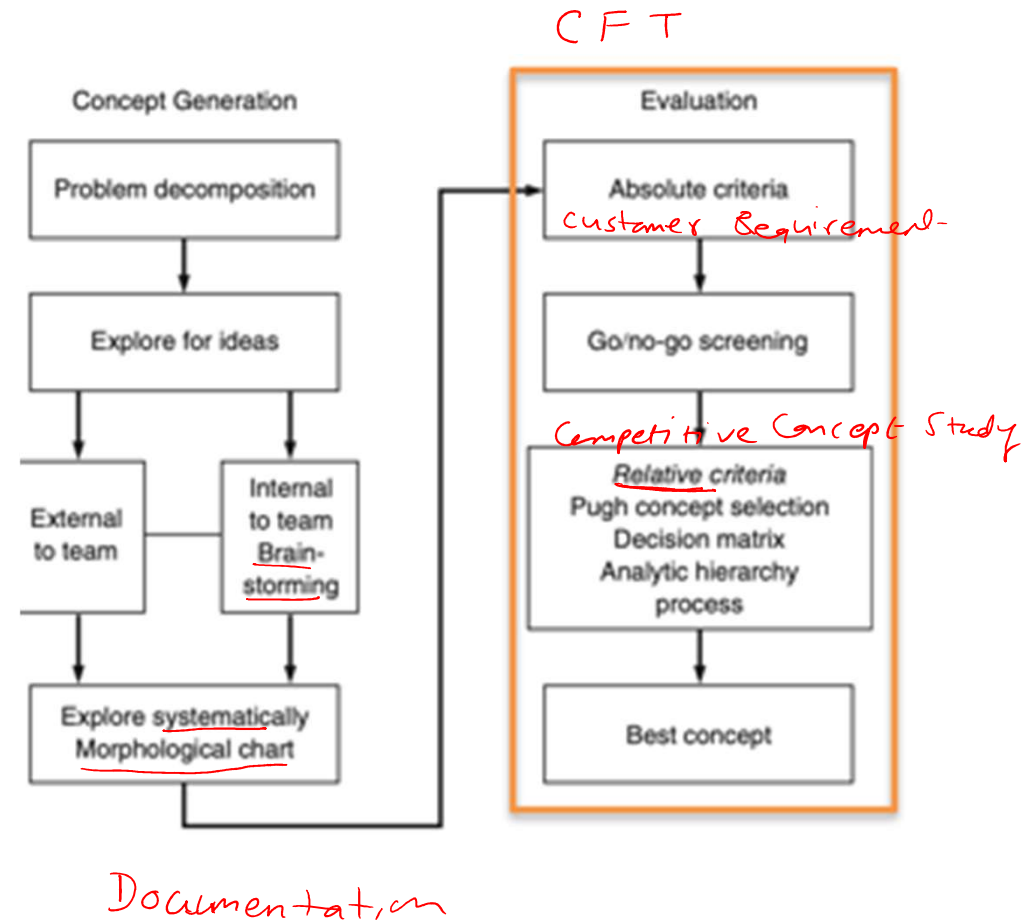
Evade conflict by procrastinating, shifting responsibility to  
someone else, and remaining inattentive to corrective information

Search in a hurried or in an "excited" way  
for an immediate problem solution

# Benefits of Structured Method



- A customer-focused product
- A competitive design
- Better product-process coordination
- Reduced time to product introduction
- Effective group decision making
- Documentation of the decision process



# Absolute and Relative Comparison



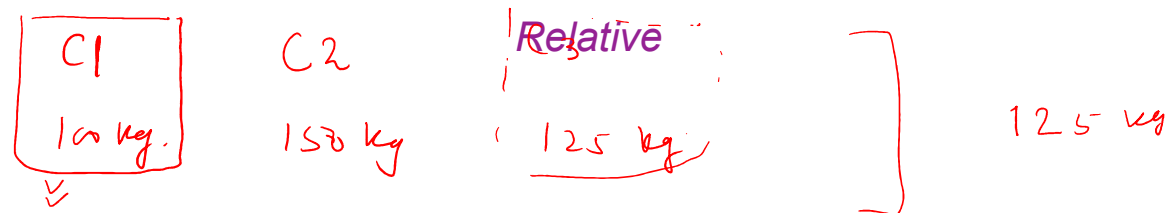
*Absolute comparison:* The concept is directly compared with a fixed and known set of requirements such as a product specification.

*Relative comparison:* The concepts are compared with each other on the basis of a metric.

## Absolute

Example 1: Checking to see if a design alternative would be under the weight limit

Example 2: Checking if the best design possible would be the lightest design



# Evaluation Process: Absolute Way



Filters used to select the design concept based on Judgement and Experience (in sequence)

- (1) Functional feasibility of the design
- (2) Technology readiness
- (3) Go/No-go screening

# Evaluation Process: Functional Feasibility



Concepts are placed into one of the three categories:

- (1) Not feasible
- (2) Feasible, under certain condition
- (3) Feasible

Expertise!  
Strong evidence!

# Evaluation Process: Go/No-go



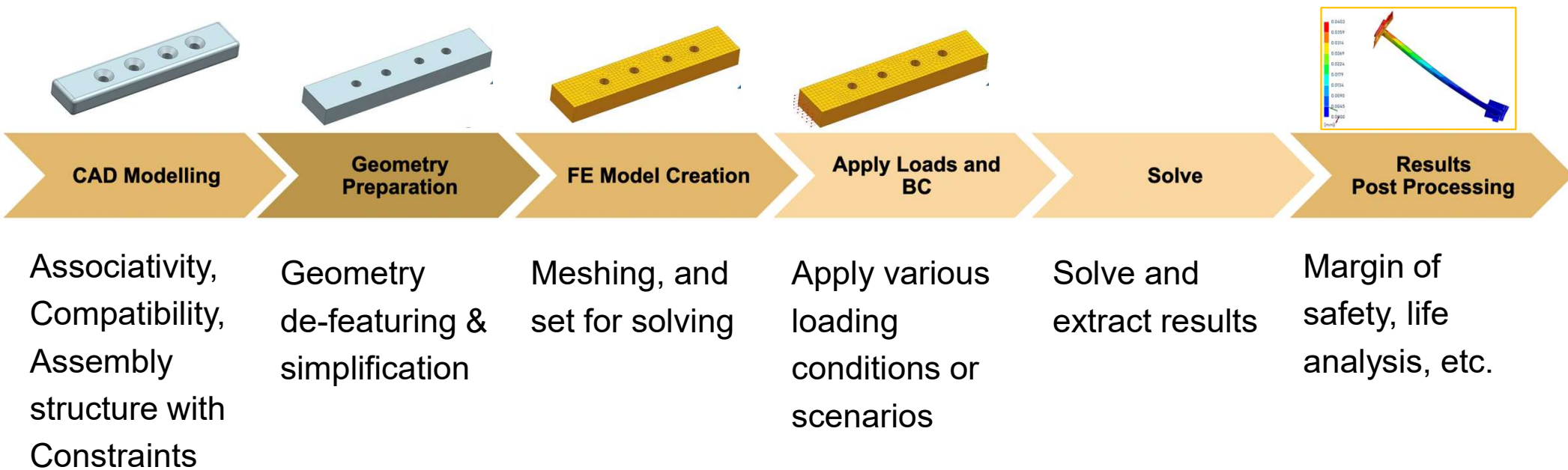
Check if the design concept satisfies the constraints of the problem, by eliminating the design concepts that are not able to meet the following:

- (1) Constraints
- (2) Minimum acceptable level of important engineering characteristics

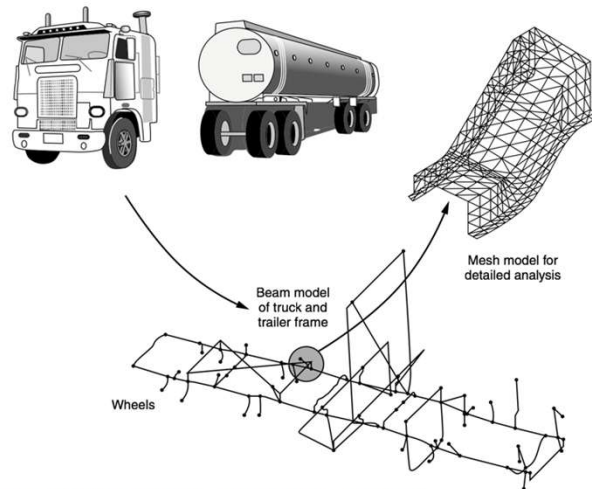
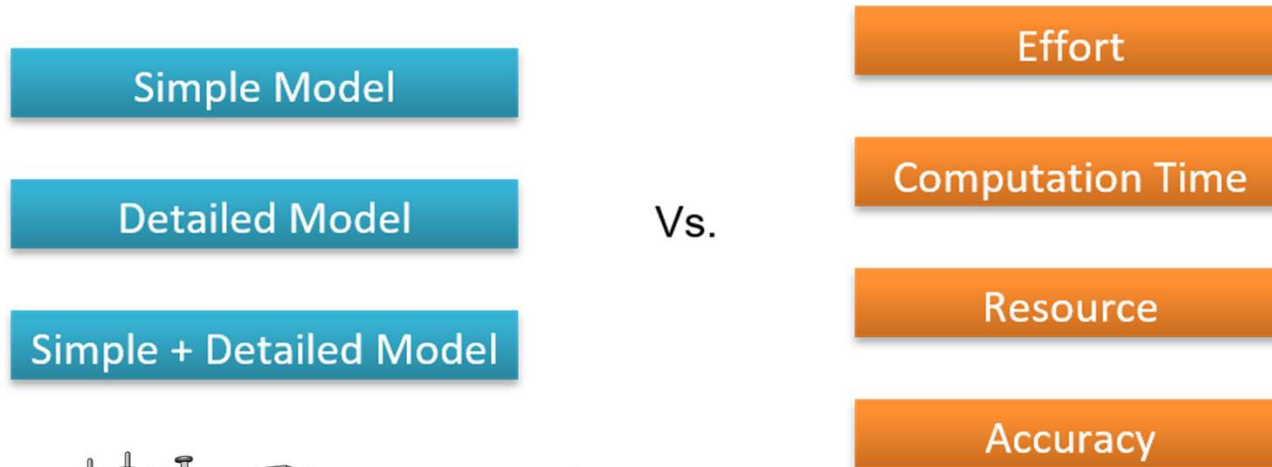


The weak areas (No-go) in a concept may be able to be fixed by borrowing ideas from another concept! >>> *Trigger a new idea*

# Geometric Modelling and FEA in Computer

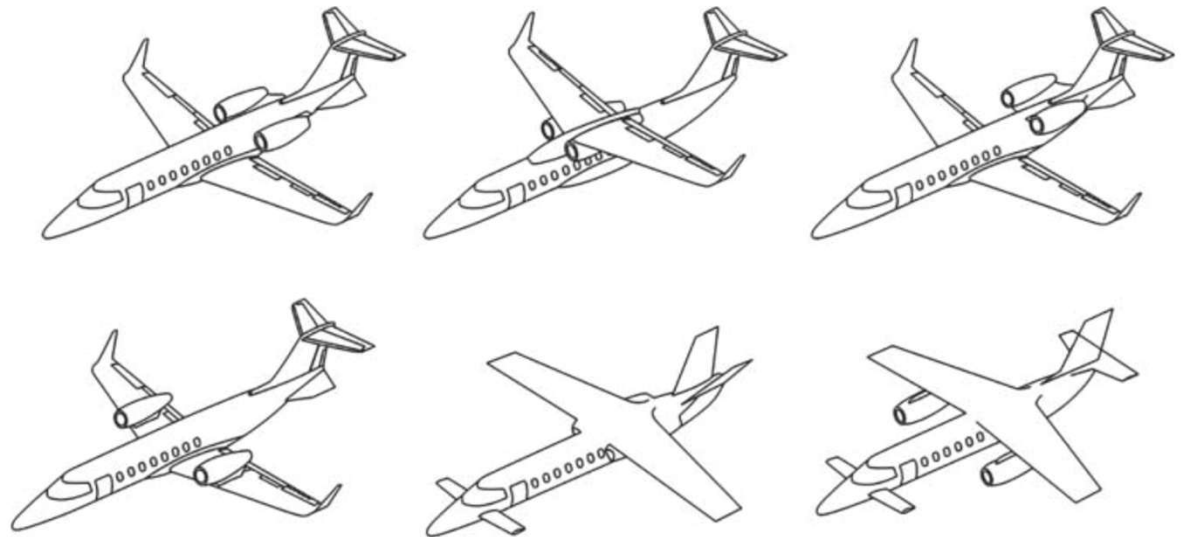


# FEA Approach





# Sample Case Study: Aircraft Design



# Aircraft Design: Design Drivers



## Market Drivers

(from operators)

Payload-range, Speed,  
Field performance,  
Comfort Level,  
Functionality  
Maintenance  
Support, etc  
Aircraft family  
Operational costs, etc

## Regulatory Drivers

(from government, etc)

Airworthiness Regulations,  
Policies (fare deregulation, etc)  
Route permission,  
Airport Fees.  
Interest rates  
Environmental issues  
Safety issues, etc

## Technology Drivers

(from industries)

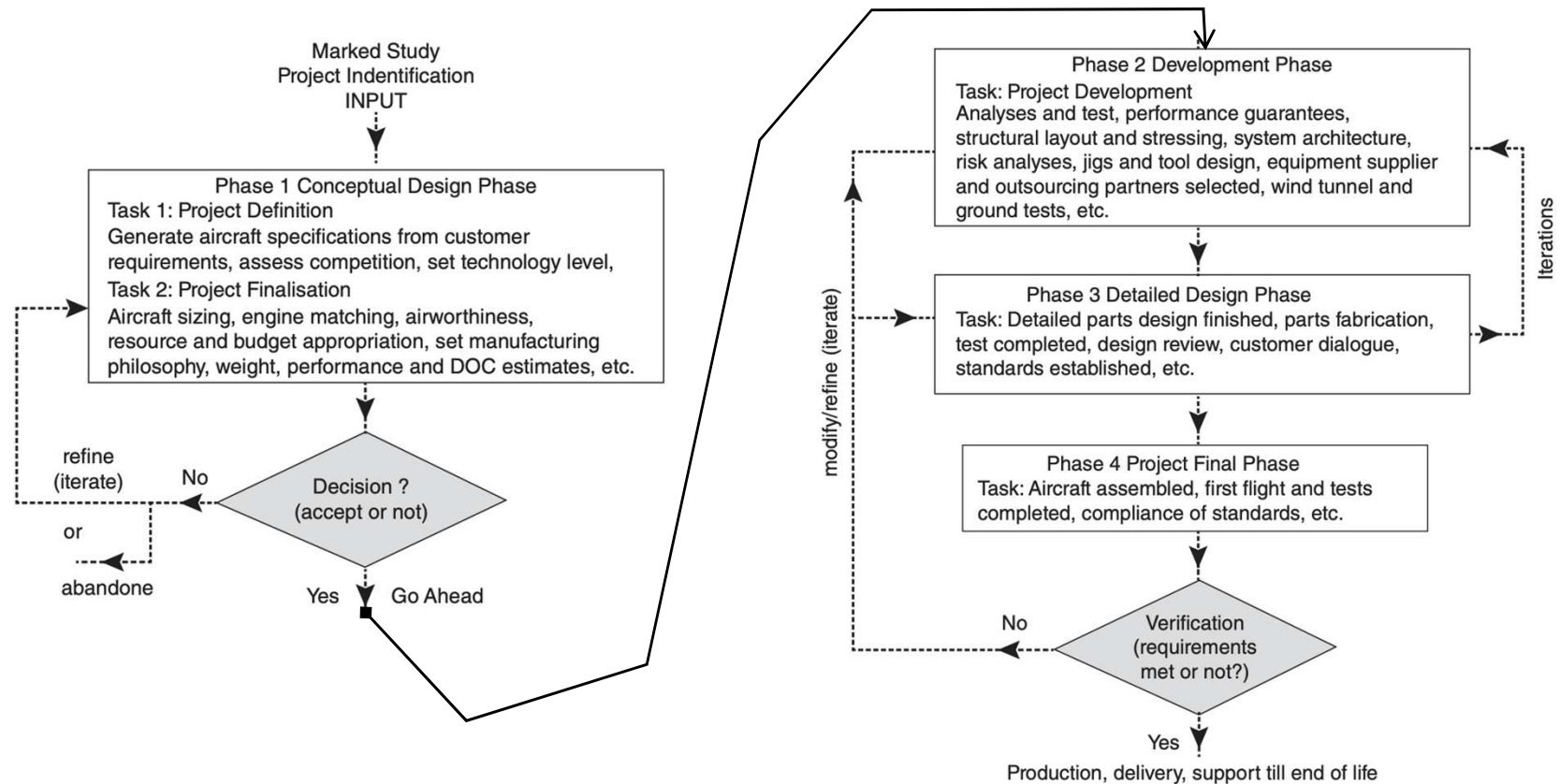
Aerodynamics, Propulsion,  
Structure, Material,  
Avionics/Electrical,  
System, Fly-by-wire, etc  
Manufacturing philosophy, etc

Iterations

The Final Design

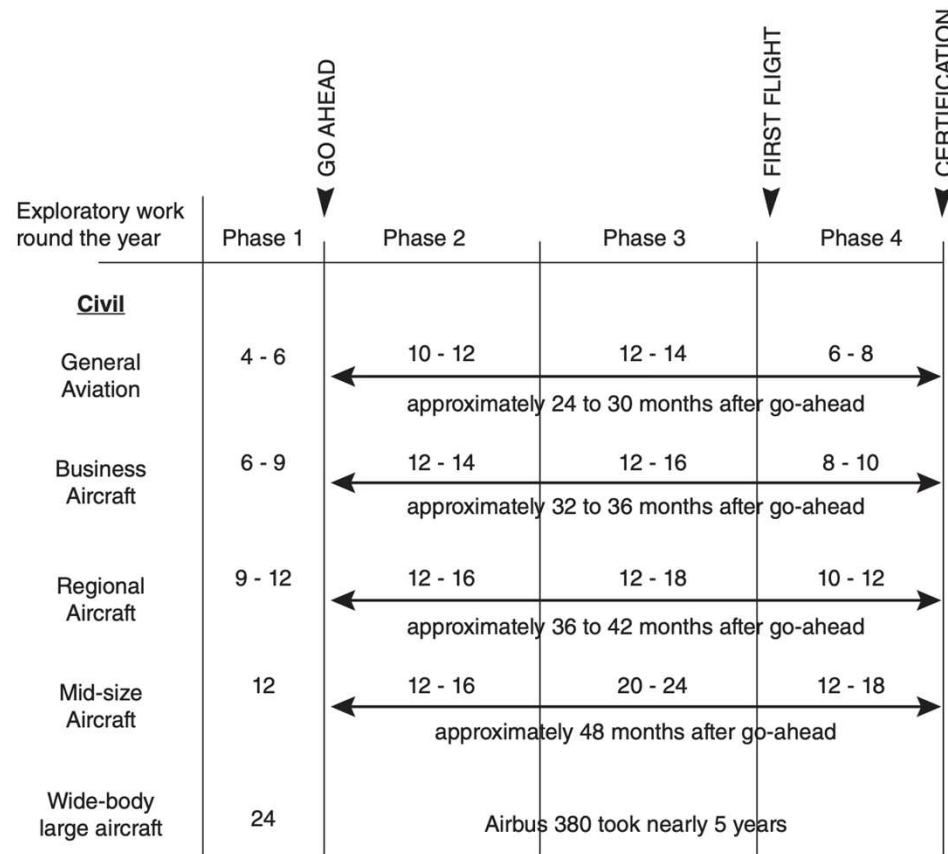
Ref: Conceptual Aircraft Design, Wiley

# Aircraft Design: Phases



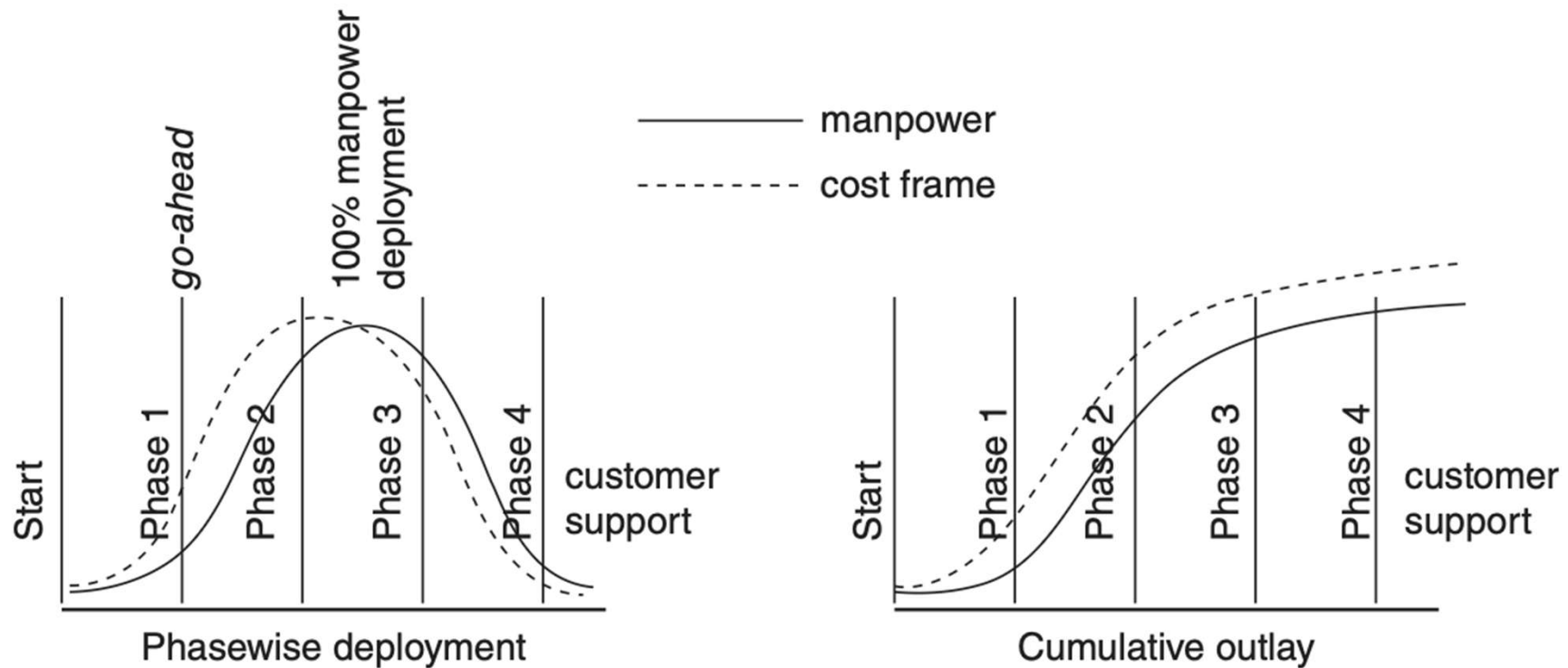
Ref: Conceptual Aircraft Design, Wiley

# Aircraft Design: Project Time Frame



Ref: Conceptual Aircraft Design, Wiley

# Aircraft Design: Manpower and Cost



Ref: Conceptual Aircraft Design, Wiley

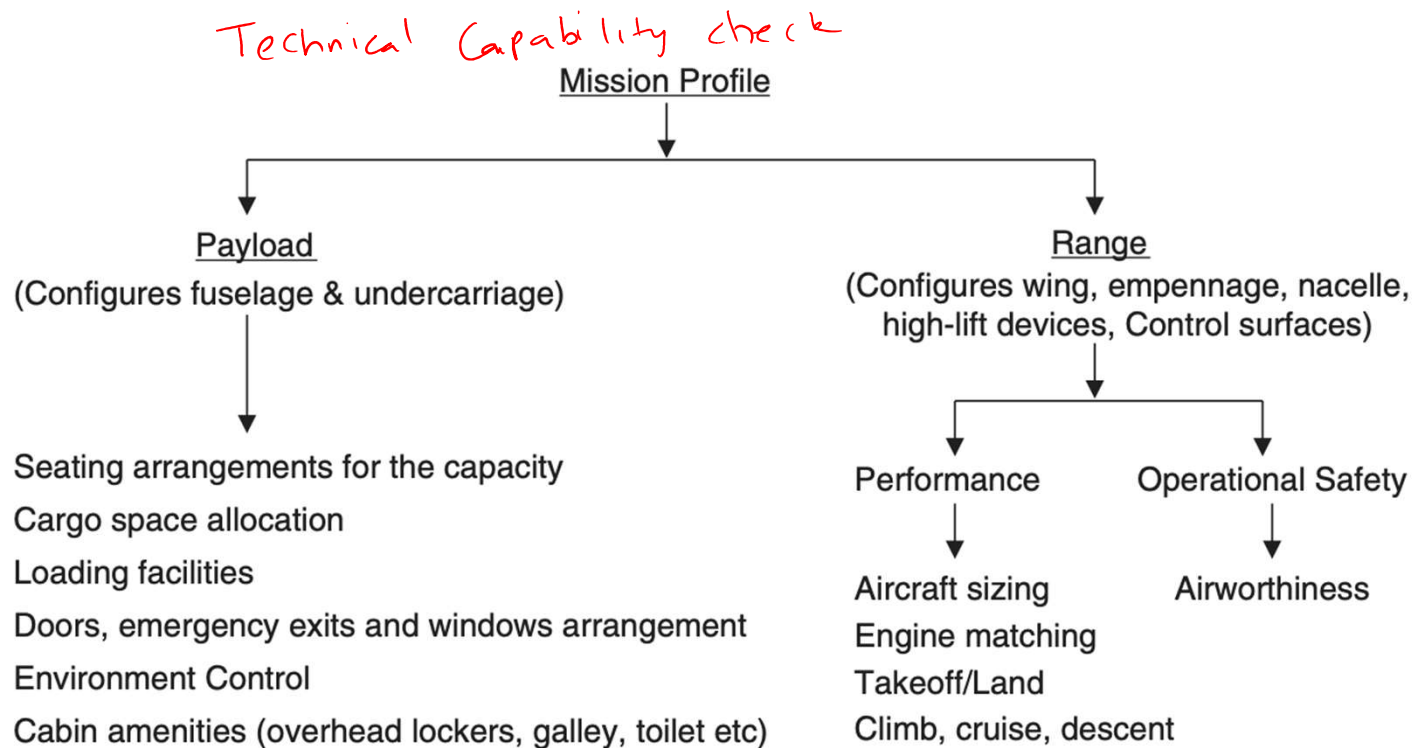
# Aircraft Design: Development Cost (2015)



| Aircraft class (turbofan)             | Development cost (US\$ <sup>a)</sup> ) | Unit cost (US\$* ) |
|---------------------------------------|----------------------------------------|--------------------|
| 6-passenger general aviation aircraft | 7–12 million                           | ≈±2 million        |
| 10-passenger business aircraft        | 25–40 million                          | 5–10 million       |
| 50-passenger regional aircraft        | 60–100 million                         | 25–30 million      |
| 150-passenger midsize aircraft        | 200–500 million                        | 50–70 million      |
| 500+ - passenger large aircraft       | 2–10 billion?                          | 200–300 million    |

Ref: Conceptual Aircraft Design, Wiley

# Aircraft Design: Top Level Definition



Ref: Conceptual Aircraft Design, Wiley

# Aircraft Design: Design Specification



## Airbus 320 Class

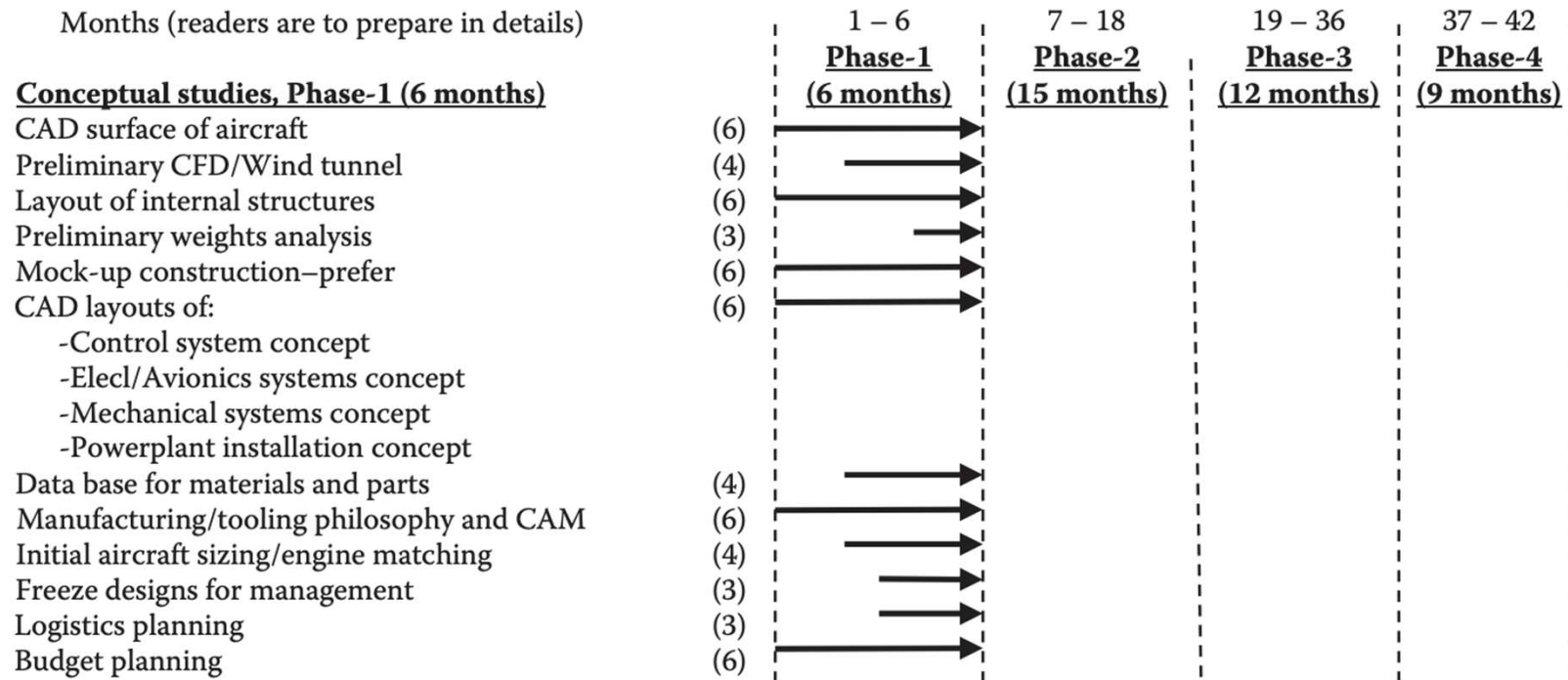
|                         |                                                            |
|-------------------------|------------------------------------------------------------|
| Payload                 | 150 passengers = $90 \times 150 = 14\,500$ kg              |
| Range                   | 2800 nm (nautical miles) + reserve                         |
| Crew                    | 2 pilots + 5 attendants                                    |
| Maximum cruise speed    | 0.75 Mach                                                  |
| Cruise altitude         | Above 30 000 ft (ceiling over 40 000 ft)                   |
| Takeoff distance        | 2000 m @ sea level to 15 m                                 |
| Landing distance        | 2000 m (at 95% takeoff weight) @ sea level from 15 m       |
| Initial rate of climb   | $14 \text{ m s}^{-1}$                                      |
| Undercarriage           | Retractable                                                |
| Cabin comfort           | Pressurised cabin with air-conditioning and oxygen supply, |
| Cabin interior diameter | =144 in.                                                   |
| Technology level        | Advanced                                                   |
| Power plant             | Turbofan engine                                            |

Ref: Conceptual Aircraft Design, Wiley



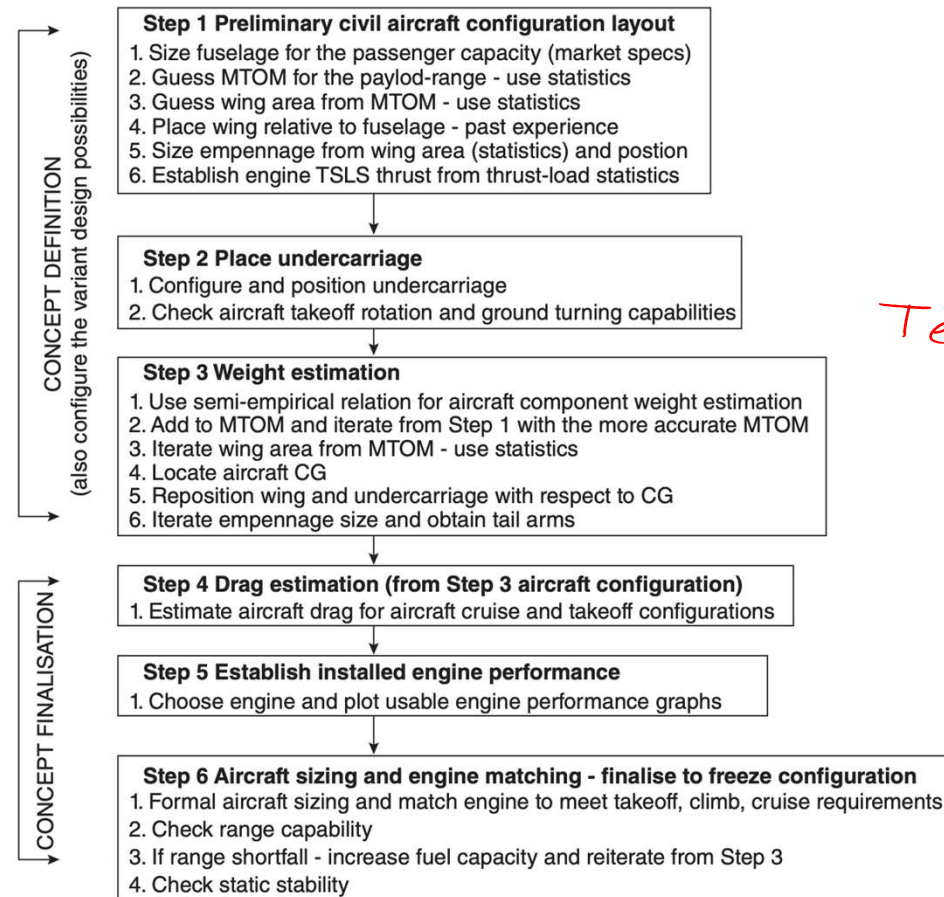
# Aircraft Design: Gantt chart for Concept Study

CAD/CAM/CAE  $\Rightarrow$  Aircraft



Ref: Conceptual Aircraft Design, Wiley

# Aircraft Design: Phase 1 Concept Study



Technical  
Parameter

~~Intuitive~~

structured design  
& evaluation

Ref: Conceptual Aircraft Design, Wiley



# THANK YOU!



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## **Product Design**

**(AE ZG531/ AEL ZG531/ DE ZG541/DM ZG541/MM ZG541)**



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# Session 7: Concept Selection and Testing

# Content



- Method for choosing a concept  $\Rightarrow$  Relative Concept Evaluation

- Concept screening (Pugh Concept Selection) method

*P matrix      P<sub>u</sub> matrix*

- Concept scoring (Weighted decision matrix) method

- Analytic Hierarchy Process (AHP)

- Concept testing process

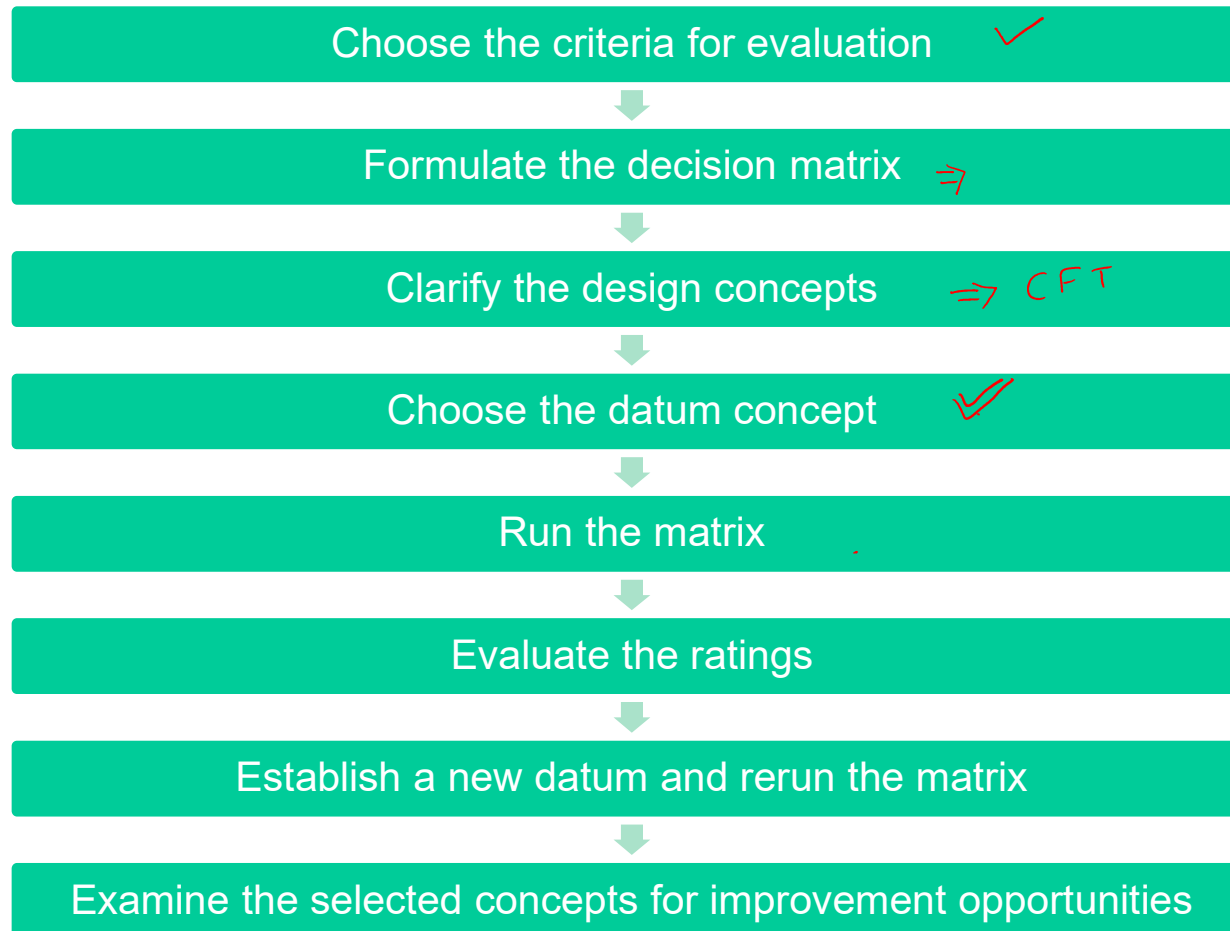
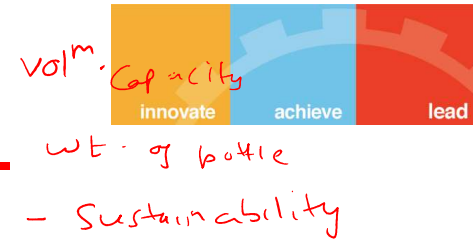
*1 Concept  $\Rightarrow$  successful product*

*as product*

| Criteria | Concept 1 | Concept 2 | Concept 3 |
|----------|-----------|-----------|-----------|
| 1        | .         | .         | .         |
| 2        | .         | .         | .         |
| 3        | .         | .         | .         |
|          |           |           |           |
|          |           |           |           |
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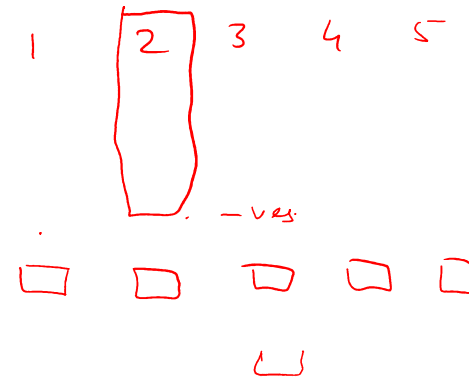
# Concept Screening (Pugh Concept Selection)

water  
bottle



5 Concepts

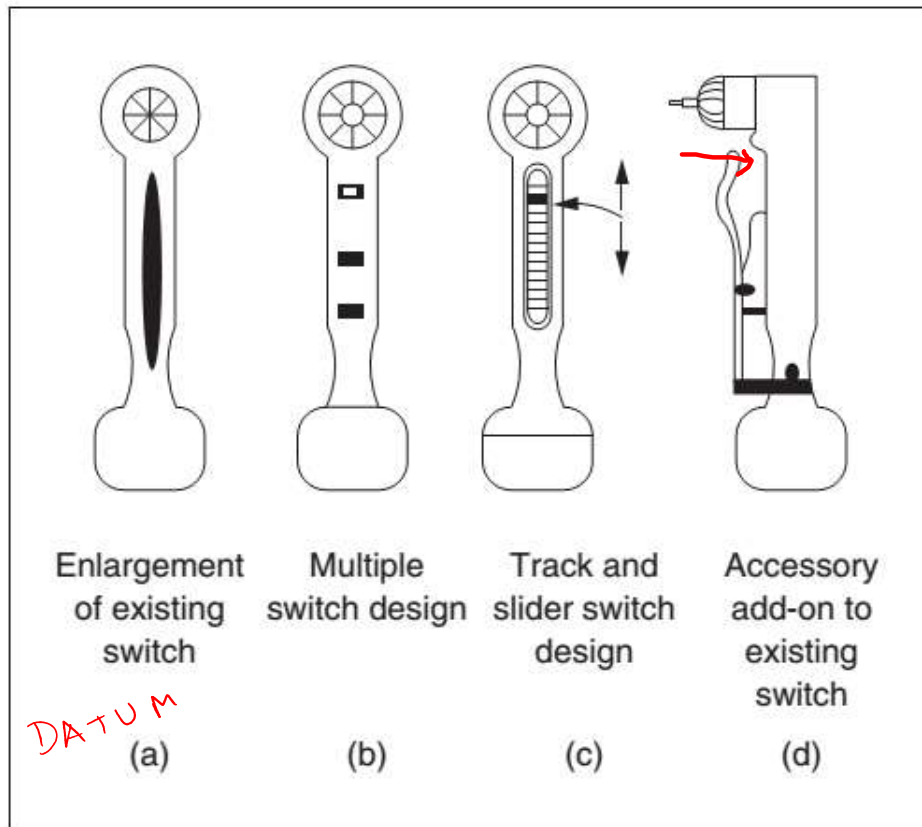
datum



# Pugh concept selection: Example



- Improvement of on/off switch in a right-angle drill:



- Concept A is a modest change to the existing switch, and will be the DATUM.
- Concept B adds three buttons for on/off/ and reverse.
- Concept C is a track and slider design.
- Concept D is an add-on accessory to make it easier to operate the existing switch.

# Pugh concept selection: Example



*Design Evaluation*

| ROW          | CRITERIA                  | CONCEPT |       |   |    |
|--------------|---------------------------|---------|-------|---|----|
|              |                           | D       | A     | C | B  |
| 1            | COST                      | +       |       | + | +  |
| 2            | ADDED FUNCTIONALITY       | =       | S     | = | -  |
| 3            | SIMPLICITY OF DESIGN      | +       | S     | = | =  |
| 4            | AVAILABILITY OF MATERIALS | +       | S     | - | -  |
| 5            | EASE OF MANUFACTURING     | +       | DATUM | + | =  |
| 6            | EASE OF ASSEMBLY          | +       |       | + | -  |
| 7            | ABILITY TO PROTOTYPE      | +       |       | = | +  |
| 8            | COMFORT                   | -       |       | + | -  |
| 9            | WEIGHT                    | +       |       | = | -  |
| 10           | AESTHETICS                | -       | S     | = | -  |
| max = PLUSES |                           | 7       | 0     | 4 | 2  |
| MINUSES      |                           | 2       | 0     | 1 | 6  |
|              |                           | 5       | 0     | 3 | -4 |

*Handwritten notes:*  
 - "baseline w.r.t." with an arrow pointing to the DATUM column.  
 - "max = PLUSES" and "MINUSES" with arrows pointing to the bottom rows.  
 - "5" in a box below the final scores.  
 - "Attacking -ve." with an arrow pointing to the final scores.

The highest-ranking design, an add-on attachment that makes it easier to operate the switch, has two negatives, poorer aesthetic appeal and poor ergonomics (comfort to the hand).

- Design D provides force amplification, but it is not easy on the ligaments in the fingers.
- The next ranking design, the track and slider design, has only a single minus for "availability of materials."

# Example: Coffee Mill



- Evaluation of cleaning function of coffee mill:

| Removable Chamber                                                                                                   | Removable Blade                                                                                                     | Washable                                                                        | Scraped                                                                                                                                        | Removable Unit                                                                                                      |
|---------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|
|                                                                                                                     |                                                                                                                     |                                                                                 |                                                                                                                                                |                                                                                                                     |
| <ul style="list-style-type: none"> <li>The joint from the unit to the chamber</li> <li>Noise suppression</li> </ul> | <ul style="list-style-type: none"> <li>Contaminants in the center hole</li> <li>Pop-off blade complexity</li> </ul> | <ul style="list-style-type: none"> <li>Sealing</li> <li>Heat buildup</li> </ul> | <ul style="list-style-type: none"> <li>Effective wipe over 3D surface</li> <li>Flatness during grinding</li> <li>Dirt along scraper</li> </ul> | <ul style="list-style-type: none"> <li>The joint from the unit to the chamber</li> <li>Noise suppression</li> </ul> |

# Example: Coffee Mill



## Coffee mill redesign evaluation criteria

|                  |                                                                                           |
|------------------|-------------------------------------------------------------------------------------------|
| Cost             | Unit manufacturing cost, not development cost, not delivery cost (\$)                     |
| Store grinder    | To put away in a cabinet out of sight (cm <sup>3</sup> )                                  |
| Put in beans     | From the time beans are in a bag until the chopper switch can be activated (s)            |
| Take out coffee  | Removing all grounds, getting the coffee into a coffee maker (s)                          |
| Power setup      | From the time the grinder is plugged in until the switch can be activated (s)             |
| Cleanable        | After coffee taken out to the point of being spotless (s or number of steps)              |
| Development risk | Difficulty getting a working alpha prototype (number of potential faults or difficulties) |

# Example: Coffee Mill

bench marked concept → Datum  
 — your own previous design → -1-



|                   | Removable Chamber                                                                 | Removable Blade                                                                   | Washable                                                                          | Scraper                                                                            | Removable Unit                                                                      |
|-------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
|                   |  |  |  |  |  |
| Cost              | S                                                                                 | +                                                                                 | -                                                                                 | +                                                                                  | S                                                                                   |
| Store Grinder     | S                                                                                 | +                                                                                 | +                                                                                 | +                                                                                  | S                                                                                   |
| Put in Beans      | S                                                                                 | S                                                                                 | -                                                                                 | -                                                                                  | S                                                                                   |
| Take Out Coffee ✓ | S                                                                                 | -                                                                                 | -                                                                                 | -                                                                                  | -                                                                                   |
| Power Setup       | S                                                                                 | S                                                                                 | S                                                                                 | S                                                                                  | S                                                                                   |
| Cleanable ✓       | S                                                                                 | -                                                                                 | S                                                                                 | -                                                                                  | S                                                                                   |
| Development Risk  | S                                                                                 | +                                                                                 | -                                                                                 | +                                                                                  | S                                                                                   |
| Σ+                | 0                                                                                 | +3                                                                                | +1                                                                                | +3                                                                                 | 0                                                                                   |
| Σ-                | 0                                                                                 | -2                                                                                | -4                                                                                | -3                                                                                 | -1                                                                                  |
| Σ                 | 0                                                                                 | +1                                                                                | -3                                                                                | 0                                                                                  | -1                                                                                  |

## Guidelines:

- Do not become too quantitative
- Do not reject before proper examination
- Examine highly rated concepts
- Look for 'positives' features
- If most concepts get the same rating on a certain criterion, review the 'clarity' and evaluation

S/o/blank

Iterative

# New Datum & Re-run



Purpose of this step is:

- To establish a new datum, usually the 'highest' rated concept, and run the matrix again
- Not to verify the selection in round-1, but to gain more insight
- Get different perspectives (due to new datum), and get more clarity on relative strengths and weaknesses *SWOT*

Examining the selected concept will help to:

- Generate new approach
- Change negatives to positives
- Identify design modifications

# Example: Bread Toaster



4 slot electric toaster



Gas grill



Testing fork and fire



Electric conveyer



**Domestic Use Purpose**

# Example: Bread Toaster



## Step 1: Identified set of criteria for evaluation

| Primary            | Secondary                                                    |
|--------------------|--------------------------------------------------------------|
| Good Toast Quality | Even Toasting<br>Good Taste<br>Repeatable<br>Quick           |
| Have Capacity      | Large Range of Bread Products<br>Multiple Slices/Units       |
| Long Life          | Reliable<br>Durable<br>Low<br>Maintenance                    |
|                    | Affordable<br>Attractive<br>Safe<br>Good Size                |
| Easy to Use        | Easy to use Controls<br>Easy to Load<br>Easy to Remove Toast |

# Example: Bread Toaster



Step 2: Select one concept as a datum

|                               |                     |                               | Design Concepts |        |                 |                      |                   |           |  |  |
|-------------------------------|---------------------|-------------------------------|-----------------|--------|-----------------|----------------------|-------------------|-----------|--|--|
|                               |                     |                               |                 | Weight | Electric 4-slot | Toasting Fork & Fire | Electric Conveyor | Gas Grill |  |  |
| Pugh Concept Selection Matrix |                     |                               |                 |        |                 |                      |                   |           |  |  |
| Selection Criteria            | Good Toast Quality  | Even Toasting                 |                 | 0      |                 |                      |                   |           |  |  |
|                               |                     | Good Taste                    |                 | 0      |                 |                      |                   |           |  |  |
|                               |                     | Repeatable                    |                 | 0      |                 |                      |                   |           |  |  |
|                               |                     | Quick                         |                 | 0      |                 |                      |                   |           |  |  |
|                               | Capacity            | Large Range of Bread Products |                 | 0      |                 |                      |                   |           |  |  |
|                               |                     | Multiple Slices/Units         |                 | 0      |                 |                      |                   |           |  |  |
|                               | Long Life           | Reliable                      |                 | 0      |                 |                      |                   |           |  |  |
|                               |                     | Durable                       |                 | 0      |                 |                      |                   |           |  |  |
|                               |                     | Low Maintenance               |                 | 0      |                 |                      |                   |           |  |  |
|                               | Physical Attributes | Affordable                    |                 | 0      |                 |                      |                   |           |  |  |
|                               |                     | Attractive                    |                 | 0      |                 |                      |                   |           |  |  |
|                               |                     | Safe                          |                 | 0      |                 |                      |                   |           |  |  |
|                               |                     | Good Size                     |                 | 0      |                 |                      |                   |           |  |  |
|                               | Easy to Use         | Easy to use Controls          |                 | 0      |                 |                      |                   |           |  |  |
|                               |                     | Easy to Load                  |                 | 0      |                 |                      |                   |           |  |  |
|                               |                     | Easy to Remove Toast          |                 | 0      |                 |                      |                   |           |  |  |
|                               |                     |                               |                 |        |                 |                      |                   |           |  |  |
| TOTAL +                       |                     |                               |                 | 0      |                 |                      |                   |           |  |  |
| TOTAL -                       |                     |                               |                 | 0      |                 |                      |                   |           |  |  |
| TOTAL SCORE                   |                     |                               |                 | 0      |                 |                      |                   |           |  |  |
| WEIGHTED SCORE                |                     |                               |                 |        |                 |                      |                   |           |  |  |

# Example: Bread Toaster



Step 3: Compared all the concepts with respect to datum concept.

|                    |                     |                               | Design Concepts               |        |                 |                      |                   |           |
|--------------------|---------------------|-------------------------------|-------------------------------|--------|-----------------|----------------------|-------------------|-----------|
|                    |                     |                               |                               |        |                 |                      |                   |           |
|                    |                     |                               | Pugh Concept Selection Matrix | Weight | Electric 4-slit | Toasting Fork & Fire | Electric Conveyor | Gas Grill |
| Selection Criteria | Good Toast Quality  | Even Toasting                 |                               | 0      | -               | 0                    | -                 |           |
|                    |                     | Good Taste                    |                               | 0      | ++              | 0                    | 0                 |           |
|                    |                     | Repeatable                    |                               | 0      | -               | +                    | -                 |           |
|                    |                     | Quick                         |                               | 0      | +               | 0                    | 0                 |           |
|                    | Capacity            | Large Range of Bread Products |                               | 0      | +               | 0                    | +                 |           |
|                    |                     | Multiple Slices/Units         |                               | 0      | -               | +                    | +                 |           |
|                    | Long Life           | Reliable                      |                               | 0      | +               | -                    | 0                 |           |
|                    |                     | Durable                       |                               | 0      | +               | 0                    | 0                 |           |
|                    |                     | Low Maintenance               |                               | 0      | -               | -                    | 0                 |           |
|                    | Physical Attributes | Affordable                    |                               | 0      | +               | -                    | +                 |           |
|                    |                     | Attractive                    |                               | 0      | +               | -                    | -                 |           |
|                    |                     | Safe                          |                               | 0      | -               | -                    | -                 |           |
|                    |                     | Good Size                     |                               | 0      | -               | -                    | -                 |           |
|                    | Easy to Use         | Easy to use Controls          |                               | 0      | -               | 0                    | +                 |           |
|                    |                     | Easy to Load                  |                               | 0      | -               | +                    | -                 |           |
|                    |                     | Easy to Remove Toast          |                               | 0      | -               | +                    | -                 |           |
|                    | TOTAL +             |                               |                               |        | 0               | 8                    | 4                 | 4         |
| TOTAL -            |                     |                               |                               | 0      | 18              | 6                    | -8                |           |
| TOTAL SCORE        |                     |                               |                               | 0      | -8              | -2                   | -4                |           |
| WEIGHTED SCORE     |                     |                               |                               |        |                 |                      |                   |           |

# Weighted Decision Matrix

Simple criteria  
Simple criteria  
Sub criteria



Weighted Decision Matrix is a method of evaluating concepts by ranking the **design criteria** with **weighting factors**, and scoring the degree to which each design concept meets the criteria.

Systematic methods of determining weighting factors are:

## 1. Direct Assignment

*Based on experience, expertise, and engineering judgement*

⇒ product champion

## 2. Objective Tree

*Hierarchical objective tree, Based on experience and expertise*

## 3. Analytic Hierarchy Process

*Least arbitrary method*

(AHP)

productivity  
cost  
maintainability  
safety

100%

0.3  
0.2  
0.15  
0.35

importance  
rating

Compare / Evaluate

Concepts

# Concept Scoring Method (Weighted Decision Matrix)



$$\sum_{i=1}^n w_i = 1.0 \text{ and } 0 \leq w_i \leq 1$$



Evaluation Scheme for Design Alternatives or Objectives

| 11-point Scale | Description                         | 5-point Scale | Description  |
|----------------|-------------------------------------|---------------|--------------|
| 0              | Totally useless solution            | 0             | Inadequate   |
| 1              | Very inadequate solution            |               |              |
| 2              | Weak solution                       | 1             | Weak         |
| 3              | Poor solution                       |               |              |
| 4              | Tolerable solution                  | 2             | Satisfactory |
| 5              | Satisfactory solution               |               |              |
| 6              | Good solution with a few drawbacks  | 3             | Good         |
| 7              | Good solution                       |               |              |
| 8              | Very good solution                  | 4             | Excellent    |
| 9              | Excellent (exceeds the requirement) |               |              |
| 10             | Ideal solution                      |               |              |

# Weighted Decision Matrix – Example

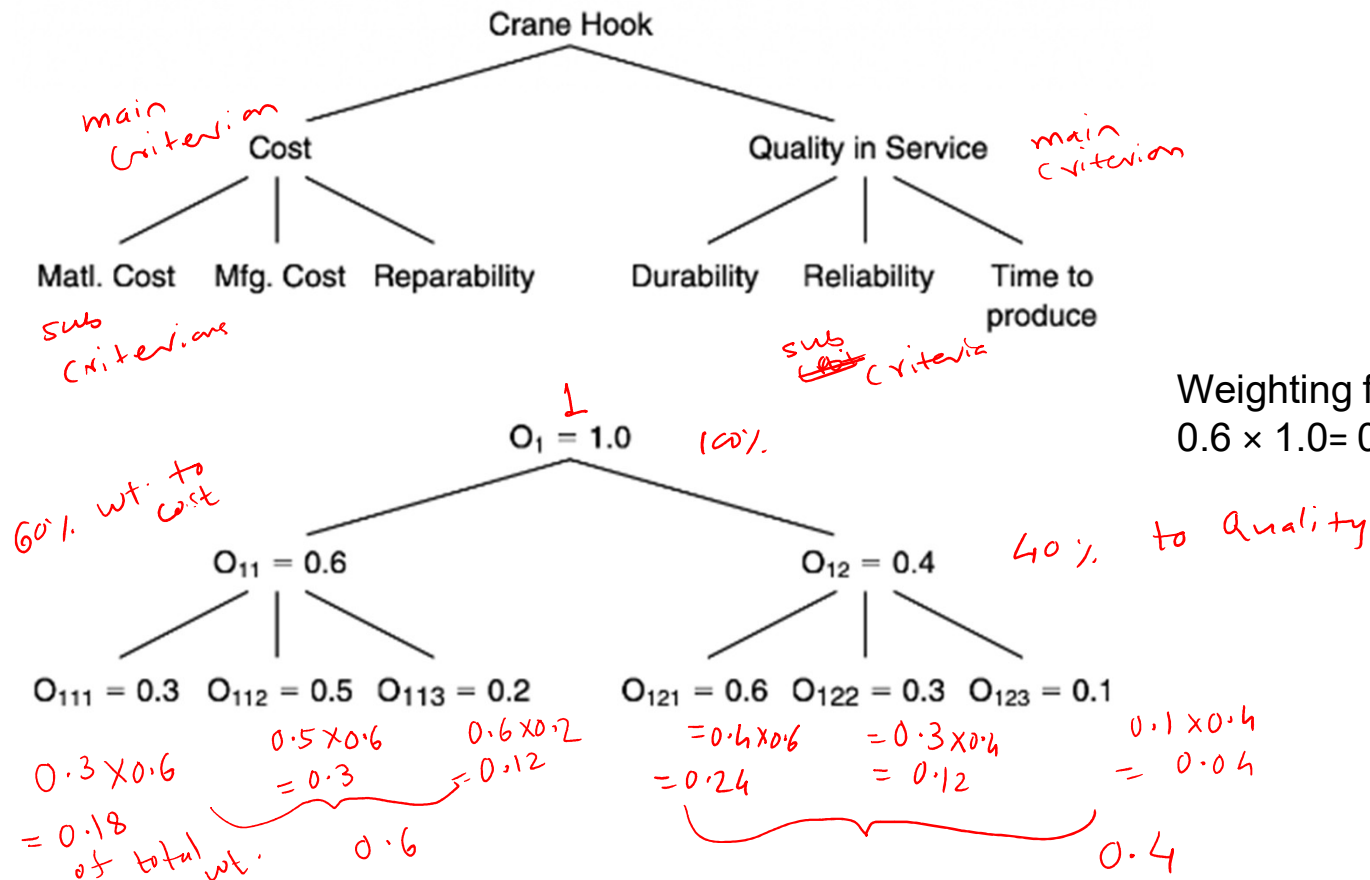


11 point scale

Weighted Decision Matrix for a Steel Crane Hook

| Design Criterion   | Weight Factor | Units      | Built-Up Plates Welded |       |                         | Built-Up Plates Riveted |       |                         | Cast Steel Hook |       |                           |
|--------------------|---------------|------------|------------------------|-------|-------------------------|-------------------------|-------|-------------------------|-----------------|-------|---------------------------|
|                    |               |            | Magnitude              | Score | Rating                  | Magnitude               | Score | Rating                  | Magnitude       | Score | Rating                    |
| Material cost      | 0.18          | c/lb       | 60                     | 8     | $0.18 \times 8$<br>1.44 | 60                      | 8     | $0.18 \times 8$<br>1.44 | 50              | 9     | $0.18 \times 9 =$<br>1.62 |
| Manufacturing cost | 0.30          | \$         | 2500                   | 7     | $0.3 \times 7$<br>2.10  | 2200                    | 9     | 2.70                    | 3000            | 4     | 1.20                      |
| Reparability       | 0.12          | Experience | Good                   | 7     | 0.84                    | Excellent               | 9     | 1.08                    | Fair            | 5     | 0.60                      |
| Durability         | 0.24          | Experience | High                   | 8     | 1.92                    | High                    | 8     | 1.92                    | Good            | 6     | 1.44                      |
| Reliability        | 0.12          | Experience | Good                   | 7     | 0.84                    | Excellent               | 9     | 1.08                    | Fair            | 5     | 0.60                      |
| Time to produce    | 0.04          | Hours      | 40                     | 7     | $0.04 \times 7$<br>0.28 | 25                      | 9     | 0.36                    | 60              | 5     | 0.20                      |
|                    |               |            | 7.42                   |       |                         | 8.58                    |       |                         | 5.66            |       |                           |

# Objective Tree for Weighting Factors



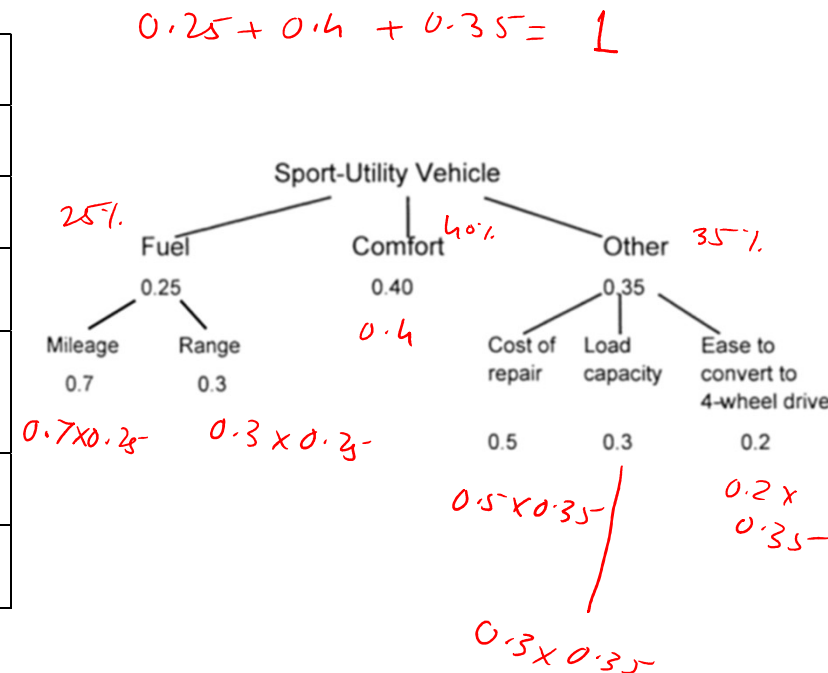
Weighting factor for material cost,  $O_{111} = 0.3 \times 0.6 \times 1.0 = 0.18$

# Weighted Decision Matrix – Example



- Four preliminary designs for sport-utility vehicles had the characteristics listed below Table. An appropriate weighing factor is selected based on the objective tree given in the below Figure. Using the weighted decision matrix, which design looks to be the most promising?

| Characteristics                  | Parameter       | Design A  | Design B  | Design C | Design D |
|----------------------------------|-----------------|-----------|-----------|----------|----------|
| Gas mileage                      | Miles per gal   | 20        | 16        | 15       | 20       |
| Range                            | Miles           | 300       | 240       | 260      | 400      |
| Ride comfort                     | Rating          | Poor      | Very Good | Good     | Fair     |
| Ease to convert to 4-wheel drive | Rating          | Very Good | Good      | Good     | Poor     |
| Load Capacity                    | Kg              | 1000      | 700       | 1000     | 600      |
| Cost of repair                   | Avg. of 5 parts | \$700     | \$625     | \$600    | \$500    |



# Weighted Decision Matrix – Example

0-10 scale

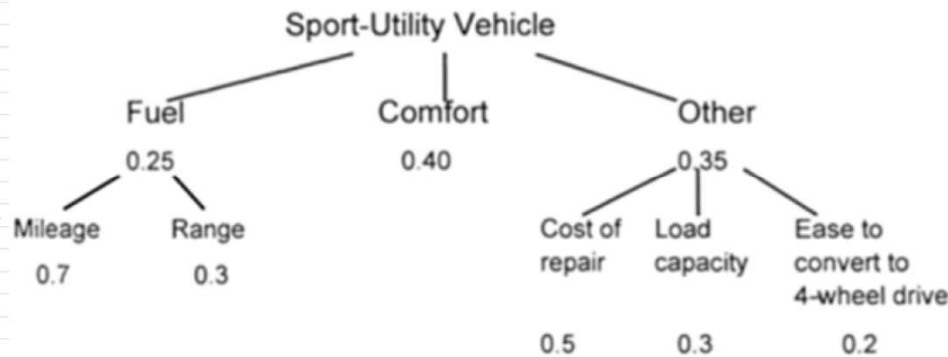


| Concept Scoring Method (Weighted Decision matrix) |                                  |                 |                           |            |         |                    |            |         |                  |            |         |                |            |         |                |
|---------------------------------------------------|----------------------------------|-----------------|---------------------------|------------|---------|--------------------|------------|---------|------------------|------------|---------|----------------|------------|---------|----------------|
| SI N                                              | Characteristics                  | Parameters      | Weight factor             | Design A   |         |                    | Design B   |         |                  | Design C   |         |                | Design D   |         |                |
|                                                   |                                  |                 |                           | Maginitude | Ratings | Weighted score     | Maginitude | Ratings | Weighted score   | Maginitude | Ratings | Weighted score | Maginitude | Ratings | Weighted score |
| 1                                                 | Gas mileage                      | Miles per gal   | $0.7 \times 0.25 = 0.175$ | ✓ 20       | 9       | $0.175 \times 9 =$ | 16         | 7       | $0.175 \times 7$ | 15         | 6       |                | 20         | 9       |                |
| 2                                                 | Range                            | Miles           | $0.3 \times 0.25 = 0.075$ | 300        | 8       | $0.075 \times 8 =$ | 240        | 5       | $0.075 \times 5$ | 260        | 6       |                | 400        | 9       |                |
| 3                                                 | Ride comfort                     | Rating          | 0.4                       | ✓ Poor     | 3       | $0.4 \times 3 =$   | Very Good  | 8       | $0.4 \times 8$   | Good       | 7       |                | Fair       | 5       |                |
| 4                                                 | Ease to convert to 4-wheel drive | Rating          | 0.07                      | Very Good  | 8       | $0.07 \times 8 =$  | Good       | 7       |                  | Good       | 7       |                | Poor       | 3       |                |
| 5                                                 | Load capacity                    | Kg              | 0.105                     | 1000       | 9       | $0.105 \times 9 =$ | 700        | 7       |                  | 1000       | 9       |                | 600        | 6       |                |
| 6                                                 | Cost of repair                   | Avg. of 5 parts | 0.175                     | ✓ \$700    | 3       | $0.175 \times 3 =$ | \$625      | 5       |                  | \$600      | 5       |                | \$500      | 9       |                |
|                                                   |                                  | Sum of Weighted | 1                         |            |         | $\Sigma$           |            |         | $\Sigma$         |            |         | $\Sigma$       |            |         | $\Sigma$       |
|                                                   |                                  | Rank            |                           |            |         | Sum of Column      | 5.405      |         |                  |            |         |                |            |         |                |

0 to 5

check

1.57  
0.6  
0.52  
0.945  
0.525  
5.405



↑ highest value

# Analytic Hierarchy Process (AHP)



- The Analytic Hierarchy Process (AHP) is a problem-solving methodology for making a choice from among a set of alternatives when the selection criteria represent multiple objectives, have a natural hierarchical structure, or consist of qualitative and quantitative measurements.
- AHP builds upon the mathematical properties of matrices for making consistent pairwise comparisons.   
*Handwritten notes: Cost, availability, Colour, Electrical Consumption, LG, Samsung, IFS, choice of alternative, selection criteria*
- AHP also defines a pairwise, comparison-based method for determining relative ratings for the degree to which each of a set of options fulfils each of the criteria.
- AHP includes the calculation of an inconsistency measurement and threshold values that determine if the comparison process has remained consistent.

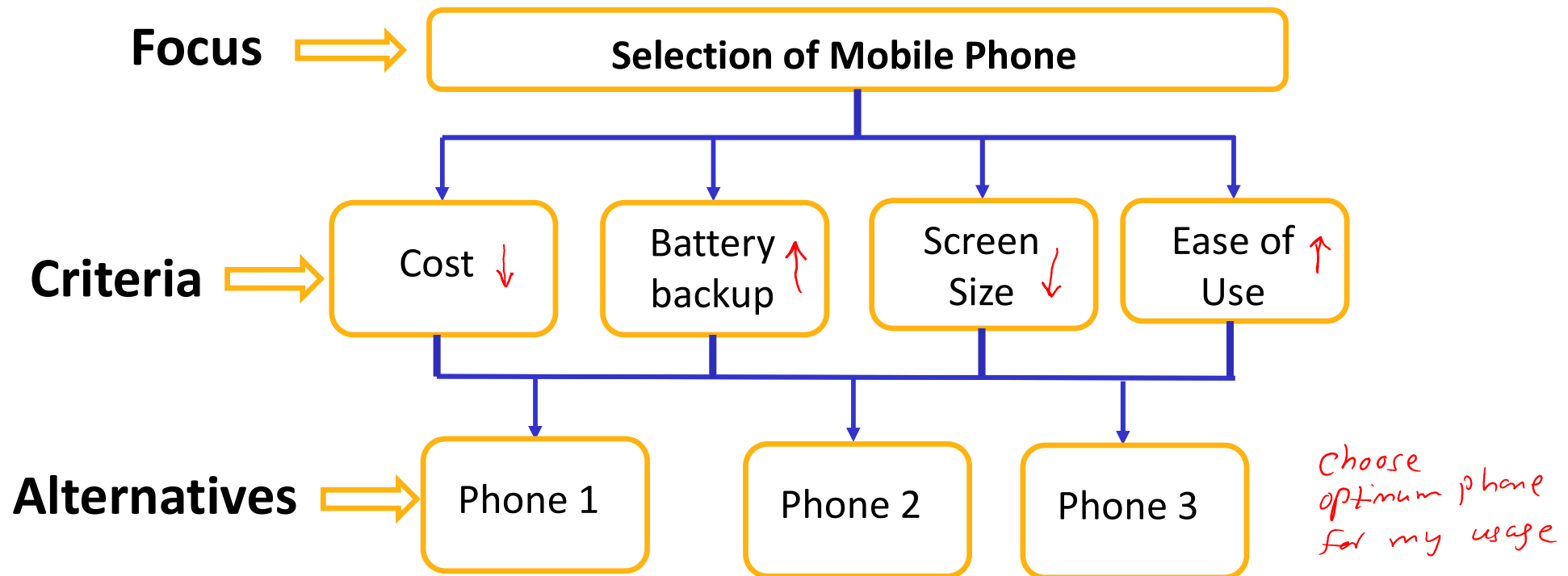
# AHP Example

## Analytic Hierarchy Process (AHP) Tool



- Apply AHP for selecting a mobile phone.

### Step 1: Develop AHP model



# AHP Example



Qualitative

|         | Cost ↓ | Battery backup ↑ | Screen size ↓ | Ease of Use ↑ |
|---------|--------|------------------|---------------|---------------|
| Phone 1 | 15000  | 4000             | 5.6           | 5             |
| Phone 2 | 11000  | 3000             | 6             | 3             |
| Phone 3 | 16000  | 2500             | 5.2           | 8             |

Spec.

|      | Cost | B.B. | S.S. | E.U. |
|------|------|------|------|------|
| Cost | 1    | 1/5  | 1/3  | 1/7  |
| B.B. | 5    | 1    | .    | .    |
| S.S. | 3    | 3    | 1    | .    |
| E.U. | 7    | 7    | .    | 1    |

matrix  
pairwise criteria  
Comparison

# Analytic Hierarchy Process (AHP)



- AHP is relevant for choice problems in the following categories:
  - Comparing untested concepts
  - Structuring a decision-making process for a new situation
  - Evaluating non commensurate trade-offs
  - Performing and tracking group decision making
  - Integrating results from different sources (e.g., analytical calculations, HOQ relative values, group consensus, and expert opinion);
  - Performing strategic decision making.

# AHP Process



**AHP's Ratings for Pairwise Comparison of Selection Criteria**

| Rating Factor | Relative Rating of Importance of Two Selection Criteria A and B                                     | Explanation of Rating                                                                                              |
|---------------|-----------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|
| 1             | A and B have equal importance.                                                                      | A and B both contribute equally to the product's overall success.                                                  |
| 3             | A is thought to be moderately more important than B.                                                | <u>A</u> is slightly more important to product success than <u>B</u> .                                             |
| 5             | A is thought to be strongly more important than B.                                                  | A is <u>strongly</u> more important to product success than B.                                                     |
| 7             | A is thought to be very much more important than B, or is demonstrated to be more important than B. | A's <u>dominance</u> over B has been <u>demonstrated</u> .                                                         |
| 9             | A is demonstrated to have much more importance than B.                                              | There is the <u>highest possible</u> degree of evidence that proves A is more important to product success than B. |

*The ratings of even numbers 2, 4, 6, and 8 are used when the decision maker needs to compromise between two positions in the table.*

2  
4  
6  
8

# AHP Process



- The rating of pair A to pair B is the reciprocal of the rating of pair B to A. That means if it is determined that A is strongly more important than B, the rating of A to B is set as 5. This makes the rating of B to A  $1/5$  or 0.20.
- AHP Process:
  - Complete criteria comparison matrix [C] using 1–9 ratings described.  $A_1$
  - Normalize the matrix [C] to give [NormC].  $A_2$
  - Average row values. This is the weight vector {W}.  $A_1 \times A_2$
  - Perform a consistency check on [C].

# AHP Example



- Derive priorities (weights) for the criteria

|                         | 1             | 2             | 3      | 4     |
|-------------------------|---------------|---------------|--------|-------|
| Attribute Or Criteria → | Price or Cost | Storage Space | Camera | Looks |

**Step 2)** Determine the relative importance of different attributes or Criteria with respect to the goal.



**Pair-wise comparison matrix**

## AHP Example Step 1 -Complete comparison matrix [C] using 1–9 ratings



### A1 Matrix

|                    |                               |
|--------------------|-------------------------------|
| 1                  | Equal importance              |
| 3                  | Moderate importance           |
| 5                  | Strong importance             |
| 7                  | Very strong importance        |
| 9                  | Extreme importance            |
| 2, 4, 6, 8         | Intermediate values           |
| 1/3, 1/5, 1/7, 1/9 | Values for inverse comparison |

Q. To be asked  
How important is the  
cost over battery  
backup?  
The answer will be from  
decision maker in the  
form of 1 to 9

|                | Cost | Battery backup | Screen Size | Ease of Use |
|----------------|------|----------------|-------------|-------------|
| Cost           | 1    | 1/7            | 1/5         | 1/3         |
| Battery backup | 7    | 1              | 2           | 3           |
| Screen Size    | 5    | 1/2            | 1           | 3           |
| Ease of Use    | 3    | 1/3            | 1/3         | 1           |



# A2 Matrix

| A                  | B                                                                 | C                         | D                                         | E               | F |
|--------------------|-------------------------------------------------------------------|---------------------------|-------------------------------------------|-----------------|---|
| Criteria Weightage | Normalize the matrix [C] to give [NormC].<br>(Column B*C*D*E)^1/4 |                           |                                           |                 |   |
| Cost               | $1 \times 1/7 \times 1/5 \times 1/3 = 0.01$                       | $\sqrt[4]{0.01} = 0.3124$ |                                           |                 |   |
| Battery backup     | $7 \times 1 \times 2 \times 3 = 42$                               | $\sqrt[4]{42} = 2.5457$   |                                           |                 |   |
| Screen Size        | $5 \times \frac{1}{2} \times 1 \times 3 = 7 \frac{1}{2}$          | $\sqrt[4]{7.5} = 1.6549$  |                                           |                 |   |
| Ease of Use        | $3 \times \frac{1}{3} \times \frac{1}{3} \times 1 = 1/3$          | $\sqrt[4]{1/3} = 0.7598$  |                                           |                 |   |
| Sum of all         |                                                                   | 5.2728                    |                                           |                 |   |
| [NormC]            | Cost                                                              | 0.0592                    | C11/C15                                   | $0.3124 / 5.27$ |   |
| A2                 | Battery backup                                                    | 0.4828                    | C12/C15                                   |                 |   |
|                    | Screen Size                                                       | 0.3138                    | C13/C15                                   |                 |   |
|                    | Ease of Use                                                       | 0.1441                    | C14/C15                                   |                 |   |
|                    |                                                                   | 1                         | If sum is 1 your calculations are correct |                 |   |



## A3 and A4 Matrix

| A                                                 | B                        | C                               | D                              | E | F | G | H | I | J |
|---------------------------------------------------|--------------------------|---------------------------------|--------------------------------|---|---|---|---|---|---|
| <b>Consistency check</b>                          |                          |                                 |                                |   |   |   |   |   |   |
| A3                                                | A1*A2                    | 0.239                           |                                |   |   |   |   |   |   |
|                                                   |                          | 1.9571                          |                                |   |   |   |   |   |   |
|                                                   |                          | 1.2835                          |                                |   |   |   |   |   |   |
|                                                   |                          | 0.5872                          |                                |   |   |   |   |   |   |
| Average row values. This is the weight vector {W} |                          |                                 |                                |   |   |   |   |   |   |
| A4                                                | A3/A2                    | 4.034                           |                                |   |   |   |   |   |   |
|                                                   |                          | 4.0545                          |                                |   |   |   |   |   |   |
|                                                   |                          | 4.0905                          |                                |   |   |   |   |   |   |
|                                                   |                          | 4.0762                          |                                |   |   |   |   |   |   |
|                                                   | Avarage                  | <b>4.0638</b>                   | $\lambda_{max}$                |   |   |   |   |   |   |
|                                                   | <b>Consistency index</b> | $\lambda_{max} - n / n-1$       |                                |   |   |   |   |   |   |
|                                                   |                          | <b>0.021266667</b>              |                                |   |   |   |   |   |   |
|                                                   | <b>Consistency ratio</b> | Consistency index /Random Index |                                |   |   |   |   |   |   |
|                                                   |                          | 0.02362963                      | If <0.1 weightages are correct |   |   |   |   |   |   |

### Random Index

- Random index (RI) is the consistency index of a randomly generated pairwise comparison matrix.
  - RI depends on the number of elements being compared (i.e., size of pairwise comparison matrix) and takes on the following values:

| n  | 1    | 2    | 3    | 4    | 5    | 6    | 7    | 8    | 9    | 10   |
|----|------|------|------|------|------|------|------|------|------|------|
| RI | 0.00 | 0.00 | 0.58 | 0.90 | 1.12 | 1.24 | 1.32 | 1.41 | 1.45 | 1.49 |



| Priority       |             |   |
|----------------|-------------|---|
| Cost           | 0.059245926 | 4 |
| Battery backup | 0.482800975 | 1 |
| Screen Size    | 0.313849276 | 2 |
| Ease of Use    | 0.144103823 | 3 |

Battery backup > screen size > ease of use > cost

|         | Cost ↓ | Battery backup ↑ | Screen size ↓ | Ease of Use ↑ |
|---------|--------|------------------|---------------|---------------|
| Phone 1 | 15000  | 4000             | 5.6           | 5             |
| Phone 2 | 11000  | 3000             | 6             | 3             |
| Phone 3 | 16000  | 2500             | 5.2           | 8             |

## Normalize the Matrix

|         | cost (Rs) (↓)                  | Battery Backup (MAh) (↑)    | Screen size (inches) (↓)   | Ease of use (↑)       |
|---------|--------------------------------|-----------------------------|----------------------------|-----------------------|
| phone 1 | $\frac{11000}{15000} = 0.7333$ | $\frac{4000}{4000} = 1.0$   | $\frac{5.2}{5.6} = 0.9285$ | $\frac{5}{8} = 0.625$ |
| phone 2 | $\frac{11000}{11000} = 1.0$    | $\frac{3000}{4000} = 0.75$  | $\frac{5.2}{6} = 0.86667$  | $\frac{3}{8} = 0.375$ |
| phone 3 | $\frac{11000}{16000} = 0.6875$ | $\frac{2500}{4000} = 0.625$ | $\frac{5.2}{5.2} = 1.0$    | $\frac{8}{8} = 1.0$   |



fractional value of how that criteria for the concept is deviating from ideal

Now, synthesize the model

|         | cost     | Battery Backup | Screen size | Ease of use |
|---------|----------|----------------|-------------|-------------|
| Weights | 0.059246 | 0.4828         | 0.31385     | 0.1441      |
| phone 1 | 0.7333   | 1.0            | 0.9285      | 0.625       |
| phone 2 | 1.0      | 0.75           | 0.86667     | 0.375       |
| phone 3 | 0.6875   | 0.625          | 1.0         | 1.0         |

phone 2  $\Rightarrow$

$$0.059246 \times 1 + 0.4828 \times 0.75 + 0.31385 \times 0.86667 + 0.1441 \times 0.375$$

fraction. = phone 1  
phone 2  
phone 3

$$\text{Phone 1} = (0.7333 \times 0.059246) + (1.0 \times 0.4828) \\ + (0.9285 \times 0.31305) + (0.665 \times 0.144)$$

|         |           |         |
|---------|-----------|---------|
| Phone 1 | $\hat{=}$ | 0.90774 |
| Phone 2 | $=$       | 0.7474  |

$$\text{Phone 3} = 0.50043$$

# Pros and cons of AHP method



Pros

- It allows **multi criteria decision making**.
- It is applicable when it is difficult to formulate criteria evaluations, i.e., it allows **qualitative evaluation** as well as quantitative evaluation.
- It is applicable for **group decision making** environments

Cons

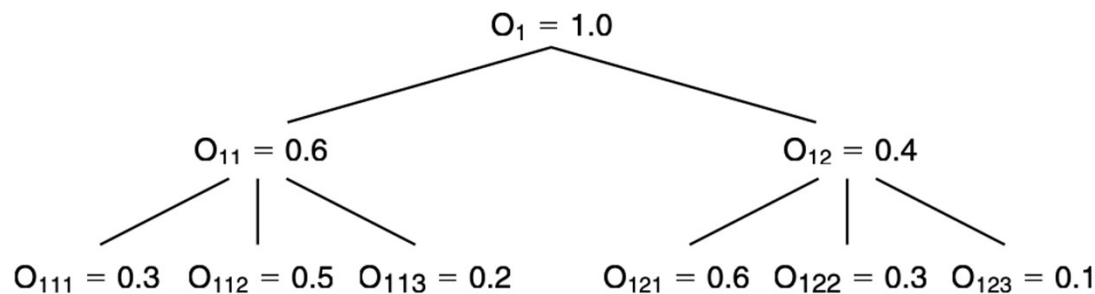
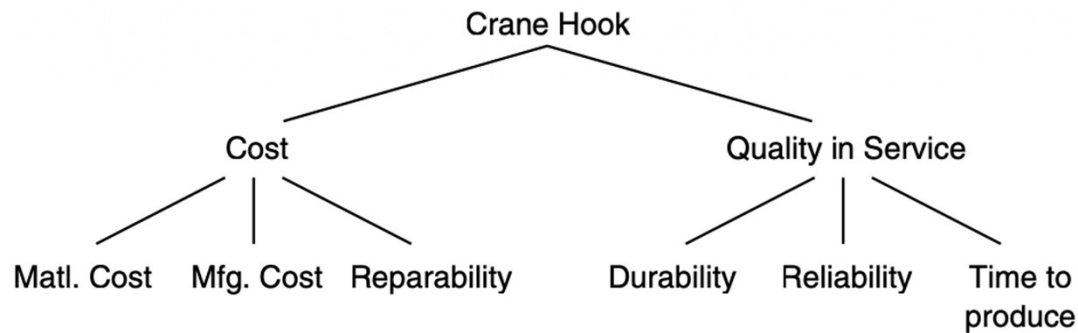
- There are hidden assumptions like **consistency**. Repeating evaluations is cumbersome.
- Difficult to use when the **number of criteria or alternatives is high, i.e., more than 7**.
- Difficult to add a **new criterion or alternative**
- Difficult to **take out an existing criterion or alternative**, since the best alternative might differ if the worst one is excluded.

Users should be trained to use AHP methodology.

Use **GDSS**  
Use **constraints** to eliminate some alternatives

Use **cost/benefit** ratio if applicable

# Objective Tree for Weighting Factors



Weighting factor for material cost,  $O_{111}$   
 $= 0.3 \times 0.6 \times 1.0 = 0.18$

# Criteria Comparison Matrix: Crane Hook



**Criteria Comparison Matrix [C]**

|                      | <b>Material<br/>Cost</b> | <b>Mfg Cost</b> | <b>Reparability</b> | <b>Durability</b> | <b>Reliability</b> | <b>Time Prod</b> |
|----------------------|--------------------------|-----------------|---------------------|-------------------|--------------------|------------------|
| <b>Material Cost</b> | 1.00                     | 0.33            | 0.20                | 0.11              | 0.14               | 3.00             |
| <b>Mfg Cost</b>      | 3.00                     | 1.00            | 0.33                | 0.14              | 0.33               | 3.00             |
| <b>Reparability</b>  | 5.00                     | 3.00            | 1.00                | 0.20              | 0.20               | 3.00             |
| <b>Durability</b>    | 9.00                     | 7.00            | 5.00                | 1.00              | 3.00               | 7.00             |
| <b>Reliability</b>   | 7.00                     | 3.00            | 5.00                | 0.33              | 1.00               | 9.00             |
| <b>Time Prod</b>     | 0.33                     | 0.33            | 0.33                | 0.14              | 0.11               | 1.00             |
| <b>Sum</b>           | 25.33                    | 14.67           | 11.87               | 1.93              | 4.79               | 26.00            |

# Normalised Criteria Comparison Matrix: Crane Hook



|                      | Material |          |              |            |             |           | Criteria    |
|----------------------|----------|----------|--------------|------------|-------------|-----------|-------------|
|                      | Cost     | Mfg Cost | Reparability | Durability | Reliability | Time Prod | Weights {W} |
| <b>Material Cost</b> | 0.039    | 0.023    | 0.017        | 0.058      | 0.030       | 0.115     | 0.047       |
| <b>Mfg Cost</b>      | 0.118    | 0.068    | 0.028        | 0.074      | 0.070       | 0.115     | 0.079       |
| <b>Reparability</b>  | 0.197    | 0.205    | 0.084        | 0.104      | 0.042       | 0.115     | 0.124       |
| <b>Durability</b>    | 0.355    | 0.477    | 0.421        | 0.518      | 0.627       | 0.269     | 0.445       |
| <b>Reliability</b>   | 0.276    | 0.205    | 0.421        | 0.173      | 0.209       | 0.346     | 0.272       |
| <b>Time Prod</b>     | 0.013    | 0.023    | 0.028        | 0.074      | 0.023       | 0.038     | 0.033       |
| <b>Sum</b>           | 1.000    | 1.000    | 1.000        | 1.000      | 1.000       | 1.000     | 1.000       |

Average Row  
Values

# Consistency Check Process in AHP



As the number of criteria increases, it is difficult to assure consistency. That is why the AHP process includes a consistency check on [C].

1. Calculate weighted sum vector,  $\{Ws\} = [C] \times \{W\}$
2. Calculate consistency vector,  $\{Cons\} = \{Ws\}/\{W\}$
3. Estimate  $\lambda$  as the average of values in  $\{Cons\}$
4. Evaluate consistency index,  $CI = (\lambda - n)/(n - 1)$
5. Calculate consistency ratio,  $CR = CI/RI$ . The random index (RI) values are the consistency index values for randomly generated versions of [C]. The values for RI are listed in Table 7.11. The rationale for this comparison is that the [C] matrix constructed by a knowledgeable decision maker will show much more consistency than a matrix randomly populated with values from 1 to 9.
6. If  $CR < 0.1$  the  $\{W\}$  is considered to be valid; *otherwise adjust [C] entries and repeat.*

# Consistency Check for Crane Hook (Criteria)



Criteria Comparison Matrix [C]

|               | Material Cost | Mfg Cost | Reparability | Durability | Reliability | Time Prod |
|---------------|---------------|----------|--------------|------------|-------------|-----------|
| Material Cost | 1.00          | 0.33     | 0.20         | 0.11       | 0.14        | 3.00      |
| Mfg Cost      | 3.00          | 1.00     | 0.33         | 0.14       | 0.33        | 3.00      |
| Reparability  | 5.00          | 3.00     | 1.00         | 0.20       | 0.20        | 3.00      |
| Durability    | 9.00          | 7.00     | 5.00         | 1.00       | 3.00        | 7.00      |
| Reliability   | 7.00          | 3.00     | 5.00         | 0.33       | 1.00        | 9.00      |
| Time Prod     | 0.33          | 0.33     | 0.33         | 0.14       | 0.11        | 1.00      |
| Sum           | 25.33         | 14.67    | 11.87        | 1.93       | 4.79        | 26.00     |

|               | Material Cost | Mfg Cost | Reparability | Durability | Reliability | Time Prod | Criteria Weights {W} |
|---------------|---------------|----------|--------------|------------|-------------|-----------|----------------------|
| Material Cost | 0.039         | 0.023    | 0.017        | 0.058      | 0.030       | 0.115     | 0.047                |
| Mfg Cost      | 0.118         | 0.068    | 0.028        | 0.074      | 0.070       | 0.115     | 0.079                |
| Reparability  | 0.197         | 0.205    | 0.084        | 0.104      | 0.042       | 0.115     | 0.124                |
| Durability    | 0.355         | 0.477    | 0.421        | 0.518      | 0.627       | 0.269     | 0.445                |
| Reliability   | 0.276         | 0.205    | 0.421        | 0.173      | 0.209       | 0.346     | 0.272                |
| Time Prod     | 0.013         | 0.023    | 0.028        | 0.074      | 0.023       | 0.038     | 0.033                |
| Sum           | 1.000         | 1.000    | 1.000        | 1.000      | 1.000       | 1.000     | 1.000                |

| # of Criteria | RI Value |
|---------------|----------|
| 3             | 0.52     |
| 4             | 0.89     |
| 5             | 1.11     |
| 6             | 1.25     |
| 7             | 1.35     |

Consistency Check

| $\{Ws\} = [C]\{W\}^T$<br>Weighted Sum Vector | $\{W\}$ Criteria Weights | $\{Cons\} = \{Ws\}/\{W\}$<br>Consistency Vector |
|----------------------------------------------|--------------------------|-------------------------------------------------|
| 0.286                                        | 0.047                    | 6.093                                           |
| 0.515                                        | 0.079                    | 6.526                                           |
| 0.839                                        | 0.124                    | 6.742                                           |
| 3.090                                        | 0.445                    | 6.950                                           |
| 1.908                                        | 0.272                    | 7.022                                           |
| 0.210                                        | 0.033                    | 6.324                                           |
| Average of {Cons} = $\lambda$                |                          | 6.610                                           |

$$\text{Consistency Index, CI} = (\lambda - n)/(n - 1) \quad 0.122$$

$$\text{Consistency Ratio, CR} = \text{CI}/\text{RI} \quad 0.098^2$$

**Is Comparison Consistent:**  $\text{CR} < 0.10$  Yes

**Next: Similar method for rating the design alternatives**

# AHP Ratings for Pairwise Comparison of Concepts



| Rating Factor | Relative Rating of the Performance of Alternative A Compared to Alternative B | Explanation of Rating                                                                                         |
|---------------|-------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|
| 1             | A = B                                                                         | The two are the same with respect to the criterion in question.                                               |
| 3             | A is thought to be moderately superior to B.                                  | Decision maker slightly favors A over B.                                                                      |
| 5             | A is thought to be strongly superior to B.                                    | Decision maker strongly favors A over B.                                                                      |
| 7             | A is demonstrated to be superior to B.                                        | A's dominance over B has been demonstrated.                                                                   |
| 9             | A is demonstrated to be absolutely superior to B.                             | There is the highest possible degree of evidence that proves A is superior to B under appropriate conditions. |

# AHP Process for Determining Ratings for Concepts



Process to determine ratings of design alternatives, with respect to their performance for each criteria, is:

- (1) Complete comparison matrix  $[C]$  using 1–9 ratings
- (2) Normalise the matrix  $[C]$  to give  $[\text{Norm}C]$
- (3) Average row values. This is the Vector Priority  $\{P_i\}$ .
- (4) Perform a consistency check on  $[C]$

# [C] & [NormC] for Crane Hook (Concepts)



## Design Alternative Ratings for Material Cost

| Material Cost Comparison [C] |             |              |            |
|------------------------------|-------------|--------------|------------|
|                              | Plates Weld | Plates Rivet | Cast Steel |
| Plates Weld                  | 1.000       | 1.000        | 0.333      |
| Plates Rivet                 | 1.000       | 1.000        | 0.333      |
| Cast Steel                   | 3.000       | 3.000        | 1.000      |
| Sum                          | 5.000       | 5.000        | 1.667      |

## Normalized Cost Comparison [NormC]

|              | Plates Weld | Plates Rivet | Cast Steel | Design Alternative Priorities {P <sub>i</sub> } |
|--------------|-------------|--------------|------------|-------------------------------------------------|
| Plates Weld  | 0.200       | 0.200        | 0.200      | 0.200                                           |
| Plates Rivet | 0.200       | 0.200        | 0.200      | 0.200                                           |
| Cast Steel   | 0.600       | 0.600        | 0.600      | 0.600                                           |
|              | 1.000       | 1.000        | 1.000      | 1.000                                           |

# Consistency Check for Crane Hook (Concepts)



| Consistency Check                            |                                     |                                                   |
|----------------------------------------------|-------------------------------------|---------------------------------------------------|
| $\{Ws\} = [C]\{P_i\}$<br>Weighted Sum Vector | $\{P_i\}$ Alternative<br>Priorities | $\{Cons\} = \{Ws\}/\{P_i\}$<br>Consistency Vector |
| 0.600                                        | 0.200                               | 3.000                                             |
| 0.600                                        | 0.200                               | 3.000                                             |
| 1.800                                        | 0.600                               | 3.000                                             |
|                                              | Average of $\{Cons\} =$             | 3.000                                             |
|                                              | Consistency Index, CI =             | 0                                                 |
|                                              | Consistency Ratio, CR =             | 0                                                 |
|                                              | <i>Is Comparison Consistent</i>     | <i>YES</i>                                        |

# Final Rating Matrix: Crane Hook (Concepts)



| Selection Criteria | Welded Plates | [FRating]<br>Riveted Plates | Cast Steel |
|--------------------|---------------|-----------------------------|------------|
| Material Cost      | 0.200         | 0.200                       | 0.600      |
| Manufacturing Cost | 0.260         | 0.633                       | 0.106      |
| Reparability       | 0.292         | 0.615                       | 0.093      |
| Durability         | 0.429         | 0.429                       | 0.143      |
| Reliability        | 0.260         | 0.633                       | 0.105      |
| Time to Produce    | 0.260         | 0.633                       | 0.106      |

Normalized Cost Comparison [NormC]

|              | Plates Weld | Plates Rivet | Cast Steel | Design Alternative<br>Priorities {P <sub>i</sub> } |
|--------------|-------------|--------------|------------|----------------------------------------------------|
| Plates Weld  | 0.200       | 0.200        | 0.200      | 0.200                                              |
| Plates Rivet | 0.200       | 0.200        | 0.200      | 0.200                                              |
| Cast Steel   | 0.600       | 0.600        | 0.600      | 0.600                                              |
|              | 1.000       | 1.000        | 1.000      | 1.000                                              |

# Alternative Value: Crane Hook



|               | Material Cost | Mfg Cost | Reparability     | Durability | Reliability | Time Prod | Criteria Weights {W} |
|---------------|---------------|----------|------------------|------------|-------------|-----------|----------------------|
| Material Cost | 0.039         | 0.023    | 0.017            | 0.058      | 0.030       | 0.115     | 0.047                |
| Mfg Cost      | 0.118         | 0.068    | 0.028            | 0.074      | 0.070       | 0.115     | 0.079                |
| Reparability  | 0.197         | 0.20     | Criteria Weights |            |             | 0.115     | 0.124                |
| Durability    | 0.355         | 0.47     |                  |            |             | 0.269     | 0.445                |
| Reliability   | 0.276         | 0.205    | 0.421            | 0.173      | 0.209       | 0.346     | 0.272                |
| Time Prod     | 0.013         | 0.023    | 0.028            | 0.074      | 0.023       | 0.038     | 0.033                |
| Sum           | 1.000         | 1.000    | 1.000            | 1.000      | 1.000       | 1.000     | 1.000                |

| Selection Criteria | Welded Plates       | [FRating]<br>Riveted Plates | Cast Steel |
|--------------------|---------------------|-----------------------------|------------|
| Material Cost      | 0.200               | 0.200                       | 0.600      |
| Manufacturing Cost | 0.260               | 0.633                       | 0.106      |
| Reparability       | Final Rating Matrix |                             | 0.093      |
| Durability         |                     |                             | 0.143      |
| Reliability        | 0.260               | 0.633                       | 0.105      |
| Time to Produce    | 0.260               | 0.633                       | 0.106      |

$$[FRating]^T \{W\} = \{Alternative Value\}$$

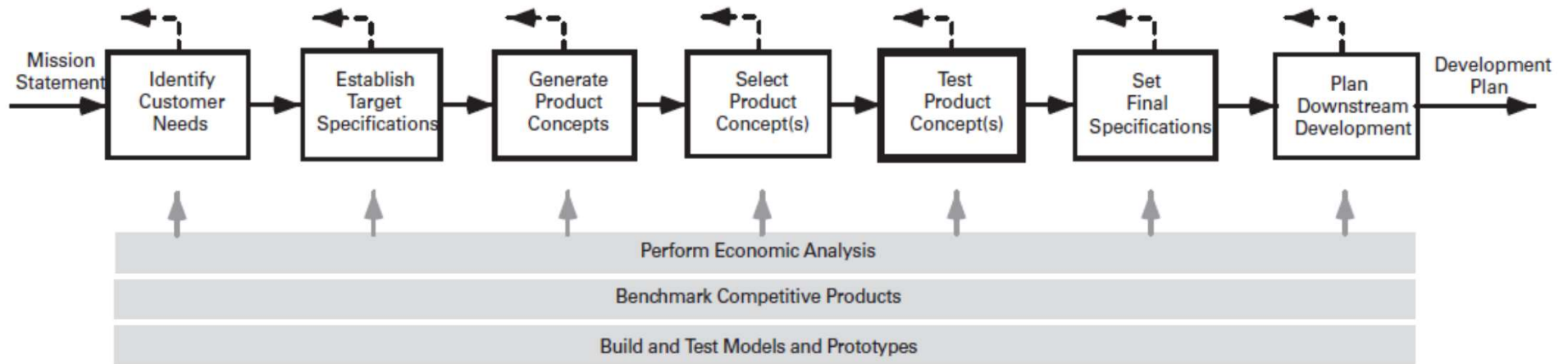
|                      | Alternative Value |
|----------------------|-------------------|
| Welded plate design  | 0.336             |
| Riveted plate design | 0.520 ✓           |
| Monolithic casting   | 0.144             |



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# Session 7: Concept Testing

# Concept Testing



# Concept Testing Process



- Concept testing is based on listening to the audience's views and exploring how viable the concept is for them – without predefined parameters or expectations.
- It's an open learning process where you can continue to engage with the same target audience and get their take on your product developments as they happen.

1. Define the purpose of the concept test. *⇒ finalise Concept-  
ip from stakeholder*
2. Choose a survey population. *writing model  
story board*
3. Choose a survey format. *CAD*
4. Communicate the concept. *Desc.  
Video  
prototyping*
5. Measure customer response. *feedback*
6. Interpret the results. *Analyse*
7. Reflect on the results and the process.

*stake holders*

*Customer  
End users  
Vendors  
partners*

# Concept Testing Process

---



- **Step 1: Define the Purpose of the Concept Test**
  - Which of several alternative concepts should be pursued?
  - How can the concept be improved to better meet customer needs?
  - Approximately how many units are likely to be sold?
  - Should development be continued?

# Concept Testing Process



- **Step 2: Choose a Survey Population**

---

## **Factors Favoring a Smaller Sample Size**

- Test occurs early in concept development process.
- Test is primarily intended to gather qualitative data.
- Surveying potential customers is relatively costly in time or money.
- Required investment to develop and launch the product is relatively small.
- A relatively large fraction of the target market is expected to value the product (i.e., many positively inclined respondents can be found without a large sample).

## **Factors Favoring a Larger Sample Size**

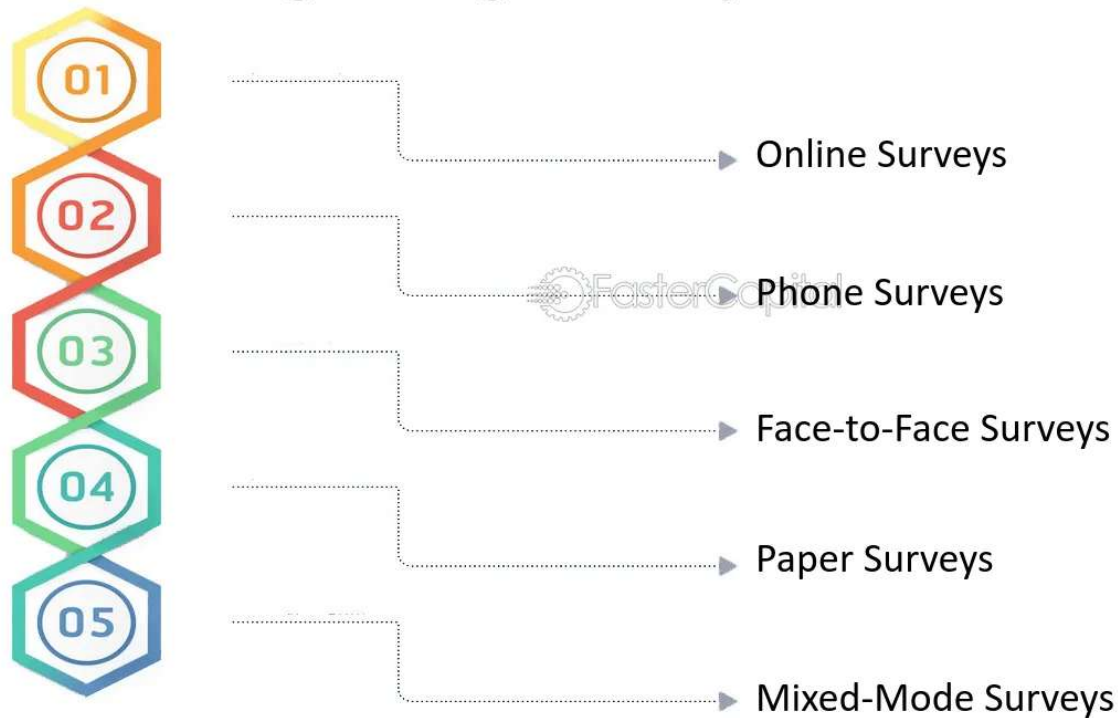
- Test occurs later in concept development process.
- Test is primarily intended to assess demand quantitatively.
- Surveying customers is relatively fast and inexpensive.
- Required investment to develop and launch the product is relatively high.
- A relatively small fraction of the target market is expected to value the product (i.e., many people have to be sampled to reliably estimate the fraction that values the product).

# Concept Testing Process



- **Step 3: Choose a Survey Format**

## Choosing the Right Survey Method and Format



# Concept Testing Process

---



- **Step 4: Communicate the Concept**

- Verbal description
- Sketch
- Photos and renderings
- Storyboard
- Video
- Simulation
- Interactive multimedia
- Physical appearance models

# Concept Testing Process

---



- **Step 5: Measure Customer Response**
- **Step 6: Interpret the Results**
- **Step 7: Reflect on the Results and the Process**

## Define the purpose (step 1)



Which of the alternative concepts should be pursued  
How can the concept be improved to better meet  
customer needs  
Approximately how much units are likely to be sold  
Should the development be continued

## Choose a survey population and sample size (step 2)



1. Sample size varies from a few to thousands
2. Factors affecting the sample size
  1. The stage of product development
  2. Cost to conduct survey
  3. Nature and intent of the survey
  4. Budget (amount) of the development project
  5. How possible to collect the intended information.
3. Possible to structure multiple surveys with different objectives at different stages.

## Choose a survey format (step 3)



### Formats

- Face to face interaction
- Telephone
- Postal mail
- Electronic mail
- Internet (a test site on the internet)

3D  
2D drawing  
Video / Animation

Each has its pros and cons

Each has its bias.

# Communicating the Concept



Sketch



Rendering



CAD Model



Story board



Beta Prototype



Working prototype





|                           | Telephone | Electronic<br>Mail | Postal<br>Mail | Internet | Face-to-<br>Face |
|---------------------------|-----------|--------------------|----------------|----------|------------------|
| Verbal description        | •         | •                  | •              | •        | •                |
| Sketch                    |           | •                  | •              | •        | •                |
| Photo or rendering        |           | •                  | •              | •        | •                |
| Storyboard                |           | •                  | •              | •        | •                |
| Video                     |           |                    |                | •        | •                |
| Simulation                |           |                    |                | •        | •                |
| Interactive multimedia    |           |                    |                | •        | •                |
| Physical appearance model |           |                    |                |          | •                |
| Working prototype         |           |                    |                |          | •                |



- **PART 1, Qualification**

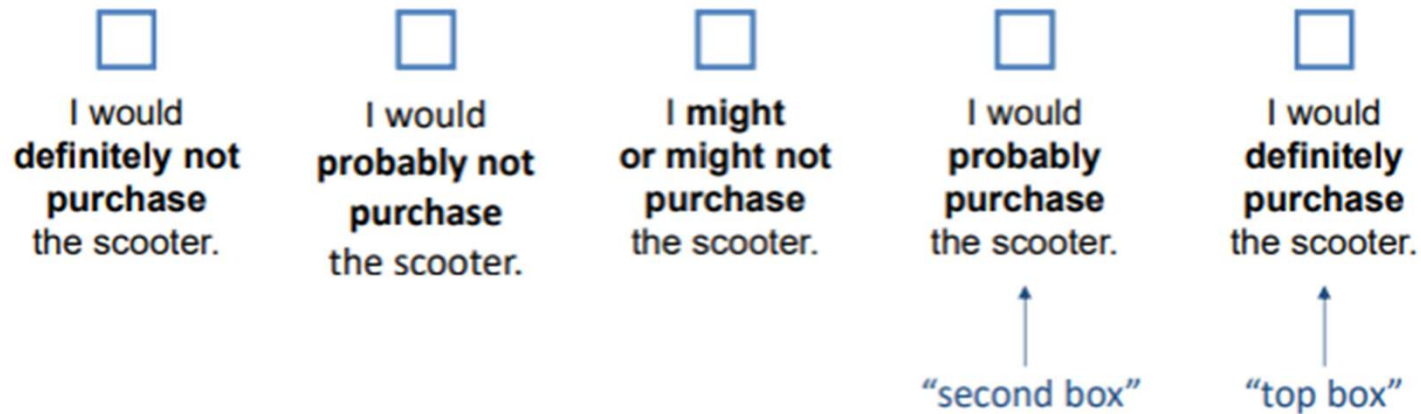
- How far do you live from campus?
  - <If not 1-3 miles, thank the customer and end interview.>
- How do you currently get to campus from home?
- How do you currently get around campus?

- **PART 2, Product Description**

The product is a lightweight electric scooter that can be easily folded and taken with you inside a building or on public transportation. The scooter weighs about 25 pounds. It travels at speeds of up to 15 miles per hour and can go about 12 miles on a single charge. The scooter can be recharged in about two hours from a standard electric outlet. The scooter is easy to ride and has simple controls — just an accelerator button and a brake

- **PART 3, Purchase Intent**

- If the product were priced according to your expectations, how likely would you be to purchase the scooter within the next year?





- **PART 4, Comments**

- What would you expect the price of the scooter to be?
- What concerns do you have about the product concept?
- Can you make any suggestions for improving the product concept?

- **Thank you**



The product is a lightweight electric scooter that can be easily folded and taken with you inside a building or on public transportation.

The scooter weighs about 25 pounds. It travels at speeds of up to 15 miles per hour and can go about 12 miles on a single charge.

The scooter can be recharged in about two hours from a standard electric outlet.

The scooter is easy to ride and has simple controls — just an accelerator button and a brake.

## Measure customer response (step 5)



### Measurement

- Measure their preferences among alternative concepts
- Understand why and how they respond to the product concepts
- Attempt to measure purchase intent (the likelihood of buying)



- *Applicable to future*
- *Considering alternate markets*
- *Was the concept communicated properly or not?*
- *Reality vs estimates (Sales)*

# Summary



- Concept selection is the process of evaluating concepts with respect to customer needs and other criteria, comparing the relative strengths and weaknesses of the concepts, and selecting one or more concepts for further investigation or development.
- All teams use some method, implicit or explicit, for selecting concepts. Decision techniques employed for selecting concepts range from intuitive approaches to structured methods.
- Successful design is facilitated by structured concept selection. Generally, it recommends a two-stage process: concept screening and concept scoring.
- Concept selection is applied not only during concept development but throughout the subsequent design and development process.
- Concept selection is a group process that facilitates the selection of a winning concept, helps build team consensus, and creates a record of the decision-making process.

# Summary

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- A concept test solicits a direct response to a description of the product concept from potential customers in the target market.
- Concept testing is distinct from concept selection in that it is based on data gathered directly from potential customers and relies to a lesser degree on judgments made by the development team.
- Concept testing can verify that customer needs have been adequately met by the product concept, assess the sales potential of a product concept, and/or gather customer information for refining the product concept.

# Summary

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- Concept testing is appropriate at several different points in the development process: when identifying the original product opportunity, when selecting which of two or more concepts should be pursued, when assessing the sales potential of a product concept, and/or when deciding whether to continue further development and commercialization of the product.



# THANK YOU!